

Disentangling Representation Effects and Normal Reference Matching in Few-Shot Industrial Anomaly Localization

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Abstract

Combining pretrained visual encoders can improve few-shot industrial anomaly localization, but an apparent fusion gain may reflect changed weights, additional information, or altered normal-reference selection. We introduce a controlled decomposition that distinguishes these effects using frozen encoders and identical normal supports. An exact duplicate control isolates reweighting; complementary representation replacements measure the value of additional descriptors at fixed slot or branch-group weights. Crossing these constructions with joint and independent reference matching yields a paired interaction. It is the difference between the representation effects under the two matching rules, and it separates absolute representation effects from their dependence on matching. For DINOv2-S/14, the two primary interactions on the Metal Parts Defect Detection (MPDD) dataset are +0.772 and +0.595 pixel average-precision percentage points, whereas their directions remain unresolved on the BeanTech Anomaly Detection (BTAD) dataset. Further validation supports positive interactions on MVTec AD and VisA, with VisA treated as in-domain. A separately specified KolektorSDD2 confirmation meets all four directional criteria for DINOv2-S/14 and WideResNet50-2. Encoder and support-set analyses reveal meaningful variation across conditions. The resulting evidence shows why representation composition and reference selection should be assessed together and provides a practical way to distinguish additional-feature value from reweighting and matching effects. Code and reproduction materials are available at <https://github.com/USEU117/reference-matching-interaction-ad>.

Keywords: few-shot anomaly localization; frozen visual encoders; normal reference matching; fixed feature fusion; representation interaction.

1 Introduction

Industrial visual inspection must locate defects whose appearance is difficult to enumerate before deployment. A new product may provide only a few verified normal images, while representative scratches, contamination, missing parts, or structural defects remain unavailable. Industrial anomaly benchmarks reflect this setting through normal training images and heterogeneous test anomalies [1, 2, 3, 4]. Few-shot anomaly localization therefore relies on a small normal support set to identify departures from expected appearance without training a target defect classifier.

Frozen pretrained encoders provide local descriptors that can be compared directly with a memory of normal support patches. AnomalyDINO demonstrates the effectiveness of DINOv2 features for this purpose [5, 6]. Combining encoders can enrich these descriptors, and recent systems already integrate visual and vision-language pretrained representations [7, 8]. However, comparing the final scores of a two-branch and a three-branch system leaves a central design question unresolved: what makes the additional representation useful?

Two effects are confounded in that comparison. First, adding an equally weighted branch also changes the weights of existing information. Copying one of two original branches adds no information, yet increases its effective weight from one half to two thirds. Replacing that copy with a different encoder is a distinct intervention whose benefit cannot be read directly from the original two-versus-three comparison. Second, the selected normal reference depends on how branch evidence is combined. Joint matching scores one shared support patch across branches before selecting the best candidate; independent matching allows each branch to select its own closest reference. A new descriptor can alter both the distance evidence and the chosen reference under a fixed matching rule.

This coupling makes two questions distinct: the absolute effect of a representation and whether that effect depends on matching. Independent matching has a lower or equal raw minimum-distance score, but average precision depends on the ranking of normal and anomalous pixels rather than score magnitude alone. Similarly, an added representation may help under both matching rules while showing little evidence that its gain depends on the rule. A controlled comparison must distinguish these possibilities instead of attributing the complete fusion gain to representation diversity.

Prior research establishes the importance of fusion design and consistent neighborhoods [9, 10, 11, 12]. We build on those ideas to isolate a specific unresolved attribution problem in dense industrial localization. With frozen encoders, identical supports and fixed distance weights, we cross two representation replacements with two reference-selection rules. The resulting paired contrasts quantify the value of the added descriptor and the change in that value when the shared-reference constraint is removed. This provides an interpretable basis for deciding whether a fusion improvement comes from reweighting, additional information, or their interaction with reference selection.

The contributions are as follows.

1. **A controlled decomposition of additional-representation value.** We construct an exact duplicate control—a comparator that changes the branch weights without adding descriptor information—to expose the weight change hidden in an ordinary two-versus-three-branch comparison. Replacing the duplicate at fixed slot weights isolates the representation change, while a complementary construction preserves the combined weight of the original visual branch and the added branch while retaining the AnomalyCLIP branch weight. Together, these controls distinguish new descriptor information from redistribution of existing evidence and make the attribution of a fusion gain testable.
2. **An interaction design and definition for representation and matching.** We cross both representation constructions with joint and independent normal-reference matching on the same support bank. The resulting difference-in-differences measures how much reference selection changes the effect of an added representation. Reporting this interaction alongside absolute representation effects separates two practically different outcomes: a useful representation with little established matching dependence, and a favorable matching interaction without a clear absolute gain. The contribution is this defined controlled comparison; it studies established matching operators through statistical contrasts while keeping the encoders frozen.
3. **Conditional empirical findings on absolute utility and matching sensitivity.** We identify a separation between an encoder's absolute utility and its sensitivity to reference selection. On MPDD, adding DINOv2-S/14 produces positive matching interactions without establishing a positive absolute effect under independent matching; on BTAD, its absolute effects are positive while the interaction direction remains unresolved. WideResNet50-2, additional datasets and a separately specified KolektorSDD2 confirmation extend the analysis beyond this contrast. These conditional findings explain why a favorable fusion score alone is insufficient to choose an encoder or a reference-sharing rule, and provide evidence for evaluating those choices together without claiming a universal encoder advantage.

The study centers on this attribution problem. The Dual-encoder baseline is the fixed reference configuration for the controlled comparisons; the results support conditional design choices rather than a universal encoder ranking.

2 Related Work

Research on industrial anomaly localization offers several ways to represent normal appearance, store reference evidence and combine pretrained features. Their differences matter for the present question: an improvement in a complete system can arise from training, reference coverage, fusion weights or reference selection. We review these choices together to explain which aspects our fixed-support comparison isolates.

Learned normal models identify anomalies through deviations from expected appearance or feature structure. SLSCG combines generative pretraining, simulated anomalies and graph-based embedding modeling [13]. RealNet uses diffusion-based anomaly synthesis with feature and residual selection [14]. PaDiM instead fits multivariate Gaussian distributions to pretrained patch embeddings [15]. These methods offer different ways to estimate normality, with corresponding training, adaptation and sample-size requirements. Learning a

normal distribution or reconstruction target can improve discrimination, but its effect is difficult to separate from encoder composition when the complete learning procedure changes. For our attribution question, this motivates freezing both the encoders and the target adaptation procedure; it does not imply that learned normal models are inferior under larger training sets.

Memory-based methods retain observed normal descriptors for local comparison. PatchCore combines a representative feature memory with nearest-neighbor scoring [16], while AnomalyDINO demonstrates effective few-shot localization with frozen DINOv2 patches [5]. Their performance depends on the descriptor, the coverage of the reference memory and the preprocessing that defines the evaluated image region.

Recent work expands these choices beyond adding encoders. FEAD enriches sparse normal patterns through conditional feature transformation and multi-frequency modeling [17]. K-NG studies few-shot online detection using an evolving Neural Gas representation [18]. PGAD combines global invariance, multiscale local evidence and score calibration [19], and FastRef refines normal prototypes at test time [20]. SubspaceAD models frozen descriptors through a principal subspace [21], while DCP-SFR preserves shallow defect information in deep representations [22]. These approaches highlight the roles of reference coverage, cue retention and adaptation. Changes to memory coverage or prototype adaptation also change which normal evidence a query can retrieve. Their overall improvements therefore answer a broader question than the effect of replacing one frozen descriptor. We hold the support identities and bank construction fixed to obtain that narrower comparison. The cost is that the study does not capture benefits from adaptive memories or target-specific feature learning.

Recent training-free methods also change how normal evidence is combined or updated: Hyper-FSAD uses sparse hyperedge aggregation [23], ReMem evolves a memory representation [24], and DuoAD reweights features across layers [25]. These mechanisms provide useful broader comparisons, while their aggregation, memory-update and layer-weighting choices remain outside the fixed operators studied here.

Contrastive Language–Image Pretraining (CLIP) provides transferable visual and textual representations [26]. WinCLIP combines prompt ensembles and local visual features for zero- and few-shot detection [27]. AnomalyCLIP learns object-agnostic normality and abnormality prompts [8], PromptAD learns from normal target samples [28], and InCTRL uses in-context residual learning with sample prompts [29]. AA-CLIP introduces anomaly-aware alignment [30], FAPrompt develops fine-grained abnormality prompts [31], and UniVAD extends training-free reference-based detection across domains [32]. Their supervision, adaptation and inference paths differ. The AnomalyCLIP visual branch uses only visual descriptors; text scores and learned prompt outputs do not enter anomaly scoring. This makes the scoring paths comparable as visual descriptor branches. The export code loads a prompt-learner checkpoint but obtains these descriptors directly from the frozen visual encoder; the learned prompt is not called on this path and does not update its visual weights. We retain the historical in-domain designation for VisA conservatively, without attributing the AnomalyCLIP visual descriptors to prompt learning or claiming an untouched-domain test.

Sea-CLIP is especially relevant because it combines CLIP and DINOv2 representations for few-shot detection and includes shared-reference-like and independent nearest-neighbor evidence [7]. M3DM integrates RGB and point-cloud features with multiple memories [33], and CIF uses hypergraph structural commonality to guide multimodal reference memories [34]. The architectural analysis in 3D-ADNAS examines conditional benefits of fusion modules and fusion stages [9]. These studies establish encoder combination and fusion design as existing approaches. Sea-CLIP is the closest practical motivation: combining CLIP and DINOv2 is already feasible, and reference selection already participates in such systems. A comparison of complete architectures, however, also changes feature processing or decoding. To interpret the contribution of an additional descriptor, its weight and support candidates must be controlled separately from those architectural choices.

Our emphasis is the attribution of the added representation's effect under controlled weights and references. We replace one descriptor while retaining support identities and slot weights, then measure how its effect changes under a different reference-sharing constraint. This differs from attributing a performance difference

between complete architectures to one added representation, because other architectural and adaptation choices remain fixed.

Multi-view anomaly detection studies how neighborhood agreement relates to anomalous behavior. MUVAD uses neighborhoods to distinguish cross-view inconsistency from instances that are unusual across views [11]. Neighborhood Consensus Networks align neighborhood structures [12], ECMOD combines contrastive learning with neighborhood-consistency information [35], and SCoNE addresses consistent local neighborhoods across views [10]. Shared and independent neighborhoods are therefore established concepts with relevance beyond industrial images. Their motivation also explains a tension in industrial localization: a shared reference can preserve correspondence across descriptors, whereas independent references can better accommodate each descriptor's nearest normal evidence. Neither rationale alone determines pixel-level ranking quality. This motivates measuring the representation effect under both rules rather than selecting a rule from the distance inequality.

Dense industrial localization introduces a specific setting: verified normal image supports, frozen patch descriptors and pixel-level evaluation. We investigate how changing the branch representation affects performance within this setting, using duplicate and balanced controls to identify the interaction with reference selection. The contribution concerns the controlled representation comparison and its empirical consequences, while the underlying neighborhood concepts come from prior work.

Taken together, prior work motivates our factorized comparison: it separates representation gain from reweighting and measures whether that gain depends on matching. This distinction allows an observed fusion improvement to be interpreted at the level of the design choice that produced it.

3 Framework and Methods

3.1 Problem Statement

For category c , the input consists of K verified normal support images and a query image $\mathbf{x}$. The support set is

$$\mathcal{X}_c = \{\mathbf{x}_i^c: i = 1, \dots, K\} \quad (1)$$

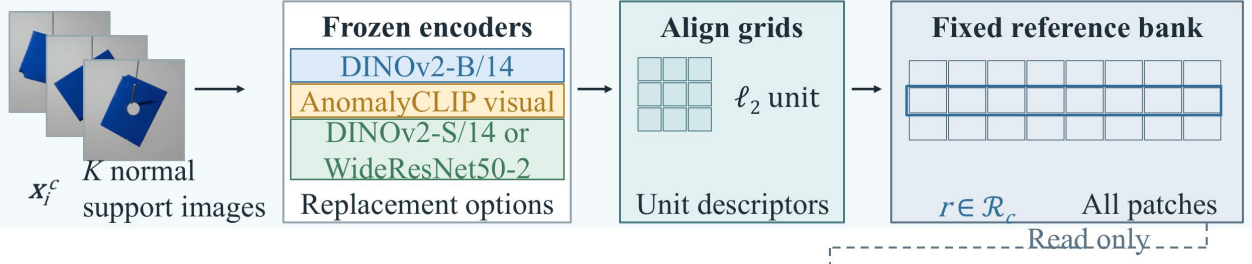
Here $\mathbf{x}_i^c$ is support image i of category c , and the query may be normal or anomalous. Category labels specify the relevant support bank; no target defect class is predicted. We use p for a query patch, r for an aligned support-patch index that contains both support-image identity and spatial position, and b for a visual branch. The same index r denotes the corresponding support location in every branch after spatial alignment. It does not require a query patch to match the same spatial coordinate in its support image.

The outputs are a continuous pixel anomaly map $\mathbf{A}_t$ and an image-level score $s_{\text{img},t}$, where the matching-rule label t is **Joint matching** or **Independent matching**. A displayed defect contour is derived from the map using a visualization threshold, as specified in Section 3.6. Test labels and masks are used only for offline evaluation and example selection. Dataset roles are distinguished in Section 4.1.1.

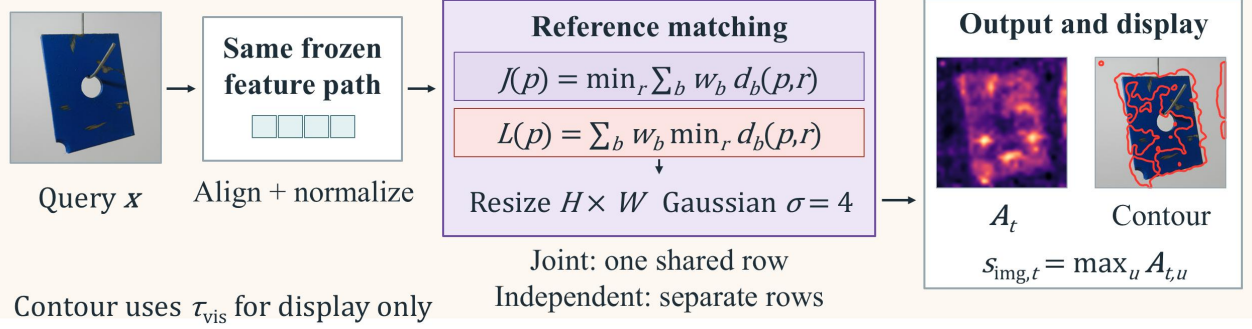
3.2 Framework Overview

Support-bank construction and query scoring use the same frozen visual paths. Each support image is encoded, spatially mapped to the common lattice, and normalized branch by branch. All support patches are retained, with one aligned row identity across branches. The bank—the stored normal patches used as references—is fixed for a category, support seed and K . A query follows the same feature path, searches that bank, and produces patch scores under one of the two matching rules. The query never updates the bank. Figure 1 shows the shared feature path, the two reference-selection rules and the weight-controlled representation comparisons.

(a) Construct the normal reference bank



(b) Score the query using the fixed bank



(c) Representation effects and matching interactions

Dual-encoder baseline	Duplicate-branch control	Equal-weight replacement	Balanced replacement
Original encoder pair	Copy existing features	Replace the duplicate	Preserve group weights
$E_{\text{Equal},t} = P(\text{Equal}_t) - P(\text{Duplicate}_t)$		$E_{\text{Balanced},t} = P(\text{Balanced}_t) - P(\text{Baseline}_t)$	
$I_{\text{Equal}} = E_{\text{Equal},L} - E_{\text{Equal},J}$		$I_{\text{Balanced}} = E_{\text{Balanced},L} - E_{\text{Balanced},J}$	
Identical supports and query images within every paired comparison			

Figure 1. Controlled framework and representation comparisons. (a) K normal support images x_i^c form a fixed, aligned reference bank. (b) The query follows the same frozen feature path. Joint matching selects one shared reference row; independent matching selects a row independently in each branch. For either matching rule, shared resizing and Gaussian smoothing produce the continuous map A_t , whose maximum is the image score. The second image panel is a thresholded display of the same map, not another learned output. (c) The duplicate-branch control isolates reweighting; the equal-weight replacement and the balanced replacement define the representation effects and their matching interactions. DINOv2-B/14 and AnomalyCLIP visual provide the baseline features; DINOv2-S/14 and WideResNet50-2 are the replacement features. The displayed map is the archived Dual-encoder baseline result under joint matching for MPDD metal_plate/test/scratches/026.png at seed 0 and $K = 1$. Multiple support thumbnails illustrate the general input, not that example's support count. The contour uses 256-bin Otsu thresholding solely for visualization.

The Dual-encoder baseline combines DINOv2-B/14 with the AnomalyCLIP visual path. Controlled variants duplicate the DINOv2-B/14 descriptor or replace that copy with a different frozen encoder. Shared resizing and smoothing convert patch scores to the continuous output map, whose spatial maximum gives the image score. Both matching rules are evaluated as fixed experimental conditions. Figure 2 shows their candidate selection. All encoders and fusion weights remain unchanged during target inference; this is a fixed experimental configuration, not a hyperparameter-free system.

3.3 Normal Reference Matching

Let F_b denote a branch feature tensor and R_b its deterministic mapping to the common spatial lattice. At location p , branch-wise normalization produces a unit descriptor $g_{b,p}$. The branch distance between a query descriptor and a support descriptor is

$$d_b(p, r) = 1 - g_{b,p}^T g_{b,r} \quad (2)$$

All reported constructions use nonnegative distance weights w_b summing to one. The descriptors in Equation (2) are nonzero; numerical norm handling follows the archived implementation. With the same aligned candidate set $\mathcal{R}_c$ in every branch, joint matching is

$$J(p) = \min_{r \in \mathcal{R}_c} \sum_b w_b d_b(p, r) \quad (3)$$

Joint matching first sums the branch distances for each single candidate r , then selects the candidate with the smallest weighted distance. Equivalently, concatenating the normalized branch descriptors with coefficients equal to the square roots of their distance weights yields a unit joint descriptor. Half its squared Euclidean distance equals the weighted cosine distance. Thus, feature coefficients and distance weights must not be confused: equal thirds in distance require equal square-root coefficients, followed by the corresponding joint normalization.

Independent matching first finds each branch's closest normal support and then combines the branch distances:

$$L(p) = \sum_b w_b \min_{r \in \mathcal{R}_c} d_b(p, r) \quad (4)$$

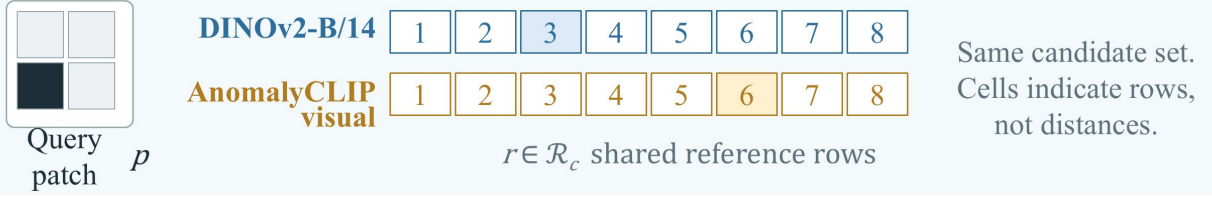
Each branch can now select a different support patch. We denote the corresponding reference indices by r_j and $r_{b,L}$ when discussing selected neighbors. Ties can select different indices with identical scores, so reference identity should be interpreted together with distance rather than as a unique explanation of every query.

The difference between the two raw patch scores is

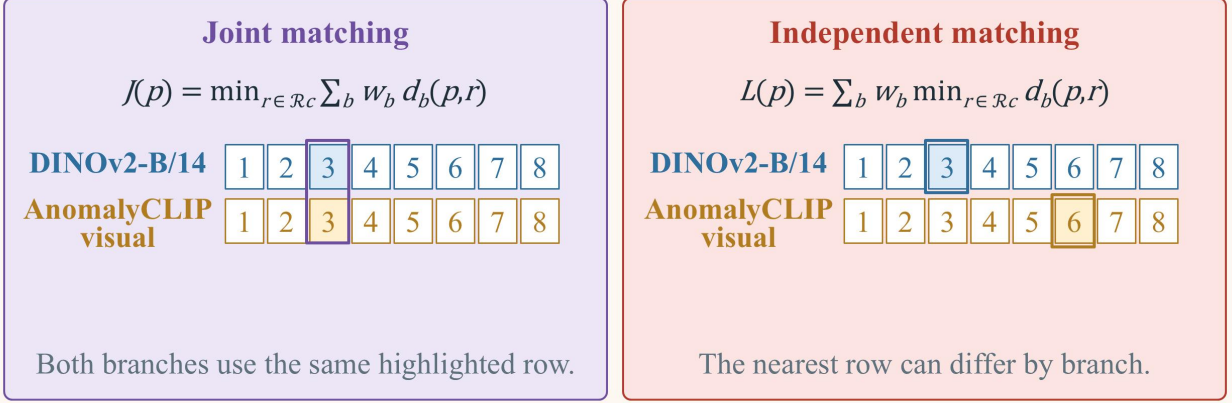
$$G(p) = J(p) - L(p) \geq 0 \quad (5)$$

For every candidate, each branch distance is at least its own minimum; weighting and summing preserves that inequality, and minimizing the weighted sum does not reverse it. Equality holds when the branch minima admit a common minimizer. A positive gap measures the distance cost of imposing a common reference. Localization quality must be assessed separately, because average precision depends on the ordering of normal and anomalous pixels. The same distinction applies after common linear resizing and smoothing.

(a) One query patch and the common candidate set



(b) Joint and independent matching



(c) The ordering constraint

$$G(p) = J(p) - L(p) \geq 0$$

$w_b \geq 0$ for every branch
The shared-row minimum cannot be lower.
Score magnitude does not determine pixel AP.

Figure 2. Joint and independent matching on the same fixed bank. (a) One query patch and the candidate reference rows $r = 1, \dots, 8$; the shaded cell in each branch row marks the candidate that branch finds closest. The cells are ordering indicators only and carry no measured value. (b) Joint matching requires a single shared reference row for all branches and then minimizes the weighted sum of distances over candidates, whereas independent matching lets each branch take its own nearest row before the weighted sum is formed. Both panels use the same aligned candidate set, the same branch descriptors and the same nonnegative weights that sum to one, so the two panels differ only in the order of combination and candidate minimization. (c) The ordering constraint $L(p)$ less than or equal to $J(p)$ follows from minimizing over a less restricted reference assignment. The gap is a raw-score difference, not an AP improvement.

3.4 Representation Constructions and Weight Controls

The four frozen encoder branches are DINOv2-B/14, DINOv2-S/14, the AnomalyCLIP visual branch and WideResNet50-2. Table 1 defines the Dual-encoder baseline, Duplicate-branch control, Equal-weight replacement and Balanced replacement. The baseline is the fixed reference configuration; each control changes a specified factor while retaining the other recorded choices. Branch and construction names, matching-rule names and descriptive subscripts are upright; scalar variables and score functions are italic; feature vectors and full maps are bold italic.

Table 1. Fixed representation constructions and the purpose of each control.

Construction	Description	Distance weights	Controlled comparison
Dual-encoder baseline	Dual-encoder baseline (DINOv2-B/14 + AnomalyCLIP visual)	DINOv2-B/14 1/2; AnomalyCLIP visual 1/2	Fixed dual-encoder baseline
Duplicate-branch control	Exact duplicate of the DINOv2-B/14 descriptor; no new encoder	DINOv2-B/14 1/3; DINOv2-B/14 copy 1/3; AnomalyCLIP visual 1/3	Reweight existing information; no new encoder
Equal-weight replacement	Equal-weight replacement (DINOv2-S/14 or WideResNet50-2 in the copied slot)	DINOv2-B/14 1/3; DINOv2-S/14 or WideResNet50-2 1/3; AnomalyCLIP visual 1/3	Replace the duplicate at fixed slot weights
Balanced replacement	Balanced replacement (DINOv2-S/14 or WideResNet50-2 in a quarter slot)	DINOv2-B/14 1/4; DINOv2-S/14 or WideResNet50-2 1/4; AnomalyCLIP visual 1/2	Preserve AnomalyCLIP visual and total non-AnomalyCLIP visual weights

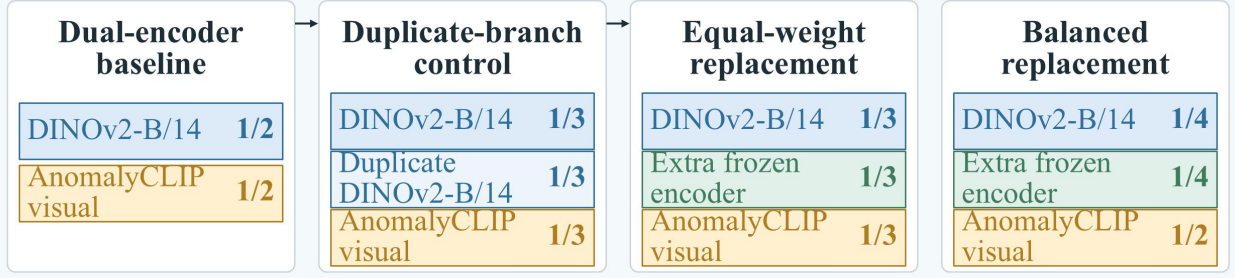
Each construction is evaluated under joint and independent matching. Bold identifies the dual-encoder baseline, not a claim of best performance. All encoder parameters are frozen. For DINOv2-S/14, balanced replacement also preserves the combined DINO-family weight.

Comparing Duplicate-branch control with Dual-encoder baseline changes only how the fixed weights are split, without adding a new representation. Duplicate-branch control is also numerically equivalent to a two-branch DINOv2-B/14 and the AnomalyCLIP visual branch construction with weights two thirds and one third. Replacing the copied DINOv2-B/14 descriptor in Duplicate-branch control with DINOv2-S/14 produces Equal-weight replacement at identical slot weights. Equal-weight replacement minus Duplicate-branch control therefore measures the effect of that representation replacement rather than the entire difference between two and three encoders.

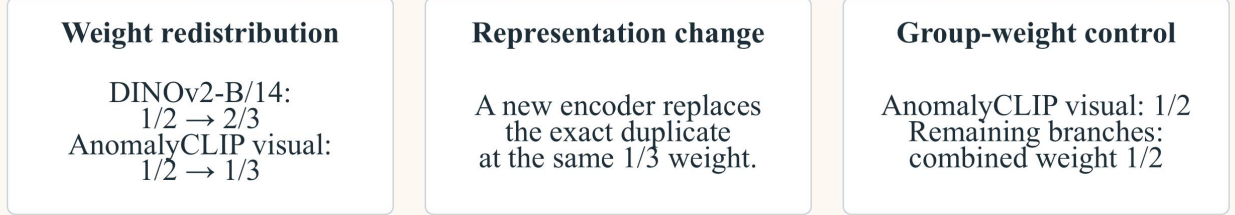
Balanced replacement provides a complementary comparison. In the DINOv2-S/14 experiment, the combined DINO family weight remains one half, while the AnomalyCLIP visual branch retains one half, allowing the DINOv2-B/14 allocation to be split between DINOv2-B/14 and DINOv2-S/14. In the WideResNet50-2 experiment, Balanced replacement preserves the total non-AnomalyCLIP visual weight but should not be called a DINO-family-preserving construction, since WideResNet50-2 is a convolutional ImageNet encoder. The weights are fixed experimental choices. Duplicate-branch control reuses the exact cached DINOv2-B/14 descriptor, ensuring that the duplicate introduces neither new information nor feature-extraction randomness.

Figure 3 summarizes the constructions and their paired contrasts.

(a) Four constructions with fixed weights



(b) What the paired comparisons isolate



Extra encoder: DINOv2-S/14 or WideResNet50-2.
Balanced replacement preserves the combined weight of the other branches.

(c) Representation effects and matching interactions



Paired contrasts use the same supports and queries under either rule.

Figure 3. Fixed representation constructions and the contrasts built from them. (a) The Dual-encoder baseline, Duplicate-branch control, Equal-weight replacement and Balanced replacement with their distance weights; a slot keeps the same colour across panels. (b) Moving from the Dual-encoder baseline to Duplicate-branch control changes only how the fixed weights are split, raising the effective DINOv2-B/14 weight from 1/2 to 2/3 and lowering the AnomalyCLIP visual weight from 1/2 to 1/3, with no new representation added, and is therefore the weight-confounding control; moving from Duplicate-branch control to Equal-weight replacement puts a real encoder into the slot at an unchanged weight; moving from the Dual-encoder baseline to Balanced replacement keeps the combined weight of the non-AnomalyCLIP visual branches and the AnomalyCLIP visual weight equal. (c) The four contrasts estimated in this paper. The Equal-weight replacement-to-Dual-encoder baseline difference moves both factors at once and is never attributed to DINOv2-S/14 alone. The extra slot is instantiated by five frozen encoders: DINOv2-S/14 and WideResNet50-2 were pre-specified, while DINO ViT-S/8, ConvNeXt-Tiny and Swin-Tiny were added after those results were known and are exploratory transfer checks. These are analysis configurations in one frozen pipeline, not trained networks.

3.5 Representation Effects and Their Matching Interaction

Let P denote category-macro pixel average precision (AP), first calculated for a specified support condition and evaluation grid. Under matching rule t , the two representation effects are

$$E_{\text{Equal},t} = P(\text{Equal}_t) - P(\text{Duplicate}_t) \quad (6)$$

$$E_{\text{Balanced},t} = P(\text{Balanced}_t) - P(\text{Baseline}_t) \quad (7)$$

The direct interaction is the difference between the same representation effect under Independent matching and Joint matching. It is the estimand for matching dependence: how much the representation effect changes when one shared reference is replaced by branch-specific references.

$$I_{\text{Equal}} = E_{\text{Equal,L}} - E_{\text{Equal,J}} \quad (8)$$

$$I_{\text{Balanced}} = E_{\text{Balanced,L}} - E_{\text{Balanced,J}} \quad (9)$$

A positive interaction means that independent matching makes the representation change more beneficial or less harmful. It does not establish that the representation effect under Independent matching is itself positive. We therefore report the absolute effects together with the interactions. Conversely, a positive representation effect with an interaction interval spanning zero supports the usefulness of the representation under the observed conditions without clearly establishing dependence on matching.

The WideResNet50-2 comparison uses the same definitions with WideResNet50-2 replacing DINOv2-S/14. To compare their interactions, both are restricted to the same seeds and support budgets. For either construction label q in Equal-weight replacement or Balanced replacement, the encoder comparison is

$$\Delta I_q = I_{q,\text{WideResNet50-2}} - I_{q,\text{DINOv2-S/14}}, q \in \{\text{Equal, Balanced}\} \quad (10)$$

All differences are formed within matched conditions and each resampling replicate before aggregation. This paired calculation directly estimates the interaction and the between-encoder difference, rather than comparing separate significance decisions.

3.6 Anomaly Outputs and Computational Cost

The patch scores form the matrix $\mathbf{a}_t$ on the common lattice, are bilinearly resized to the H by W output canvas, and are smoothed with a Gaussian of standard deviation four pixels; u indexes an output pixel. The image score is the maximum map response:

$$\mathbf{A}_t = \text{Gauss}_{\sigma=4}(\text{Resize}_{H \times W}(\mathbf{a}_t)), s_{\text{img},t} = \max_u \mathbf{A}_{t,u} \quad (11)$$

The controlled evaluation restores patch scores to the retained image canvas, with dimensions H by W ; it is 448 by 448 for square images, 448 by 588 for BTAD category 03, and 630 pixels high by 224 pixels wide for KolektorSDD2. Ground truth follows the same resized and cropped image extent. Cross-method evaluation uses original-image coordinates, as described in Section 4.1.3. Pixel metrics use continuous scores. For illustrative contours only, thresholding produces

$$\mathbf{M}_{\text{vis},t}(u) = 1[\mathbf{A}_{t,u} \geq \tau_{\text{vis}}] \quad (12)$$

Here τ_{vis} is a visualization rule, not a universal fixed operating threshold. The archived qualitative examples use min-max normalization followed by 256-bin Otsu thresholding. No test mask determines this threshold. Equation (12) defines a display operation; the evaluated model outputs remain the continuous anomaly map and image score. A deployment threshold would require separate calibration.

If a bank contains N_c normal patches, a query contains n patches, and the combined descriptor dimension is d , exhaustive matching requires order $nN_c d$ distance work and order $N_c d$ feature storage. Joint matching can use one weighted concatenation index, whereas independent matching requires the corresponding branch-wise searches. Implementations can have different overheads even when the arithmetic order is the same. Larger support budgets increase reference storage, and an actual additional encoder incurs feature-extraction cost; reusing a duplicate does not incur that encoder cost. Here, training-free denotes the absence of target optimization; reference preparation and inference still incur computation.

3.7 Model Configuration

Table 2 records the frozen feature paths and the shared processing choices; Figure S1 shows the corresponding descriptor geometry. The DINOv2-B/14 and DINOv2-S/14 encoders use DINOv2 patch features [6]. AnomalyCLIP visual uses a CLIP Vision Transformer (ViT-L/14) visual backbone with AnomalyCLIP's diagonally prominent attention map (DPAM) path [8, 26]. Its 518-pixel input yields a 37 by

37 grid; the final returned patch tensor is retained after projection, excluding the class token. Only the visual descriptor path contributes to inference.

The common lattice follows the DINOv2-B/14 canvas. Inputs are resized with preserved aspect ratio to a short side of 448 pixels and cropped from the top left to dimensions divisible by 14. The AnomalyCLIP visual grid is mapped to the retained image extent. On BTAD category 03, the resized extent is 448 by 597 pixels and the retained canvas is 448 by 588. The corrected BTAD analysis maps both predictions and masks through the corresponding coordinates; extensions using the historical canonical mask convention are identified separately. Geometric consistency does not imply identical receptive fields or learned semantic alignment between encoders.

The WideResNet50-2 branch was specified before its results were generated. It uses ImageNet-pretrained WideResNet50-2, bilinearly resamples layer2 and layer3 to the DINOv2-B/14 lattice, concatenates their 512- and 1024-channel outputs, and normalizes the resulting 1536-dimensional descriptor. Its query-feature cache is stored in float16 and converted to float32 for normalization and scoring; this storage choice is recorded in the reproduction manifest. Branch dimensions are listed separately in Table 2. Exploratory slots additionally use original DINO ViT-S/8, ConvNeXt-Tiny (**ConvNeXt-Tiny**), and Swin-Tiny (**Swin-Tiny**). They follow the same mapping and unit-normalization steps. KolektorSDD2 uses its prespecified 630 by 224 canvas for DINOv2-B/14, DINOv2-S/14 and WideResNet50-2; AnomalyCLIP visual retains its own input path before spatial mapping. All encoder parameters remain frozen, so target optimizer, learning rate, training epochs, and training-loss curves are not applicable.

Table 2. Frozen feature paths and shared model configuration.

Encoder or component	Input and retained feature path	Descriptor / target training
DINOv2-B/14 visual encoder (frozen)	DINOv2 ViT-B/14; final patch features; short side 448	768 dimensions; frozen
DINOv2-S/14 visual encoder (frozen)	DINOv2 ViT-S/14; final patch features; short side 448	384 dimensions; frozen
AnomalyCLIP visual branch (frozen)	CLIP ViT-L/14 visual + DPAM; 518×518 ; retain layer 24 after projection	768 dimensions; frozen
WideResNet50-2 branch (frozen)	WideResNet50-2; ImageNet1K V1; layer2 + layer3 on DINOv2-B/14 canvas	$512 + 1024 = 1536$; frozen
DINO ViT-S/8 encoder (frozen, exploratory)	Original DINO ViT-S/8; final patch features	384 dimensions; frozen
ConvNeXt-Tiny encoder (frozen, exploratory)	ConvNeXt-Tiny; two retained stages	576 dimensions; frozen
Swin-Tiny encoder (frozen, exploratory)	Swin-Tiny; two retained stages	576 dimensions; frozen
Shared spatial alignment	Shared canvas alignment; common DINOv2-B/14 grid; bilinear mapping; align_corners false	32×32 for a 448×448 canvas
Per-branch normalization	Shared per-branch normalization; per-branch unit normalization; joint coefficients are square roots of distance weights	No learned scale or fusion parameter
Normal reference memory and scoring	Shared memory and scoring; all aligned support patches; exact nearest-neighbor matching	Fixed bank per category, seed and K
Anomaly map and image score	Shared output mapping; resize to retained $H \times W$ canvas; Gaussian standard deviation 4 pixels	Continuous map; maximum image score
Target-domain optimization	No target optimization; no target loss, optimizer, learning rate or training epochs	0 target-trainable parameters

AnomalyCLIP visual requests layers 6, 12, 18 and 24, uses a DPAM layer setting of 20 and retains the final patch tensor. No text scores enter the model. The fixed KolektorSDD2 canvas is 630 high $\times$ 224 wide; other inputs follow the DINOv2-B/14 canvas. WideResNet50-2 is prespecified; DINO ViT-S/8, ConvNeXt-Tiny and Swin-Tiny are exploratory.

4 Experimental Evaluations

4.1 Experimental Design

4.1.1 Datasets and Reference Protocol

We evaluate the representation-by-matching design on the Metal Parts Defect Detection dataset (MPDD), the BeanTech Anomaly Detection dataset (BTAD), MVTec Anomaly Detection (MVTec AD), Visual Anomaly (VisA) and KolektorSDD2 [1, 2, 3, 4, 36]. Table 3 distinguishes their roles and support scopes. MPDD informed development. BTAD and MVTec AD serve as external validation datasets for the frozen comparison, although both had been explored before this analysis. VisA remains in-domain frozen validation: a VisA-trained prompt checkpoint was loaded in the historical export setup, although the active visual-only path neither calls that prompt learner nor loads its weights into the visual encoder. VisA therefore does not provide unseen-domain evidence, and the loaded checkpoint is not evidence that it changes the AnomalyCLIP visual descriptors. Only the KolektorSDD2 directional confirmation was specified before encoding that dataset's features.

Table 3. Dataset roles and support scopes.

Dataset	Classes	Role	DINOv2-S/14 conditions	WideResNet50-2 conditions
MPDD	6	Development	12: seeds 0–2	4: seeds 0–1; $K = 1, 4$
BTAD	3	External frozen validation	8: seeds 0–1	4: seeds 0–1; $K = 1, 4$
MVTec AD	15	External frozen validation	12: seeds 0–2	Not evaluated
VisA	12	In-domain frozen validation	12: seeds 0–2	Not evaluated
KolektorSDD2	1	Separate confirmation	12: seeds 0–2	12: seeds 0–2; all budgets

All DINOv2-S/14 studies use $K = 1, 2, 4$ and 8 . KolektorSDD2 uses the same four budgets for WideResNet50-2. Supports are nested within seed and shared across paired methods. Extra-encoder comparisons on MPDD/BTAD use the four WideResNet50-2 conditions; their wider twelve-condition runs are exploratory. VisA retains the conservative in-domain role; the loaded prompt learner does not enter the visual-only AnomalyCLIP visual path (Section 4.1.1).

For each category and seed, verified normal training images form a fixed support sequence; smaller K values are nested prefixes. The same supports and query images are used by every method in a paired comparison. Seeds 0, 1 and 2 crossed with K equal to 1, 2, 4 and 8 yield twelve dataset-level support conditions; the primary BTAD analysis uses seeds 0 and 1 and therefore eight conditions. These are repeated evaluations on common test data, not independent datasets. KolektorSDD2 supplies 1004 test images, including 110 with nonempty defect masks and 894 without defects; only verified normal training images enter the support bank.

The original WideResNet50-2 extension and the matched five-encoder comparison use seeds 0 and 1 with K equal to 1 and 4 on MPDD and BTAD. Additional DINO ViT-S/8, ConvNeXt-Tiny and Swin-Tiny results on twelve conditions are reported as exploratory scope checks. A separate eight-seed extension uses seeds 0 through 7 and all four budgets. Its BTAD category-03 masks follow the canonical convention, so its estimates are not substituted for the corrected-geometry primary estimates.

4.1.2 Controlled Constructions and External Baselines

The primary matrix includes single-branch DINOv2-B/14, DINOv2-S/14 and AnomalyCLIP visual and both matching rules for Dual-encoder baseline, Duplicate-branch control, Equal-weight replacement and Balanced replacement. Single-branch paths diagnose descriptor quality, while the eight combined configurations identify the two interaction contrasts. The WideResNet50-2 extension adds WideResNet50-2-only scoring and its Equal-weight replacement/Balanced replacement variants. DINO ViT-S/8, ConvNeXt-Tiny and Swin-Tiny entered after the DINOv2-S/14 and WideResNet50-2 results and are exploratory encoder substitutions. They test sensitivity to the added representation.

AnomalyDINO and PatchCore provide native-method context [5, 16]. AnomalyDINO is evaluated with and without support rotation. PatchCore includes the 224-pixel input and 1024-dimensional projected-feature configuration as well as an earlier local 128-pixel and 256-dimensional configuration. Both are retained to reveal configuration sensitivity. Each method keeps its own preprocessing, features and search implementation, while support seeds and budgets are matched where its protocol defines them. This comparison assesses practical configurations and does not isolate the matching factor or support a comprehensive cross-method ranking. The size of this comparison set follows the formal baseline list; the number of compared methods is not itself a target, and an approximate count such as ten is not a hard criterion.

Every configuration evaluated here omits gradient-based adaptation on the target data: encoders run in evaluation mode with fixed fusion weights and no target loss, optimizer or training epochs. This does not erase prior learning. The pretrained visual encoders and inherited checkpoints retain their upstream training provenance, and reference construction, feature encoding and scoring remain preparation computation. PatchCore builds a feature memory and coreset; the controlled pipeline, AnomalyDINO and SubspaceAD construct normal-reference statistics from the supplied supports; and WinCLIP+ uses frozen vision-language

alignment and prompt ensembles. The separate zero-shot AnomalyCLIP baseline uses an upstream auxiliary-domain-trained prompt learner and evaluates the target domain zero-shot, following the upstream rule that a prompt learner is not evaluated on its own training domain. The verified upstream source archive supplies the thirty epoch checkpoint files used by this baseline; they were not trained by this project. In contrast, the AnomalyCLIP visual branch in the controlled Dual-encoder baseline pipeline uses visual descriptors only: prompt weights and text scores do not enter its visual anomaly score. Thus, the absence of a target-domain gradient update should not be read as evidence of no previous training, no preparation work or no configuration choices; resolutions, retained layers, support budgets, smoothing and visualization thresholds remain fixed choices recorded in Table 2 and the protocol tables.

4.1.3 Metrics and Statistical Inference

Pixel average precision (AP) is the primary localization metric. Scores and labels are pooled over test images within each category, then category APs are averaged equally. Dataset-level estimates average the applicable support conditions equally. This differs from pooling all categories into a single AP or averaging per-image AP. Pixel area under the receiver operating characteristic curve (AUROC) is a secondary diagnostic. Performance tables use AP on the 0–1 scale; effects and interactions use AP percentage points, equal to 100 times an AP difference.

Three spatial evaluation conventions remain separate. The main bootstrap analysis samples pixels on a regular stride-eight grid of the retained output canvas. Full-pixel estimates use every output-canvas pixel; finer stride-four checks test sensitivity to the sampled grid. External-method comparisons instead map predictions into original-image coordinates and evaluate their common valid intersection. Geometry conventions, pixel strides and support scopes are stated with each table; scores from these settings are not interchangeable.

We use 1000 paired image-level bootstrap replicates. Images are sampled with replacement within each fixed category, without additional normal/abnormal stratification. Each sampled image contributes its evaluated pixels as a block. The same deterministic replicate stream, based on seed 20260913 together with dataset identity, category identity and replicate index, is shared across methods and support conditions. Contrasts are formed within each replicate after the required category and condition aggregation. Intervals are conditional on observed categories and support manifests; they do not estimate uncertainty over unseen datasets or arbitrary future support sets.

The primary four DINOv2-S/14 interactions, four WideResNet50-2 interactions and four matched WideResNet50-2-minus-DINOv2-S/14 contrasts each have their own exploratory four-comparison family, reported with 98.75% percentile intervals. The extended four-dataset DINOv2-S/14 table has eight cells and reports 99.375% intervals; these are approximate Bonferroni adjustments. Encoder-transfer comparisons retain a separate four-cell family per encoder, with no simultaneous guarantee over all exploratory analyses. With 1000 replicates, extreme percentile endpoints have limited precision.

KolektorSDD2 retains its specified conjunction criterion: all four DINOv2-S/14 interactions and all four WideResNet50-2 interactions must be positive and their individual 95% intervals must exclude zero. This conjunction uses individual intervals and does not imply simultaneous 95% coverage. Excluding zero supports an interaction direction under the stated protocol; practical value also depends on the effect size and absolute representation gain. An interval spanning zero leaves the direction unresolved.

Observed condition-averaged point estimates and bootstrap means are labelled separately. Full-pixel estimates and bootstrap intervals provide resolution checks for the DINOv2-S/14 analysis on MPDD and canonical-mask BTAD. They are supplied with the reproduction materials rather than entering the main tables; the canonical BTAD masks differ from the corrected geometry used for the primary BTAD results. The eight-seed study describes support sensitivity separately rather than treating its between-seed variation as part of the primary conditional interval.

4.1.4 Implementation and Reproducibility

The recorded machine has an NVIDIA GeForce RTX 3060 Laptop GPU with 6 GB VRAM, an Intel Core i9-12900H processor and 16 GB RAM. Encoders run in evaluation mode, with cached query features reused across controlled variants. Exact nearest-neighbor search implements Section 3, using Facebook AI Similarity Search (FAISS) where applicable [37]. Feature extraction, reference construction and score evaluation are distinct computational stages; cached-stage times cannot be interpreted as complete-process deployment latency.

The primary BTAD tables use corrected category-03 coordinates. The four-dataset and eight-seed extensions retain their archived canonical-mask convention and are reported separately. Geometric audits verify the retained image extent and support nesting. Duplicate descriptors reuse the exact cached values. Reproduction uses support manifests, frozen branch specifications and paired bootstrap streams; no target loss is optimized and no test-mask feedback selects a normal reference.

Resource reporting is restricted to measurements whose stages and devices are identifiable. VisA's extension includes mixed CPU/GPU execution and is excluded from speed or GPU-memory rankings. A separate synchronized benchmark now measures all six configurations on six MPDD units, including query feature extraction. Its reference-preparation and query-scoring boundary is stated in Section 4.2.4; the older stage records remain separate.

4.2 Results and Analysis

4.2.1 Main Effects and Interactions

Overall controlled matrix and reweighting. Table 4 reports the eight configurations required for the DINOv2-S/14 interaction. Across the complete DINOv2-S/14 support scope, the Dual-encoder baseline reaches pixel AP of 0.3729 with joint matching and 0.3791 with independent matching on MPDD. On corrected BTAD, the corresponding values are 0.6408 and 0.6498. Independent matching has a positive average effect for the Dual-encoder baseline, but these averages alone do not establish that another encoder benefits from that change. That question requires comparing representation effects within each matching rule.

Table 4. Primary controlled matrix with the DINOv2-S/14 representation.

Configuration / matching rule	MPDD pixel AP	MPDD SD	BTAD pixel AP	BTAD SD
Dual-encoder baseline / Joint matching	0.3729	0.0424	0.6408	0.0134
Dual-encoder baseline / Independent matching	0.3791	0.0434	0.6498	0.0129
Duplicate-branch control / Joint matching	0.3649	0.0438	0.6316	0.0136
Duplicate-branch control / Independent matching	0.3681	0.0445	0.6379	0.0133
Equal-weight replacement / Joint matching	0.3626	0.0433	0.6471	0.0108
Equal-weight replacement / Independent matching	0.3736	0.0463	0.6529	0.0086
Balanced replacement / Joint matching	0.3678	0.0409	0.6505	0.0113
Balanced replacement / Independent matching	0.3799	0.0437	0.6583	0.0090

Pixel AP on the stride-eight canvas grid. SD is the sample standard deviation across the 12 MPDD or 8 BTAD fixed support conditions; it is not a confidence interval or variation across independent test datasets. BTAD uses corrected coordinates. The table contains the eight configurations required by the interaction; single-branch diagnostics are retained in the archive. Bold identifies the Dual-encoder baseline, not a claim of best performance.

The configurations do not admit one uniform ordering. On MPDD, Equal-weight replacement remains below the Dual-encoder baseline under both matching rules, even though Equal-weight replacement improves over its Duplicate-branch control, which has the same branch slots and weights, under Independent matching.

Balanced replacement under Independent matching is close to Dual-encoder baseline under Independent matching. On BTAD, both Equal-weight replacement and Balanced replacement exceed their corresponding controls, while Balanced replacement has the highest observed AP among these eight configurations. These are conditional readings of the fixed controlled matrix, not evidence that the selected weights or branch set are globally optimal.

Table 5 isolates the change that the rest of the study is built on, the matching effect, defined as the AP under Independent matching minus the AP under Joint matching. It differs from the baseline comparison only in the selection rule, because the support identities, descriptors and weights are all held fixed. On MPDD the anchor gains +0.618 AP percentage points and on corrected BTAD +0.900, and both intervals exclude zero; the other three constructions move in the same direction, from +0.321 for the duplicate control to +1.213 for Balanced replacement on MPDD. A rule that raises the score of every construction does not yet say whether a newly added representation benefits from that rule, which requires the representation-effect and interaction contrasts below.

Table 5. Matching effect (AP under Independent matching minus AP under Joint matching) for the four constructions under identical support conditions.

Dataset / construction	Point	Bootstrap mean	95% interval
MPDD / Dual-encoder baseline	+0.618	+0.637	[+0.337, +0.977]
MPDD / Duplicate-branch control	+0.321	+0.349	[+0.137, +0.580]
MPDD / Equal-weight replacement	+1.093	+1.111	[+0.701, +1.544]
MPDD / Balanced replacement	+1.213	+1.253	[+0.839, +1.722]
BTAD / Dual-encoder baseline	+0.900	+0.916	[+0.732, +1.172]
BTAD / Duplicate-branch control	+0.631	+0.638	[+0.518, +0.785]
BTAD / Equal-weight replacement	+0.582	+0.617	[+0.373, +0.889]
BTAD / Balanced replacement	+0.780	+0.826	[+0.541, +1.171]

Pixel AP percentage points, stride eight. The matching difference is formed within one construction, so it changes only the selection rule: a positive value means independent matching scores higher than joint matching. Values are condition-averaged observed differences and are shown next to, never replaced by, the bootstrap means. MPDD uses 12 support conditions (seeds 0, 1, 2) and corrected BTAD uses 8 (seeds 0, 1). Bold identifies the Dual-encoder baseline, not a claim of best performance. These are descriptive summaries of the matching rule and are not part of the three adjusted families defined in Section 4.1.3, so no family-adjusted interval is claimed for them.

Single-branch diagnostics provide a simpler reference point. On MPDD, where these runs share the primary geometry revision, each branch alone scores below the fused Dual-encoder baseline: 0.348 for DINOv2-B/14, 0.326 for DINOv2-S/14 and 0.283 for AnomalyCLIP visual, against 0.373 and 0.379 for Dual-encoder baseline under the two rules. Combining descriptors therefore helps relative to every single branch, which is why the study keeps a fused Dual-encoder baseline rather than asking whether one encoder is enough. The corresponding BTAD diagnostics are not reported here: the corrected BTAD-03 geometry rescore covers only the eight interaction configurations, so placing the two revisions in one comparison would confound geometry with the encoder.

The exact duplicate control shows why reweighting must be separated from new information. Duplicate-branch control lowers mean AP relative to Dual-encoder baseline under both matching rules in both datasets

(Table 4). On MPDD, the observed differences are approximately -0.80 AP percentage points under Joint matching and -1.10 under Independent matching. Thus, increasing the apparent number of branches from two to three can change performance without introducing new information. The duplicated DINOv2-B/14 descriptor simply increases the effective DINOv2-B/14 weight while reducing the AnomalyCLIP visual weight.

Comparing the equal-weight replacement directly with the dual-encoder baseline does not isolate the contribution of DINOv2-S/14. This comparison changes both the effective weights and the descriptor set. Comparing the equal-weight replacement with the duplicate-branch control instead holds the branch slots and their distance weights fixed. The balanced replacement provides a complementary comparison with the baseline: it preserves the AnomalyCLIP visual weight and the combined weight of the remaining branches. These controls do not establish that equal two-branch weights are universally preferable. They show that the weight change affects the present results and must therefore be separated from the value of the new representation.

Absolute representation effects. Table 6 reports the DINOv2-S/14 effects in AP percentage points. On MPDD, Equal-weight replacement minus Duplicate-branch control changes from -0.224 under Joint matching to $+0.548$ under Independent matching. The latter 95% interval spans -0.092 to $+1.160$, so a positive absolute gain remains uncertain. Balanced replacement minus Dual-encoder baseline changes from -0.510 to $+0.085$, again with an interval spanning zero. Independent matching makes the observed representation change more favorable, but this should not be rewritten as an established absolute improvement from adding DINOv2-S/14 on MPDD.

Table 6. Absolute effects of replacing or adding DINOv2-S/14 under each matching rule.

Dataset	Effect	Point	95% interval
MPDD	$E_{\text{Equal},J}$	-0.224	$[-0.968, +0.478]$
	$E_{\text{Equal},L}$	$+0.548$	$[-0.092, +1.160]$
	$E_{\text{Balanced},J}$	-0.510	$[-1.135, +0.123]$
	$E_{\text{Balanced},L}$	$+0.085$	$[-0.434, +0.708]$
BTAD	$E_{\text{Equal},J}$	$+1.553$	$[+0.637, +2.465]$
	$E_{\text{Equal},L}$	$+1.503$	$[+0.585, +2.495]$
	$E_{\text{Balanced},J}$	$+0.974$	$[+0.177, +1.803]$
	$E_{\text{Balanced},L}$	$+0.854$	$[+0.057, +1.753]$

Pixel AP percentage points, stride eight. Values are condition-averaged observed differences, not bootstrap means. MPDD uses 12 support conditions and BTAD uses 8; each dataset name is printed once for its group. Intervals are exploratory; the primary adjusted interaction intervals appear in Table 7.

BTAD illustrates a different situation. The Equal-weight replacement effects are $+1.553$ points under Joint matching and $+1.503$ under Independent matching; their exploratory 95% intervals are both above zero. Balanced replacement also has positive estimated effects under both rules. Here, additional DINOv2-S/14 information is useful relative to the specified controls, yet the very similar Equal-weight replacement effects suggest that its usefulness need not depend strongly on the matching rule. Direct interaction inference, rather than separate effect intervals, resolves that distinction.

The first page of Figure 4 plots the eight DINOv2-S/14 effects by dataset, allowing the two rules to be compared within each contrast.

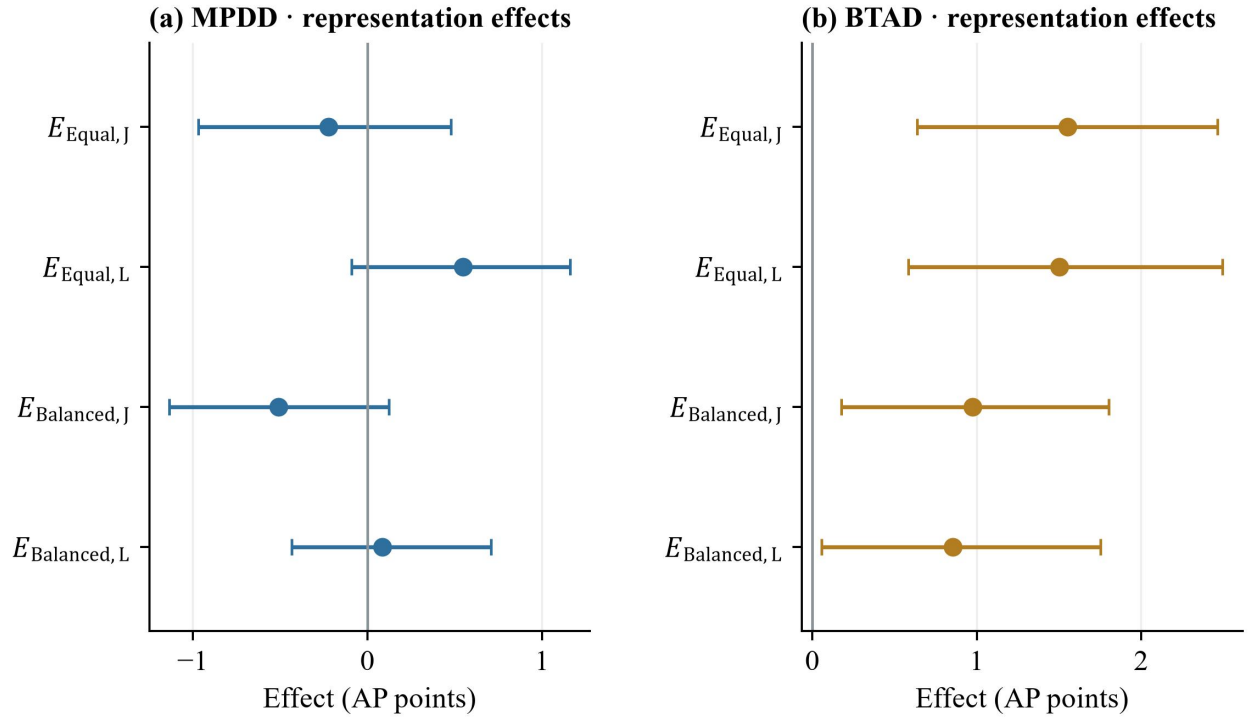


Figure 4. Absolute representation effects for DINOv2-S/14. Panels show observed effects in MPDD and corrected-geometry BTAD with individual 95% paired image-bootstrap intervals. MPDD uses seeds 0–2 and BTAD seeds 0, 1, with $K = 1, 2, 4, 8$. Units are AP percentage points. These absolute effects differ from the matching interactions shown on the continuation page.

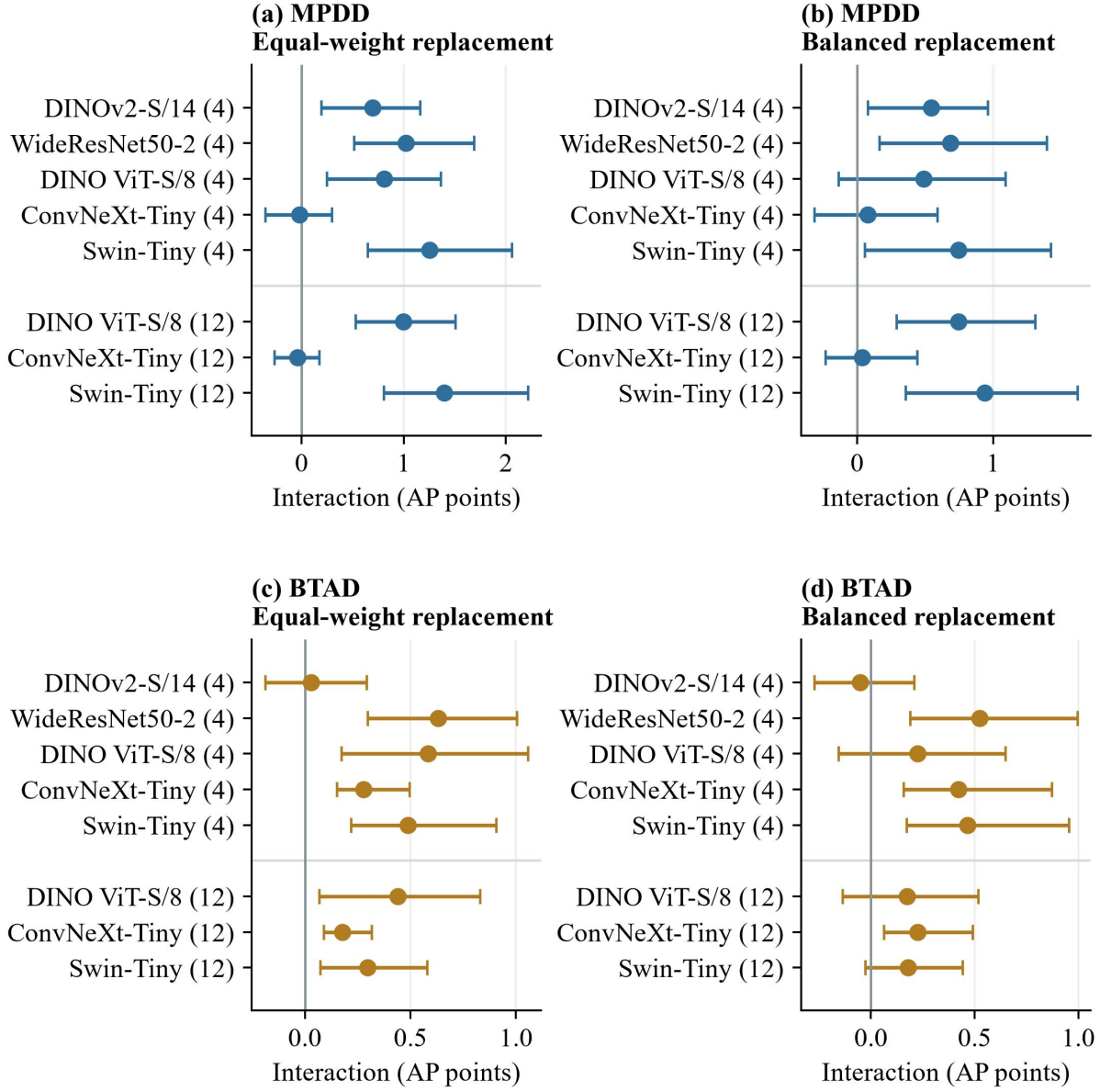


Figure 4 continued. Encoder interactions under the shared four-condition scope and the wider twelve-condition scope. Rows marked (4) pool seeds 0 and 1 with $K = 1$ and 4; rows marked (12) pool seeds 0, 1 and 2 with support budgets 1, 2, 4 and 8. Points are bootstrap replicate means and bars are 98.75% paired intervals, a separate four-comparison family for each encoder and scope. DINO ViT-S/8, ConvNeXt-Tiny and Swin-Tiny are exploratory substitutions; no family adjustment covers these additional encoders as a group. Comparisons across encoders use the shared four-condition scope, while the twelve-condition rows describe the wider evaluation scope.

Direct interaction and encoder dependence. The direct DINOv2-S/14 interactions on MPDD—the change in the representation effect from Joint matching to Independent matching—are +0.772 points for Equal-weight replacement and +0.595 for Balanced replacement (Table 7). Their adjusted 98.75% intervals are $[+0.346, +1.211]$ and $[+0.189, +1.030]$, respectively. Both exclude zero. Their point estimates exceed the descriptive 0.5-point scale, but their interval lower bounds do not, so the data do not establish an effect uniformly above that practical reference magnitude. The continuation of Figure 4 compares all five encoders under the matched four-condition scope and also shows the wider twelve-condition rows; broader dataset evidence appears in the validation theme below.

Table 7. Direct representation-by-matching interactions for DINOv2-S/14.

Dataset / contrast	Point	95% interval	98.75% interval	Full-pixel point
MPDD / I_{Equal}	+0.772	[+0.426, +1.118]	[+0.346, +1.211]	+0.787
MPDD / I_{Balanced}	+0.595	[+0.322, +0.927]	[+0.189, +1.030]	+0.601
BTAD / I_{Equal}	−0.050	[−0.192, +0.186]	[−0.235, +0.244]	−0.044
BTAD / I_{Balanced}	−0.119	[−0.265, +0.110]	[−0.306, +0.167]	−0.117

AP percentage points. The point estimates and intervals use stride eight; the last column uses all pixels and has no interval. The four stride-eight interactions form one Bonferroni-adjusted exploratory family. A positive interaction alone does not imply a positive absolute representation effect.

On BTAD, the DINOv2-S/14 interactions are −0.050 and −0.119 points, with both adjusted intervals spanning zero. The small negative points are not evidence of a general advantage for joint matching. Combined with Table 6, the result is that DINOv2-S/14 has positive representation effects under the tested rules while the current data do not clearly resolve a difference between those effects. This is distinct from saying that matching is irrelevant or that the two rules are equivalent.

The prespecified WideResNet50-2 extension provides a second representation condition. Table 8 reports its full-pixel localization results under the restricted four-condition scope. On MPDD, Equal-weight replacement with WideResNet50-2 under Independent matching reaches AP 0.4107 compared with 0.3554 for Duplicate-branch control under Independent matching; on BTAD, it reaches 0.6591 compared with 0.6335. The corresponding full-pixel representation effects are approximately +5.53 and +2.57 AP percentage points. These are absolute representation effects, not the much smaller interaction estimates. WideResNet50-2 alone has lower pixel AP than these fused constructions, showing that the fusion result cannot be explained simply by selecting WideResNet50-2's standalone score. Table 8 reports the full-pixel values supporting these WideResNet50-2 effects.

Table 8. Full-pixel localization for the prespecified WideResNet50-2 extension.

Configuration / matching rule	MPDD pixel AP	MPDD pixel AUROC	BTAD pixel AP	BTAD pixel AUROC
WideResNet50-2 only	0.3452	0.9708	0.4449	0.9542
Dual-encoder baseline / Joint matching	0.3622	0.9658	0.6361	0.9745
Dual-encoder baseline / Independent matching	0.3687	0.9688	0.6455	0.9755
Duplicate-branch control / Joint matching	0.3510	0.9626	0.6270	0.9734
Duplicate-branch control / Independent matching	0.3554	0.9647	0.6335	0.9742
Equal-weight replacement / WideResNet50-2 / Joint matching	0.3960	0.9731	0.6464	0.9743
Equal-weight replacement / WideResNet50-2 / Independent matching	0.4107	0.9768	0.6591	0.9758
Balanced replacement / WideResNet50-2 / Joint matching	0.3942	0.9736	0.6401	0.9749
Balanced replacement / WideResNet50-2 / Independent matching	0.4076	0.9773	0.6545	0.9766

All output-canvas pixels, seeds 0 and 1, $K = 1$ and 4. Each number averages category metrics and the four support conditions. No full-pixel confidence intervals were computed. The WideResNet50-2-only row does not use the AnomalyCLIP visual branch. Bold identifies Dual-encoder baseline, not a claim of best performance.

All four WideResNet50-2 interactions are positive, with adjusted intervals excluding zero (Table 9): +0.974 and +0.628 points on MPDD, and +0.622 and +0.501 on BTAD. The full-pixel points retain these directions. This establishes that a positive interaction in the observed setting is not confined to adding a smaller DINOv2 encoder. It does not establish the same result for arbitrary convolutional backbones or for additional datasets.

Table 9. Direct matching interactions with WideResNet50-2 replacing DINOv2-S/14.

Dataset / contrast	Point	95% interval	98.75% interval	Full-pixel point
MPDD / Equal-weight replacement	+0.974	[+0.602, +1.472]	[+0.514, +1.689]	+1.033
MPDD / Balanced replacement	+0.628	[+0.279, +1.218]	[+0.162, +1.394]	+0.694
BTAD / Equal-weight replacement	+0.622	[+0.373, +0.917]	[+0.298, +1.003]	+0.624
BTAD / Balanced replacement	+0.501	[+0.239, +0.883]	[+0.188, +0.995]	+0.496

AP percentage points. All four support conditions use seeds 0 and 1 and $K = 1$ and 4. Intervals are stride-eight paired percentile intervals, with the four WideResNet50-2 contrasts forming their own adjustment family. The final column has no interval.

Directly comparing the full-scope DINOv2-S/14 effects with the restricted-scope WideResNet50-2 effects would confound encoder choice with support-condition coverage. Table 10 instead restricts DINOv2-S/14 to the WideResNet50-2 seeds and budgets and directly estimates the difference between WideResNet50-2 and DINOv2-S/14 interaction difference. The observed differences on BTAD are +0.623 and +0.585 points, with adjusted intervals [+0.255, +0.945] and [+0.204, +1.060]. On MPDD, the observed differences are +0.207

and +0.081, and the intervals span zero. The bootstrap means differ from the observed points, particularly on MPDD, and are shown in a separate column rather than being mislabeled as point estimates.

Table 10. Matched difference between WideResNet50-2 and DINOv2-S/14 interactions.

Dataset / contrast	Observed difference	Bootstrap mean	98.75% interval	Full-pixel difference
MPDD / Equal-weight replacement	+0.207	+0.326	[-0.296, +1.118]	+0.264
MPDD / Balanced replacement	+0.081	+0.139	[-0.372, +0.806]	+0.075
BTAD / Equal-weight replacement	+0.623	+0.603	[+0.255, +0.945]	+0.616
BTAD / Balanced replacement	+0.585	+0.576	[+0.204, +1.060]	+0.575

AP percentage points; WideResNet50-2 interaction minus DINOv2-S/14 interaction. DINOv2-S/14 is restricted to the identical seeds 0 and 1 and $K = 1$ and 4. Intervals come from within-replicate differences, not differences of interval endpoints. Four encoder comparisons form a separate exploratory adjustment family. Full-pixel differences have no intervals.

These comparisons support encoder dependence on BTAD under the matched scope. On MPDD, the uncertainty does not resolve a difference between the interactions for DINOv2-S/14 and WideResNet50-2; it is not evidence that the encoders are equivalent. More generally, a larger interaction does not mean that an encoder is better in every absolute metric. The available image-level metrics illustrate this separation: under the four-condition BTAD scope, Balanced replacement with WideResNet50-2 under Independent matching has image AP 0.9372 versus 0.9468 for Dual-encoder baseline under Independent matching even though its full-pixel AP is higher. Image-score pooling and pixel ranking answer different evaluation questions.

Support budget and category conditions. Figure 5(a) shows the seed-averaged DINOv2-S/14 interaction against the nested support budget. On MPDD both contrasts strengthen as K grows: the Equal-weight replacement interaction rises from +0.41 percentage points at one support to +0.75, +0.88 and +1.01 at two, four and eight supports, and Balanced replacement rises from +0.37 to +0.83. On the corrected BTAD revision the same contrast weakens and changes sign, moving from +0.20 points at one support to +0.09 at two, -0.14 at four and -0.23 at eight, where the interval no longer includes zero. The curves therefore support a budget-dependent description, not a monotone law that more normal references consistently amplify the benefit of independent matching.

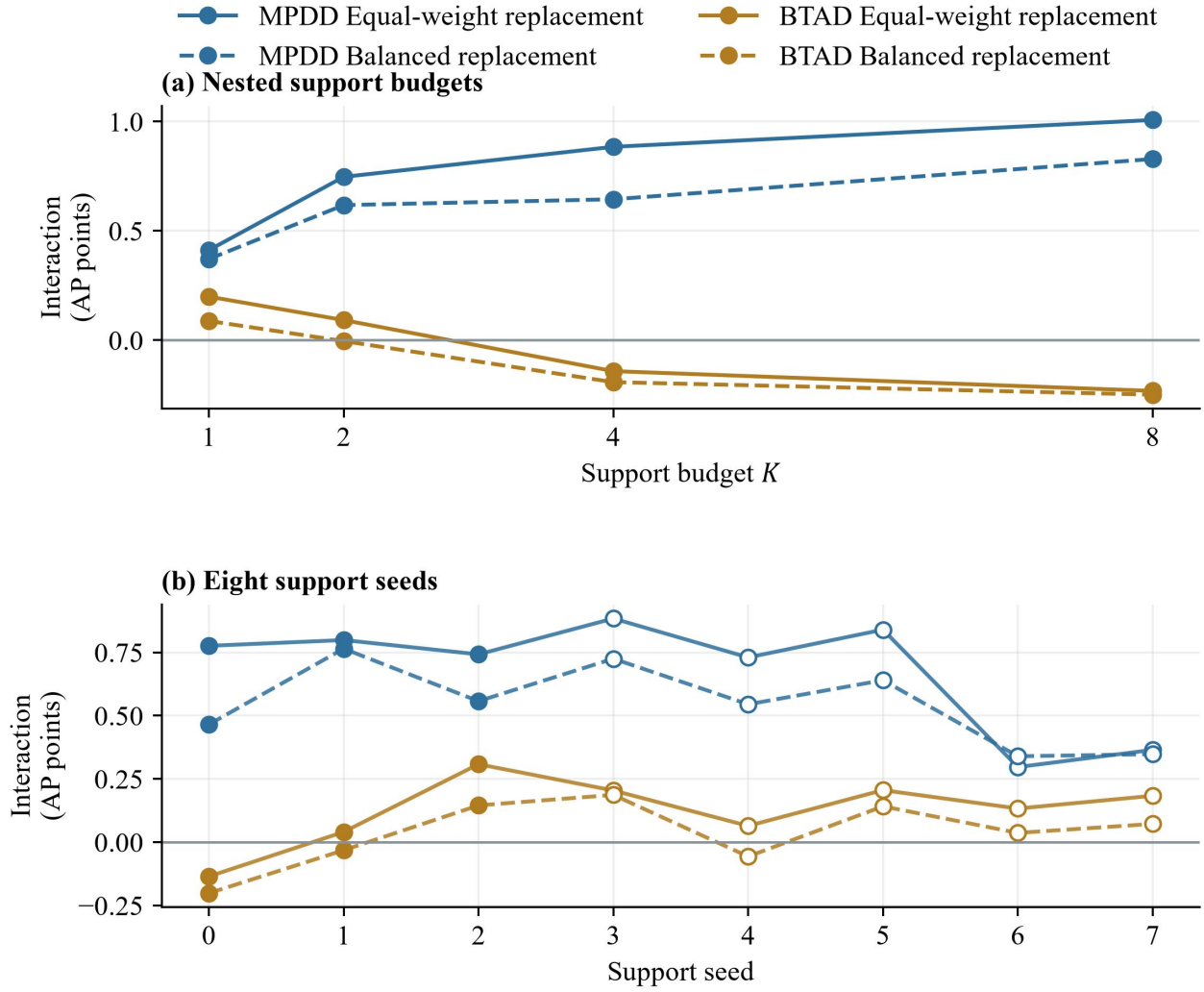


Figure 5. Sensitivity to support budget and seed. (a) Seed-averaged bootstrap means across nested K budgets, using the MPDD study revision and corrected BTAD revision. (b) Observed interaction points for eight support seeds; seeds 0–2 have separate query blocks and seeds 3–7 share a cached block. The seed extension uses canonical BTAD masks. Filled and open markers distinguish these groups. Lines connect evaluated settings and do not imply independent samples or a fitted trend. Units are AP percentage points.

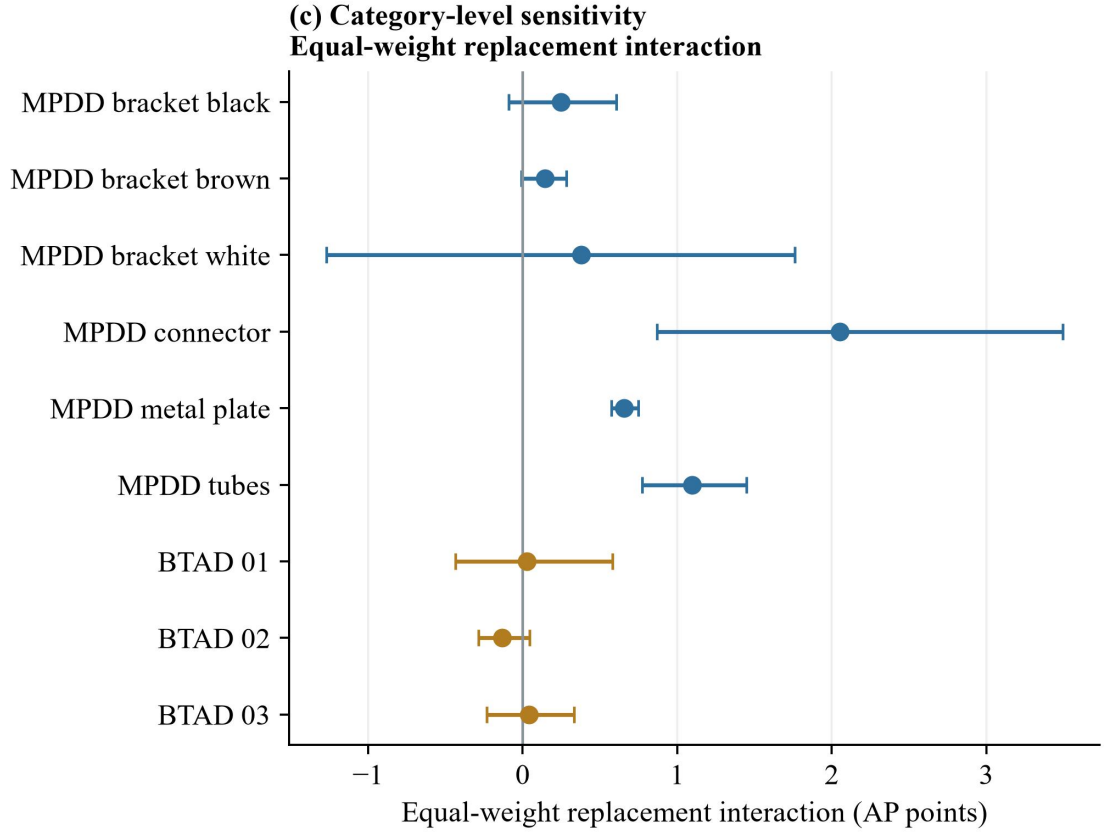


Figure 5 continued. (c) Category-level Equal-weight replacement interactions as bootstrap means with individual 95% intervals. MPDD uses the study revision and BTAD the corrected revision. This exploratory comparison has no multiplicity adjustment across categories.

Adding normal candidates cannot increase the exact minimum raw distance when the bank is nested and the descriptors remain fixed. However, AP depends on score ordering, and an interaction is a difference of four performance values. Neither AP nor the interaction inherits a monotonicity guarantee from the minimum-distance operation. The distinction also explains why a numerical constraint gap should not be treated as a defect-performance measure.

Category analyses further limit the aggregate interpretation. Figure 5(c) resolves the Equal-weight replacement interaction by category: all six MPDD categories carry a positive estimate, led by connector and tubes, whereas the three BTAD categories stay close to zero and category 02 is slightly negative. Omitting any one MPDD category leaves the Equal-weight replacement aggregate interaction positive in the recorded leave-one-category analysis. On corrected BTAD, omitting category 02 changes the Equal-weight replacement aggregate direction. With only three BTAD categories, the average is sensitive to category composition. Per-category and per-budget intervals are exploratory follow-up analyses and do not inherit the primary family's multiplicity guarantee. They identify conditions worth testing on new data rather than proving a causal role for category identity or defect size.

Full-pixel evaluation checks whether sparse pixel sampling changes the estimated interactions, particularly for small defects. The completed stride-one analysis covers the DINOv2-S/14 constructions under 12 MPDD conditions and 8 BTAD conditions, using canonical-mask geometry for BTAD. The two primary MPDD interactions remain positive with 98.75% intervals excluding zero; their intervals are narrower than at stride eight. The MPDD balanced-replacement effect under joint matching excludes zero at strides one and four but spans zero at stride eight, so not every secondary inference is insensitive to resolution. For BTAD, both canonical-mask stride-one interaction intervals span zero. The corrected-geometry stride-eight intervals also span zero, but these are not a like-for-like grid comparison; we therefore do not infer a BTAD interval-width change. The full-pixel results are supplied with the reproduction materials and do not replace the corrected-

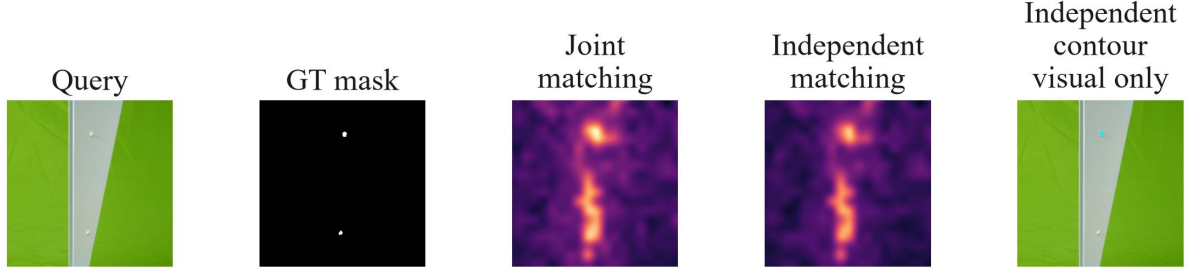
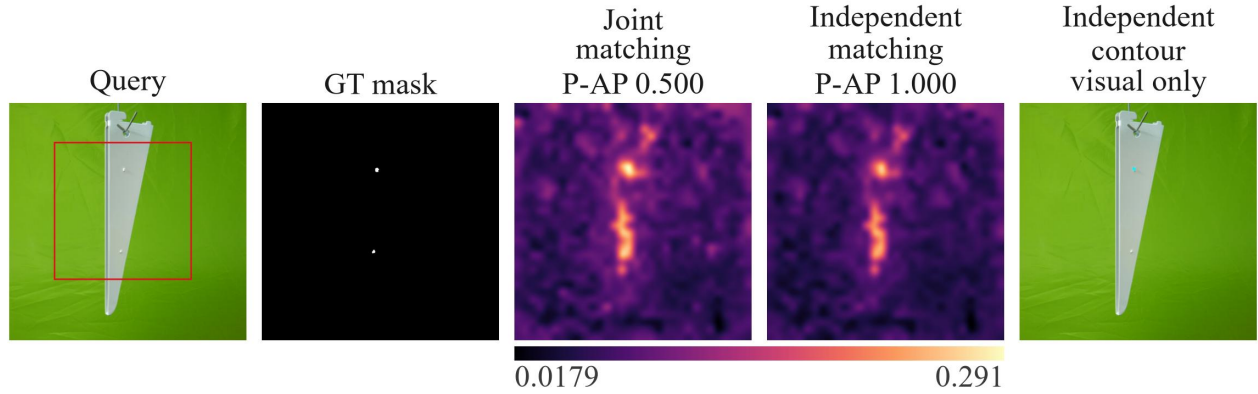
BTAD main tables. Sparse sampling remains relevant to the selected qualitative examples: their per-image AP can change markedly when only a few sampled pixels are positive.

4.2.2 Validation and Confirmation

Qualitative localization and predicted contours. Figures 6 and 7 show five MPDD cases from seed 0 and four normal supports, with three improvements and two degradations for Dual-encoder baseline under Independent matching versus Dual-encoder baseline under Joint matching. The cases are selected by extreme stored per-image AP changes from the fixed closeout candidate list. Selection is based on localization performance, not on the average reduction in raw score. These panels illustrate the Dual-encoder baseline's matching behavior; they are not a substitute for the direct representation interaction in Tables 7 and 9.

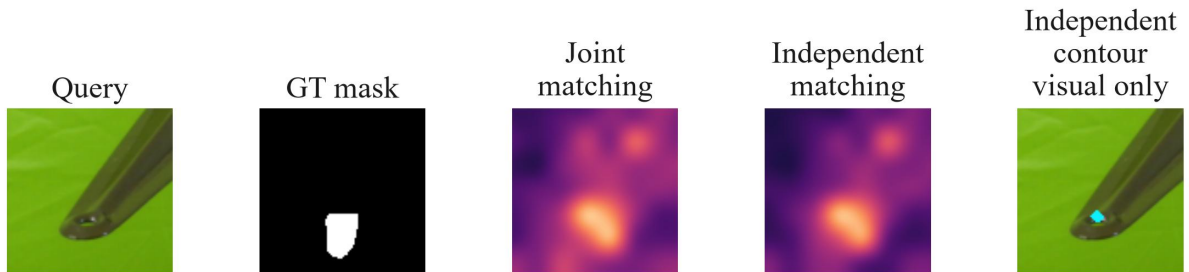
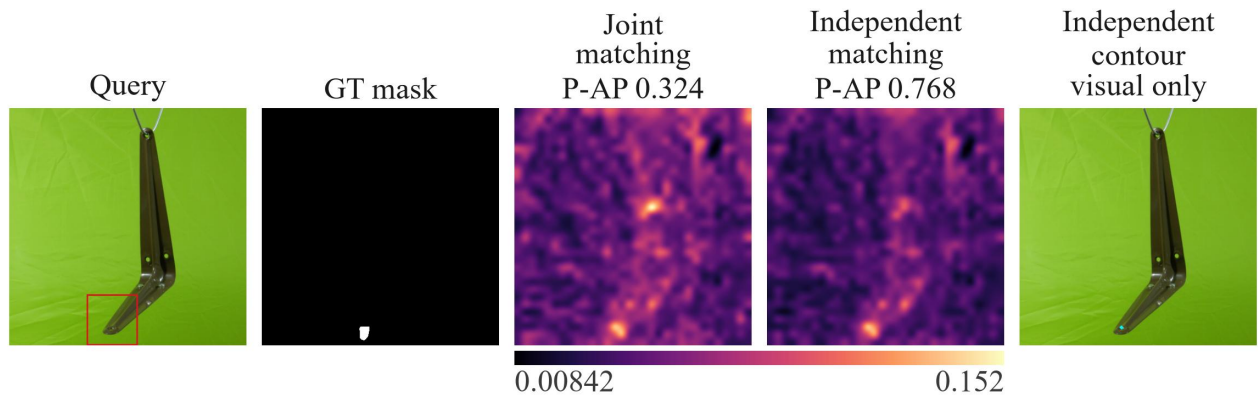
Dual-encoder baseline

I1 bracket white / 007.png AP change +0.500 (improvement)



Each column enlarges the same boxed region; the contour line is a display only.

I2 bracket brown / 007.png AP change +0.443 (improvement)

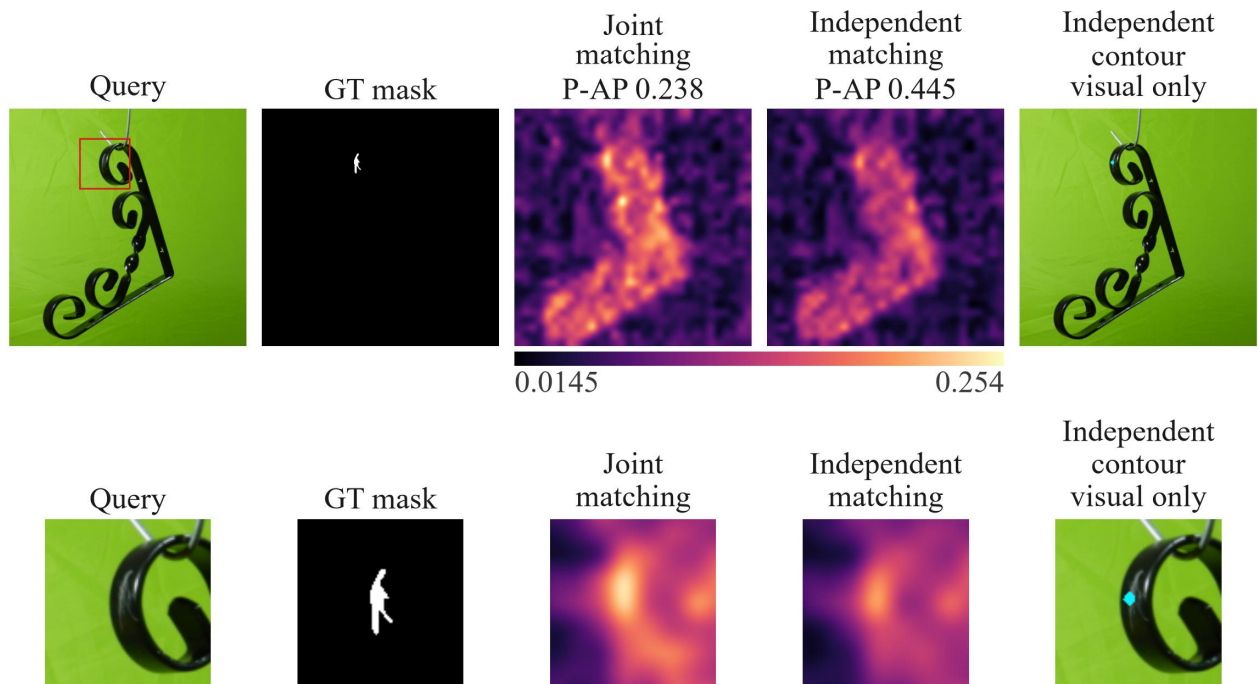


Each column enlarges the same boxed region; the contour line is a display only.

Figure 6. Three selected MPDD improvements for the Dual-encoder baseline under independent matching relative to the same baseline under joint matching at seed 0 and $K = 4$. Values are stored stride-eight per-image AP, rather than the category-pooled AP used in the main tables. Joint and independent heatmaps share one full-map min-max range per row. The cyan independent-matching contour uses 256-bin Otsu thresholding on that normalized score. Red hollow rectangles define identical GT-centered visual crops, enlarged below. GT is used for evaluation and crop placement, never to generate the predicted contour. These are selected extremes, not a random test sample.

Dual-encoder baseline

I3 bracket black / 033.png AP change +0.207 (improvement)



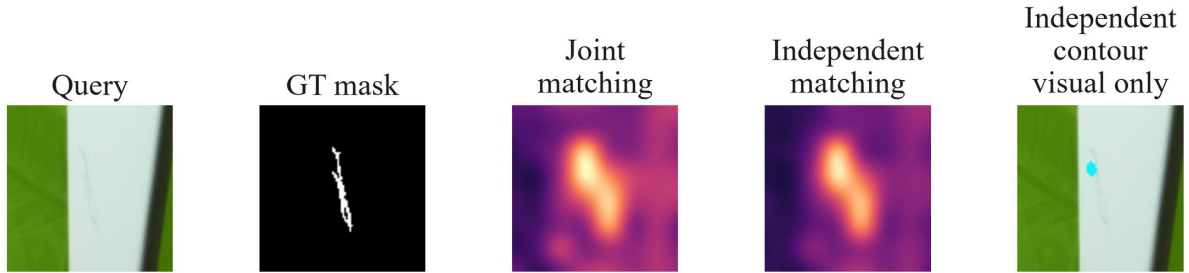
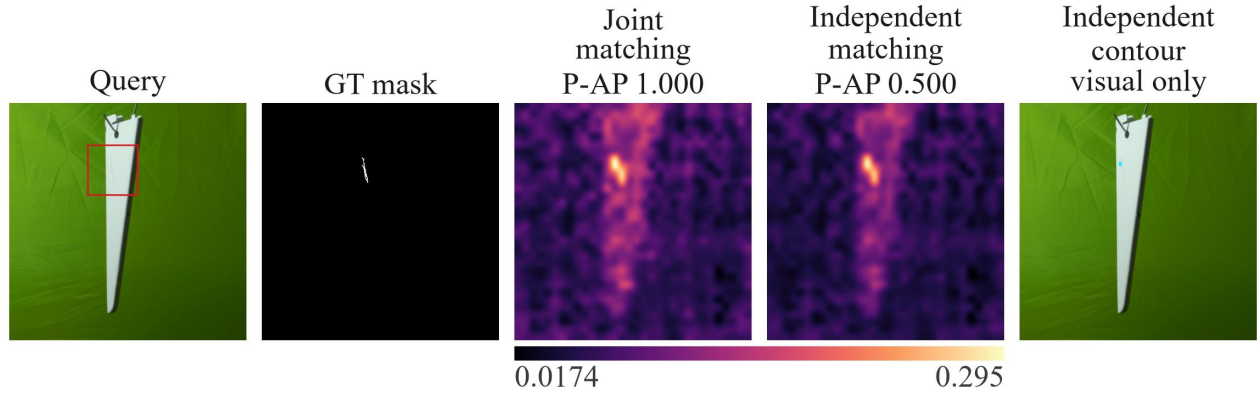
Each column enlarges the same boxed region; the contour line is a display only.

Figure 6 continued. The third selected improvement, with the same scale, crop, and contour conventions.

The heatmaps for each case use one common minimum and maximum over its Joint matching and Independent matching maps after resizing and smoothing. The Independent matching map is quantized into 256 bins, and Otsu's between-class-variance criterion selects a visualization threshold, taking the smallest maximizing bin if tied. The cyan contour traces the resulting predicted mask. It is derived entirely from the model score. For visual inspection only, a square crop is centered on the bounding box of the ground-truth defect mask, with side length clipped between 96 and 260 pixels after a 1.8-fold expansion; the same crop is applied to all columns. The ground truth determines this display region and the evaluation labels, but not the predicted mask.

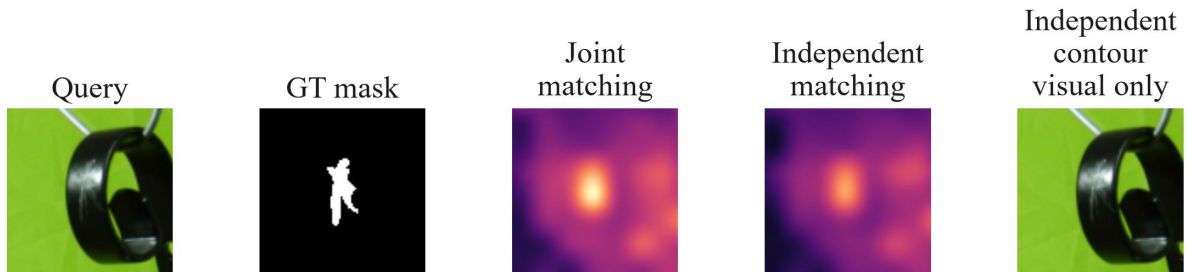
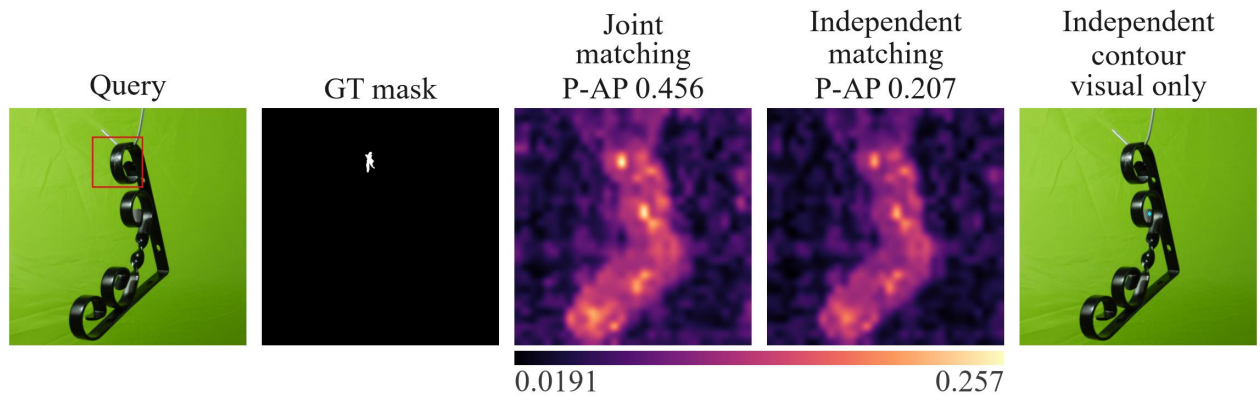
Dual-encoder baseline

D1 bracket white / 006.png AP change -0.500 (degradation)



Each column enlarges the same boxed region; the contour line is a display only.

D2 bracket black / 032.png AP change -0.249 (degradation)



Each column enlarges the same boxed region; the contour line is a display only.

Figure 7. Two selected MPDD degradations under the Dual-encoder baseline comparison and display protocol of Figure 6. Lower raw scores under independent matching do not guarantee improved pixel ordering or a complete contour. These cases illustrate matching behaviour at the baseline, rather than the representation interaction. Only these two cases and the three Figure 6 cases include the thresholded independent-matching contour; the continuation pages and category appendix compare continuous maps.

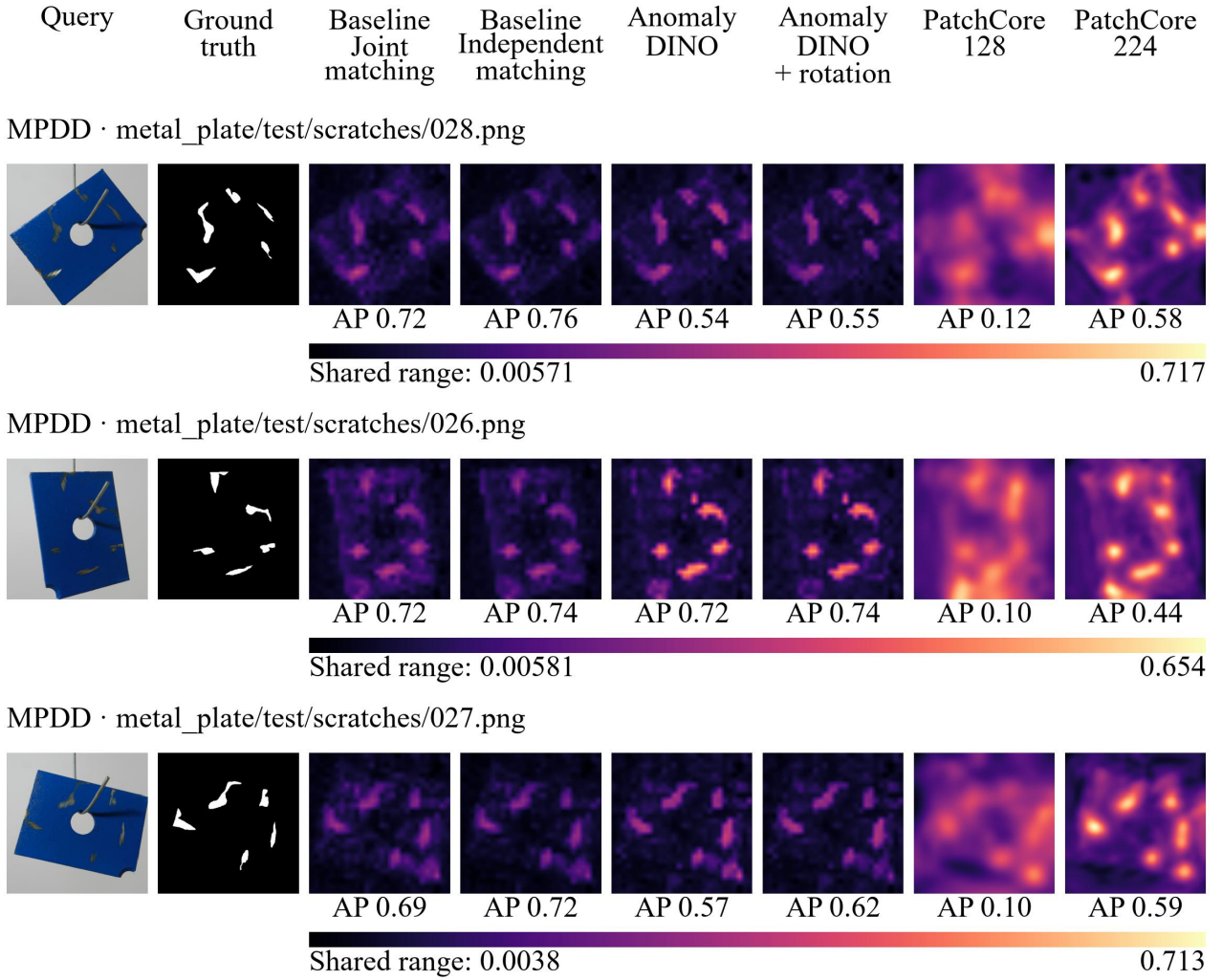


Figure 7 continued. MPDD metal plates with scratches. Six configurations share the original-image common valid region of Table 11 at seed 0 and $K = 4$. Query and GT are followed immediately by the two Dual-encoder baseline columns, then AnomalyDINO without/with reference rotation and the local 128-pixel/official 224-pixel PatchCore configurations. P-AP denotes per-image pixel AP, not category-pooled AP. Each row uses a shared score range across all six maps. Categories were selected to illustrate distinct surface, texture and circuit examples; within each category the three archived samples have the largest AP spread across configurations, with ties broken by sample identity. These selected extremes show contrasting responses, not representative performance or a method ranking. No method-specific thresholded contours are supplied on this page.

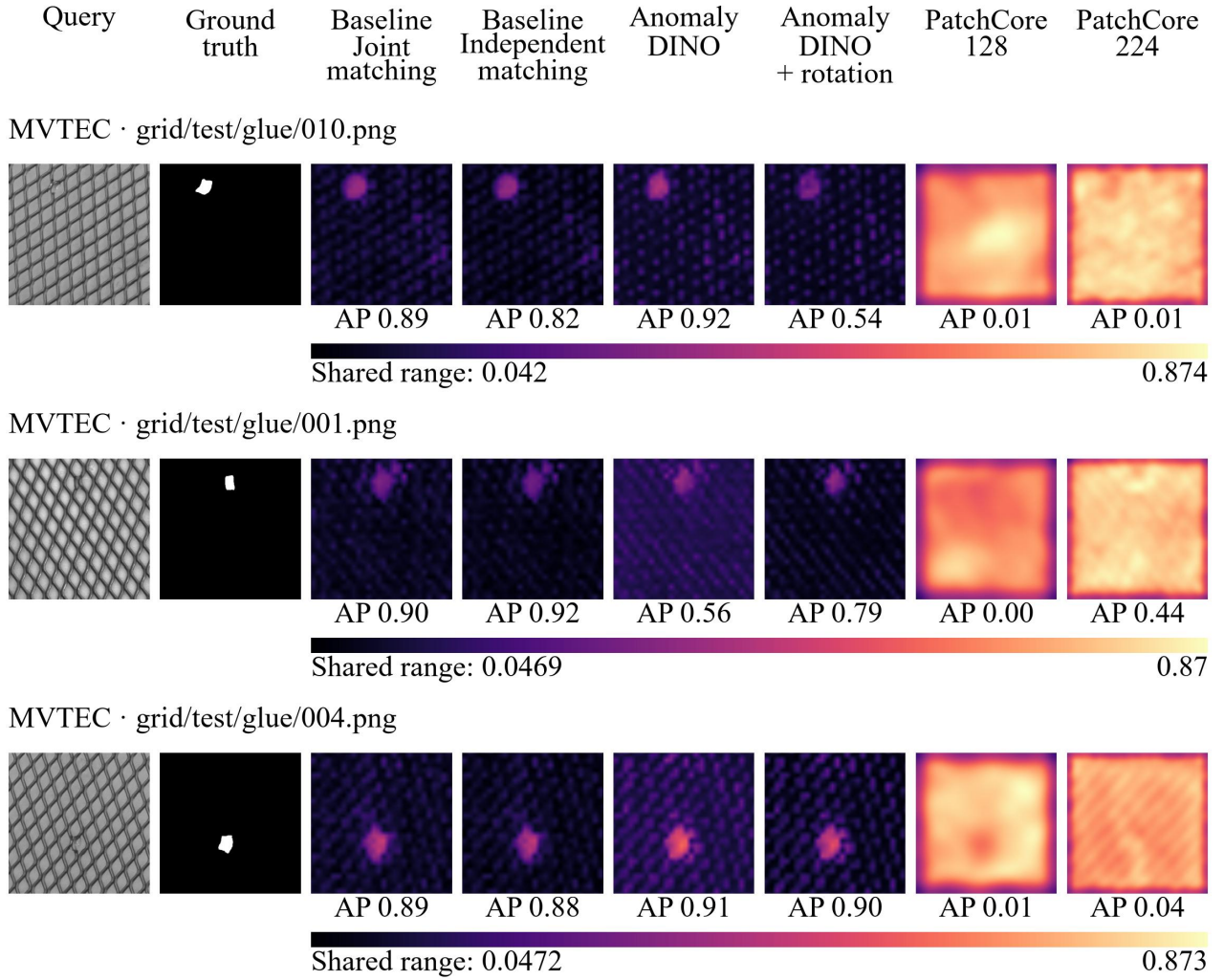


Figure 7 continued. MVTEC AD grid textures with glue defects. Six configurations share the original-image common valid region of Table 11 at seed 0 and $K = 4$. Query and GT are followed immediately by the two Dual-encoder baseline columns, then AnomalyDINO without/with reference rotation and the local 128-pixel/official 224-pixel PatchCore configurations. P-AP denotes per-image pixel AP, not category-pooled AP. Each row uses a shared score range across all six maps. Categories were selected to illustrate distinct surface, texture and circuit examples; within each category the three archived samples have the largest AP spread across configurations, with ties broken by sample identity. These selected extremes show contrasting responses, not representative performance or a method ranking. No method-specific thresholded contours are supplied on this page.

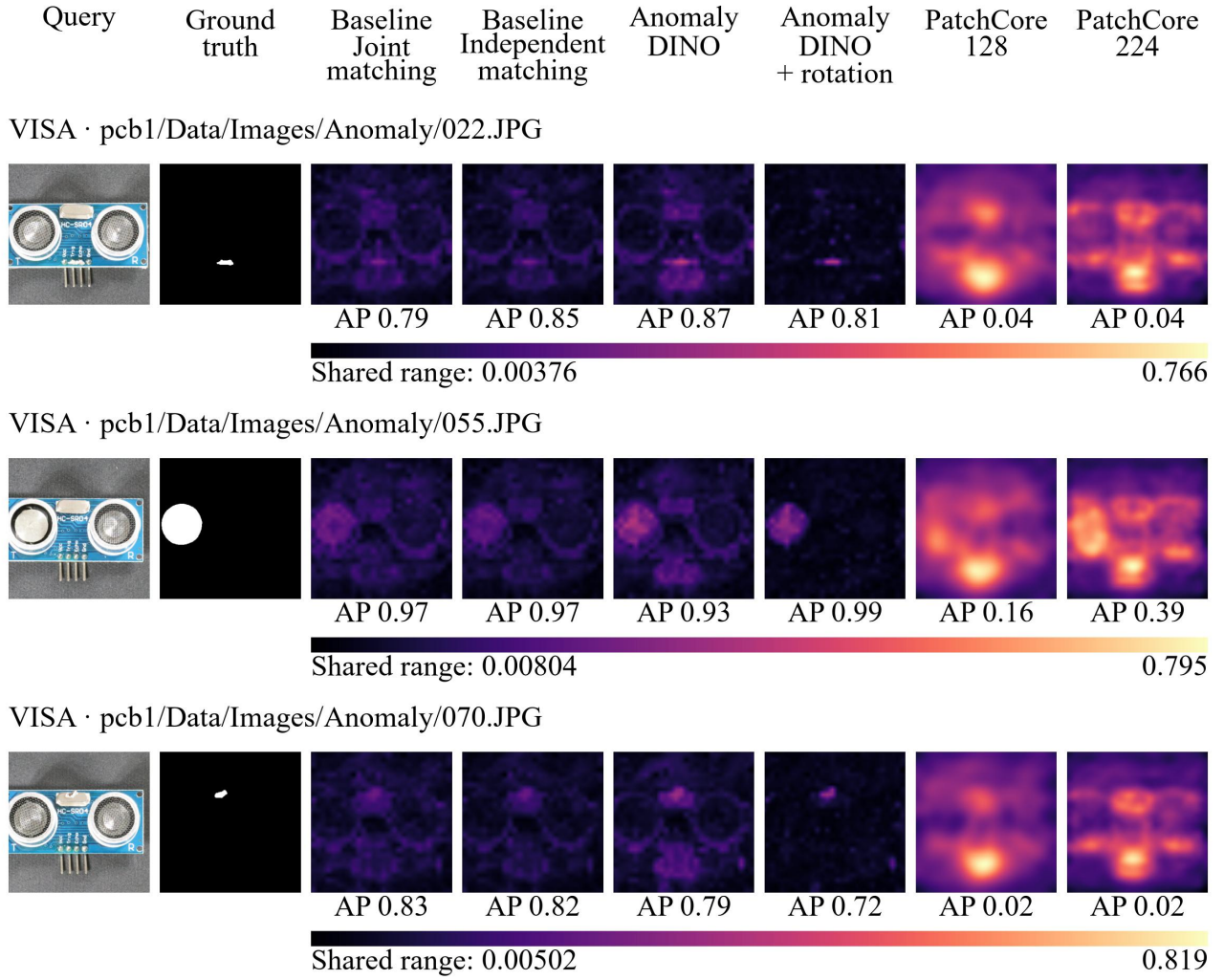


Figure 7 continued. VisA printed circuit boards. Six configurations share the original-image common valid region of Table 11 at seed 0 and $K = 4$. Query and GT are followed immediately by the two Dual-encoder baseline columns, then AnomalyDINO without/with reference rotation and the local 128-pixel/official 224-pixel PatchCore configurations. P-AP denotes per-image pixel AP, not category-pooled AP. Each row uses a shared score range across all six maps. Categories were selected to illustrate distinct surface, texture and circuit examples; within each category the three archived samples have the largest AP spread across configurations, with ties broken by sample identity. These selected extremes show contrasting responses, not representative performance or a method ranking. No method-specific thresholded contours are supplied on this page.

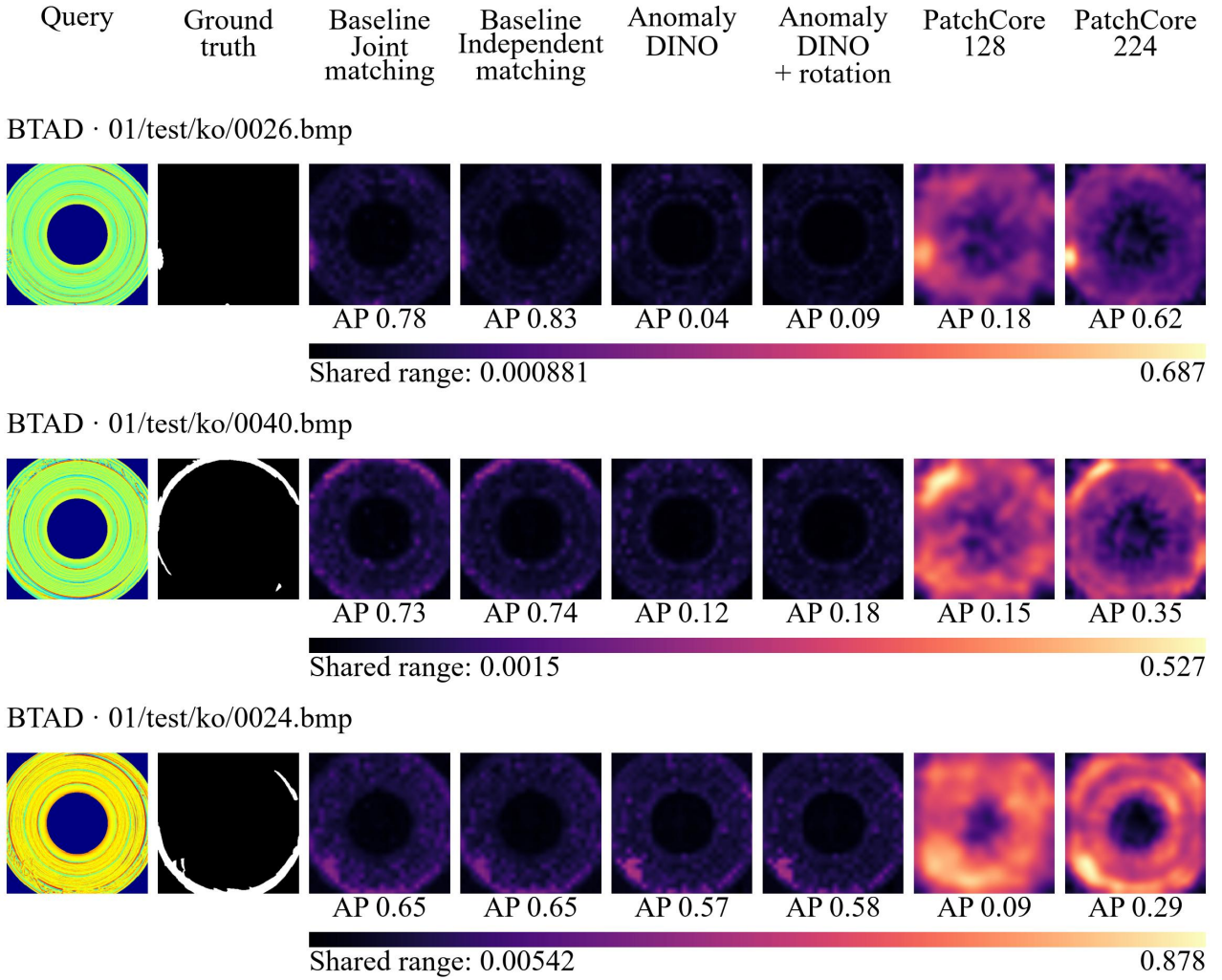


Figure 7 continued. BTAD category 01. Six configurations share the original-image common valid region of Table 11 at seed 0 and $K = 4$. Query and GT are followed immediately by the two Dual-encoder baseline columns, then AnomalyDINO without/with reference rotation and the local 128-pixel/official 224-pixel PatchCore configurations. P-AP denotes per-image pixel AP, not category-pooled AP. Each row uses a shared score range across all six maps. This page extends the four-dataset comparison of Table 11 to BTAD, whose aggregate interaction direction remains unresolved; within the category the three archived samples have the largest AP spread across configurations, with ties broken by sample identity. These selected extremes show contrasting responses, not representative performance or a method ranking. No method-specific thresholded contours are supplied on this page.

The initial five Dual-encoder baseline cases are the only main-text examples that show both a continuous map and a predicted contour. Figure 7 continues with four multi-method category panels from MPDD metal_plate, MVTEC AD grid, VisA pcb1 and BTAD category 01 at seed 0 and $K = 4$. Each panel retains three samples selected within its category by the largest per-image AP spread across the six displayed configurations, with ties broken by sample identity; it shows the query, ground truth and score maps, with the two Dual-encoder baseline columns next to the ground truth. Every row is evaluated on the common valid region and uses one shared magma minimum–maximum range across its method columns. The continuation pages therefore make the visual comparison inspectable across the four datasets of Table 11 and their defect types, but are selected illustrative examples rather than a representative performance estimate or a cross-method ranking. The remaining 32 category panels stay in the supplementary figure deck and show heatmaps only.

Improved AP does not imply an accurate defect boundary at an arbitrary operating threshold. The small contours in the five Dual-encoder baseline examples may cover only the strongest part of a scratch or mismatch, while secondary activations can remain elsewhere in the map. The degradation cases in Figure 7 likewise show that independent matching can change the relative responses unfavorably even though its raw patch distances are no larger. These observations motivate keeping continuous localization evidence,

thresholded visualization, and deployment segmentation accuracy as separate claims. No ground-truth outline is substituted for a predicted output.

External-method context. Table 11 evaluates six configurations on the intersection of their actual valid image regions across four datasets. Dual-encoder baseline under Independent matching has the highest observed AP on MPDD, BTAD and VisA among these configurations. On MVTec AD, AnomalyDINO without and with rotation reaches 0.5643 and 0.5637, respectively, above Dual-encoder baseline under Independent matching at 0.5570. The rotation-enabled AnomalyDINO configuration is higher than its non-rotation counterpart on MPDD, BTAD and VisA, but it is lower on MVTec AD (0.5643 to 0.5637). The 224-pixel PatchCore configuration is higher than the earlier local 128-pixel variant on all four displayed datasets. These within-method configuration changes show why an inadequately specified baseline can give a misleading impression of practical advantage.

Table 11. External-method context from six configurations on a common valid region.

Configuration	MPDD	BTAD	MVTec AD	VisA
Dual-encoder baseline / Independent matching	0.3698	0.6474	0.5570	0.3763
Dual-encoder baseline / Joint matching	0.3611	0.6380	0.5530	0.3713
AnomalyDINO + rotation	0.3214	0.5840	0.5637	0.3469
AnomalyDINO	0.3130	0.5614	0.5643	0.3291
PatchCore 224 / 1024	0.2275	0.3760	0.4955	0.3111
PatchCore 128 / 256	0.1649	0.2886	0.3966	0.2554

Category-macro pixel AP, averaged over seeds 0 and 1 and $K = 1$ and 4. All 864 method-category-condition rows are included. Average retained coverage is 76.56%, 70.49%, 76.56% and 59.07%, respectively. Methods differ in backbone, resolution and augmentation. Each configuration is reported under its own native protocol, keeping its own input resolution, evaluation canvas, rotation setting and reference-bank construction, so the table provides context and is not a ranking. Bold identifies Dual-encoder baseline, not a statistical superiority claim. Per-method protocol (input geometry, canvas, rotation and reference-bank construction) is detailed in Table S2.

The common region covers 76.56% of MPDD, 70.49% on average for BTAD, 76.56% of MVTec AD and 59.07% of VisA; BTAD category 03 has a smaller intersection of approximately 58.36%. Restricting evaluation avoids inventing predictions in cropped-away borders, but also removes those borders from the evaluation task. Different backbones, reference augmentation, resolutions and post-processing remain. Consequently, these point summaries provide local context rather than proof that Dual-encoder baseline exceeds complete published methods under every native or full-image protocol. AnomalyDINO also uses the DINOv2-S/14 backbone employed in the controlled study, so it is not an independent test of a wholly unrelated representation family.

Table 12 repeats the six frozen configurations and adds three further external families under their own recorded protocols. It remains a context table rather than a ranking: the added configurations differ in training provenance, input geometry and support protocol, and the zero-shot AnomalyCLIP row is not paired with the four-condition rows. The companion method-protocol table at the end of this section makes those distinctions explicit.

Table 12. Extension of Table 11 with three further external families under their own native protocols.

Configuration	Protocol	MPDD	BTAD	MVTec AD	VisA
Dual-encoder baseline / Independent matching	controlled, frozen	0.3698	0.6474	0.5570	0.3763
Dual-encoder baseline / Joint matching	controlled, frozen	0.3611	0.6380	0.5530	0.3713
AnomalyDINO + rotation	native, frozen normal modelling	0.3214	0.5840	0.5637	0.3469
AnomalyDINO	native, frozen normal modelling	0.3130	0.5614	0.5643	0.3291
PatchCore 224 / 1024	native, frozen normal modelling	0.2275	0.3760	0.4955	0.3111
PatchCore 128 / 256	native, frozen normal modelling	0.1649	0.2886	0.3966	0.2554
SubspaceAD 256 fp16	native, frozen normal modelling	0.3194	0.5869	0.4988	0.3203
WinCLIP+ 240	native, vision-language few-shot	0.1887	0.1137	0.3079	0.1112
AnomalyCLIP zero-shot 518	native, zero-shot on the target domain (upstream auxiliary-domain-trained prompt learner)	0.2724	0.4108	0.4262	0.1938

Category-macro pixel AP on the intersection of the regions the participating methods actually score. The six upper configurations are the frozen values of Table 11, copied row by row and not recomputed; the three added families keep their native input resolutions and protocols, stated in the Protocol column, and are therefore not compared under the controlled support protocol. Adding a method can change the common region, and a method whose scored extent is a subregion of the others would change the frozen values above when the table is recomputed; for the three families added here the joint recomputation was carried out and reproduced the frozen rows to the digit (864 rows, zero mismatches; 36 of 36 region rectangles identical), because each of them covers the full original image. The AnomalyCLIP row is a single native configuration with no seed or support-budget loop and is not paired with the four-condition rows. It uses an auxiliary-domain-trained prompt learner and is zero-shot on the target domain: the learner was fitted on an auxiliary dataset (VisA for the MVTEC AD column and MVTEC AD for the other three datasets, following the upstream convention of never fitting a learner on the dataset it evaluates), with no target-dataset fitting. Its checkpoint is supplied with the upstream AnomalyCLIP source archive (commit 3911738c0867544f545a076ad78f3f1d9ecbfdf, ZIP SHA256 533ED87B6658CDB247D063A249CEFEA54AB81623CB11683C6F02345B9A6CEAFE) and is an upstream-published auxiliary-domain-trained weight, not a checkpoint trained by this project. Methods differ in backbone, resolution, augmentation and training provenance, so the table provides context and is not a ranking. Per-method protocol is detailed in Table S2; SubspaceAD uses 256-pixel fp16 inputs rather than the upstream 672-pixel setting, recorded as a documented local deviation in the supplementary protocol section.

Figure S6 visualizes protocol sensitivity in the native external-method records. It is not the harmonised 448-pixel subset: its connected points are two configurations of the same method in the frozen native records. The displayed 0.100 macro pixel-AP difference between PatchCore configurations, the 0.026 comparison between SubspaceAD 256 fp16 and AnomalyDINO canvas, and the 33.3% configuration-switch reversal count are read-only aggregations of stored per-unit pixel AP, not new experimental estimates. The figure makes no ordering, interval, significance-test or state-of-the-art claim.

Table 13 reports the runtime stages for which instrumentation is available. Reference rotation increases AnomalyDINO's recorded processing time along with its AP on three of the four displayed datasets. The WideResNet50-2 extension separately records approximately 55.4 seconds of query encoding, 8.3 seconds of reference encoding, and 266.3 seconds of controlled scoring across its archived scope, with feature reuse

between configurations. Its recorded peak allocated GPU memory is approximately 417.7 MiB. Those stage totals are partially instrumented and do not constitute a synchronized complete-run latency or the simultaneous memory footprint of all encoders.

Table 13. Recorded native-method runtime stages and process GPU memory.

Dataset	Configuration	Processing s	Evaluation s	GPU peak MiB
MPDD	AnomalyDINO no rotation	44.47	8.38	111.6–112.5
	AnomalyDINO rotation	82.74	8.22	111.6–112.5
	PatchCore 224	99.75	23.40	Unavailable
BTAD	AnomalyDINO no rotation	53.19	16.39	118.8
	AnomalyDINO rotation	135.38	16.89	118.8
	PatchCore 224	115.45	40.85	Unavailable

Mean seconds per complete dataset condition over seeds 0 and 1 and $K = 1$ and 4; MPDD has 458 queries and BTAD 741. Processing is bank plus retrieval for AnomalyDINO and the combined native stage for PatchCore. GPU cells report the range of process peaks across conditions; each dataset name is printed once for its group. Cached feature extraction and other unrecorded stages are not added; these are not end-to-end per-image latencies. Dual-encoder baseline has no comparable complete historical timing.

Figure 8 shows the historical processing and evaluation stages for the two datasets in Table 13. These records remain distinct from the synchronized resource measurement in Section 4.2.4.

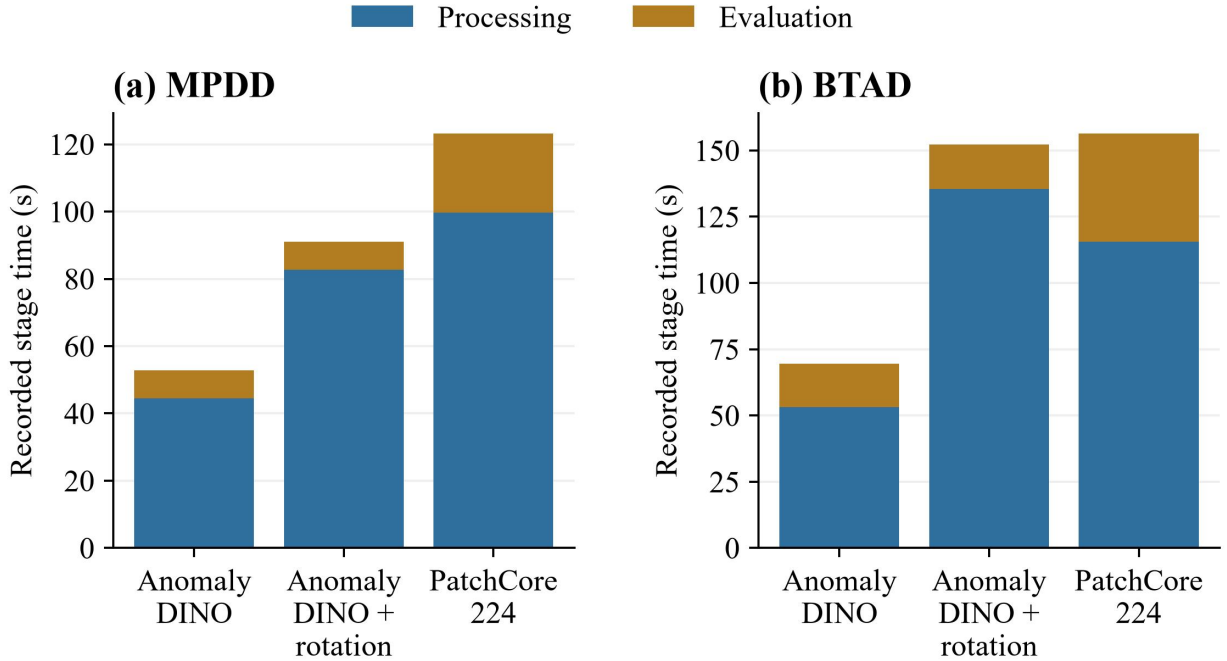


Figure 8. Historical processing and evaluation times from Table 13. Bars are mean seconds per complete dataset condition over seeds 0, 1 and $K = 1, 4$. Processing includes bank construction and retrieval for AnomalyDINO and an inseparable combined stage for PatchCore. Feature extraction is incompletely covered. These totals are historical stage measurements, not per-image latency; Figure S5 provides separately measured inference stages on three MPDD categories.

The historical records omit some stages and cannot be reconstructed as complete per-image latency. The synchronized measurement below supplies a separate measurement of timed reference-preparation and query-scoring stages on a restricted MPDD scope. Increasing the encoder set adds feature-extraction work

and descriptor storage, while increasing the support budget enlarges the bank. Whether an AP improvement justifies these costs remains application-dependent.

Separately specified confirmation. KolektorSDD2 (KSDD2) provides a separately specified confirmation beyond the previously explored datasets. Before any KSDD2 feature file existed, the specification frozen at 11:59 UTC on 18 September 2026 required all four DINOv2-S/14 and WideResNet50-2 interactions to be positive with individual 95% intervals excluding zero. Success was defined by this conjunction; a single interval spanning zero would fail the specified criterion.

The dataset contributes one category, a single production item, so the category macro average reduces to that category. Its positivity rule is fixed as "a sample is positive if and only if its ground-truth mask has at least one non-zero pixel", which gives 110 positive and 894 negative test images out of 1004. The frozen canvas is 630 pixels high by 224 pixels wide, giving a 45 by 16 patch grid with patch size and stride 14. The WideResNet50-2 branch reuses that same fixed canvas, so its input extent equals the DINOv2-B/14 canvas exactly; this resolves one ambiguity in the WideResNet50-2 branch text, which was written for the two older datasets. The AnomalyCLIP visual branch keeps its own 518-pixel preprocessing and is re-gridded onto the DINOv2-B/14 canvas as everywhere else. The scope is seeds 0, 1 and 2 with K equal to 1, 2, 4 and 8, giving twelve support conditions.

Table 14 reports the four individual 95% intervals specified in advance. This conjunction criterion is retained rather than changed after observing the results. The intervals do not provide a simultaneous 95% coverage guarantee, and the confirmation set remains separate from the exploratory adjustment families.

Table 14. Confirmation-set interactions on KolektorSDD2, whose expected direction was frozen before any of its features were encoded.

Branch	Contrast	Point	95% interval	Excludes zero
DINOv2-S/14	I_{Equal}	+0.539	[+0.299, +0.823]	Yes
	I_{Balanced}	+0.343	[+0.129, +0.579]	Yes
WideResNet50-2	I_{Equal}	+0.499	[+0.212, +0.851]	Yes
	I_{Balanced}	+0.367	[+0.128, +0.653]	Yes

AP percentage points, stride eight, 1000 paired replicates. One category and twelve support conditions (seeds 0, 1, 2 with $K = 1, 2, 4$ and 8). The frozen specification fixes the judgement as the 95% interval, so no family adjustment is applied to this confirmation set; the same four cells also exclude zero at 98.75%, which is recorded for transparency only. The Point column is the condition-averaged observed difference, the same convention as the point columns of the primary interaction tables; the replicate mean of the same four contrasts, which Figure S4 plots for this dataset, is a different summary of the same estimand and differs from this column by at most 0.005 points. Each branch name is printed once for its two rows.

All four point estimates are positive and all four 95% intervals exclude zero, so the frozen expectation is met on this dataset. We record for transparency that the same four cells also exclude zero at 98.75%, but the frozen judgement remains the 95% interval. The confirmation set deliberately does not enter the four-dataset table below, whose adjustment family was fixed over exploratory cells.

Validation across four datasets. The same two contrasts were later run on MVTec AD and VisA under the study's own aggregation rule, so that the validation datasets can be read beside MPDD and BTAD. Table 15 reports all four datasets at the 95%, 98.75% and 99.375% levels; the last level matches the eight-cell family that this table actually covers, whereas the 98.75% level remains the four-cell family of the primary DINOv2-S/14 analysis.

Table 15. The two interactions on all four datasets, with the interval levels that the wider family requires.

Dataset (role)	Contrast	Estimate	95%	98.75%	99.375%
MPDD (development)	I_{Equal}	+0.762	[+0.426, +1.118]	[+0.346, +1.211]	[+0.324, +1.250]
	I_{Balanced}	+0.615	[+0.322, +0.927]	[+0.189, +1.030]	[+0.152, +1.081]
BTAD (external frozen validation)	I_{Equal}	−0.020	[−0.192, +0.192]	[−0.237, +0.238]	[−0.245, +0.261]
	I_{Balanced}	−0.087	[−0.258, +0.109]	[−0.300, +0.179]	[−0.313, +0.201]
MVTec AD (external frozen validation)	I_{Equal}	+0.432	[+0.342, +0.530]	[+0.324, +0.553]	[+0.309, +0.570]
	I_{Balanced}	+0.402	[+0.313, +0.503]	[+0.297, +0.529]	[+0.287, +0.534]
VisA (in-domain frozen validation)	I_{Equal}	+0.927	[+0.788, +1.064]	[+0.754, +1.112]	[+0.738, +1.114]
	I_{Balanced}	+0.810	[+0.673, +0.959]	[+0.631, +1.011]	[+0.611, +1.018]

AP percentage points, stride eight, 1000 paired bootstrap replicates. The estimate is the mean of the replicate distribution, not the condition-averaged observed difference used in Table 7; on MPDD the two differ by 0.01 point. The 98.75% level is a Bonferroni adjustment over the four MPDD and BTAD cells and the 99.375% level over the eight cells this table covers. The BTAD row pools the study’s canonical-mask series rather than the corrected geometry used in Section 4.2.1, which moves the point estimate by about 0.03 point without changing the judgement. Stride-one point estimates are now available for all four datasets (+0.787 and +0.601 points on MPDD, −0.045 and −0.115 on BTAD, +0.423 and +0.386 on MVTec AD, +0.896 and +0.760 on VisA, each pair ordered as I_{Equal} then I_{Balanced}). KolektorSDD2 is reported separately in Table 14 and is not part of this family. Each dataset name is printed once for its two rows.

The four datasets do not behave alike, and we keep their readings separate instead of summarizing them as a replication. On MPDD both interactions are positive and exclude zero at the widest level reported here. On MVTec AD both are positive, with the narrowest intervals in the table, and they also exclude zero at the widest level; their point estimates, about +0.4 points, are the smallest of the three positive datasets. On VisA both are positive and the largest in the table, which does not provide independent unseen-domain evidence because VisA retains the conservative in-domain role specified in Section 4.1.1. The visual-only AnomalyCLIP visual path does not use the learned prompt outputs. On BTAD the point estimate is close to zero and both intervals span zero at all three levels, so the current data are insufficient to determine the direction of the interaction there; the point estimates are an order of magnitude smaller than those of the positive datasets, and the table must not be read as four consistent significant replications. This is not evidence of equivalence between the two rules; it means only that no direction is established there.

Three qualifications belong with the table. First, MVTec AD and VisA are frozen validation sets, not untouched confirmation sets; only KolektorSDD2 was frozen before its features were encoded. Second, the BTAD row pools the study’s canonical-mask series, whereas the primary BTAD tables use the corrected geometry for category 03; the two differ by about 0.03 point in the point estimate and both fail to exclude zero, so the reading is unchanged, but the numbers should not be quoted interchangeably. Third, the estimate column is the mean of the replicate distribution rather than the condition-averaged observed difference used in Table 7, a difference of 0.01 point on MPDD; stride-one point estimates now exist for all four datasets (+0.787 and +0.601 points on MPDD, −0.045 and −0.115 on BTAD, +0.423 and +0.386 on MVTec AD, +0.896 and +0.760 on VisA).

We use the role labels recorded by the table: development for MPDD, external frozen validation for BTAD and MVTec AD, and in-domain frozen validation for VisA. The frozen encoder specification records BTAD as a holdout; both strings describe the same fact, that BTAD had been examined in the project before these runs.

4.2.3 Sensitivity and Robustness

Further frozen encoders and support-set variation. Three further frozen encoders entered the extra slot after the DINOv2-S/14 and WideResNet50-2 results were known, so they are exploratory transfer checks rather than prespecified confirmations. DINO ViT-S/8 is the original DINO self-distillation ViT-S/8 with 384-dimensional descriptors; ConvNeXt-Tiny is ConvNeXt-Tiny with 576; Swin-Tiny is Swin-Tiny with 576. Each is mapped to the DINOv2-B/14 canvas and normalized exactly like the other branches, and the same Equal-weight replacement and Balanced replacement contrasts are formed at the frozen slot weights. Table 16 lists each encoder's interaction in the four-condition scope shared with DINOv2-S/14 and WideResNet50-2 and in the wider twelve-condition scope recorded for the additional encoders.

Table 16. Interactions of the five frozen encoders, in the four-condition scope shared with DINOv2-S/14 and WideResNet50-2 and in the wider twelve-condition scope.

Encoder (conditions)	MPDD I_{Equal}	MPDD I_{Balanced}	BTAD I_{Equal}	BTAD I_{Balanced}
DINOv2-S/14 (4)	+0.695 [+0.193, +1.160]	+0.547 [+0.078, +0.962]	+0.029 [-0.189, +0.293]	-0.052 [-0.273, +0.208]
WideResNet50-2 (4)	+1.020 [+0.514, +1.689]	+0.686 [+0.162, +1.394]	+0.631 [+0.298, +1.003]	+0.524 [+0.188, +0.995]
DINO ViT-S/8 (4)	+0.811 [+0.246, +1.362]	+0.488 [-0.136, +1.091]	+0.583 [+0.172, +1.056]	+0.224 [-0.158, +0.649]
DINO ViT-S/8 (12)	+0.995 [+0.529, +1.504]	+0.747 [+0.290, +1.309]	+0.441 [+0.067, +0.829]	+0.174 [-0.138, +0.516]
ConvNeXt-Tiny (4)	-0.018 [-0.358, +0.298]	+0.076 [-0.315, +0.588]	+0.278 [+0.152, +0.494]	+0.422 [+0.158, +0.872]
ConvNeXt-Tiny (12)	-0.038 [-0.266, +0.170]	+0.036 [-0.235, +0.443]	+0.178 [+0.090, +0.315]	+0.226 [+0.063, +0.491]
Swin-Tiny (4)	+1.253 [+0.644, +2.056]	+0.745 [+0.055, +1.425]	+0.488 [+0.217, +0.906]	+0.465 [+0.171, +0.955]
Swin-Tiny (12)	+1.397 [+0.803, +2.215]	+0.939 [+0.355, +1.620]	+0.298 [+0.073, +0.578]	+0.180 [-0.029, +0.442]

AP percentage points, stride eight; each cell gives the replicate mean followed by the 98.75% paired interval, the Bonferroni level for that encoder's four cells in that scope. Cells marked (4) pool seeds 0 and 1 with $K = 1$ and 4, the primary scope shared with DINOv2-S/14 and WideResNet50-2, so their paired differences against DINOv2-S/14 are same-condition pairs; cells marked (12) pool seeds 0, 1, 2 with $K = 1, 2, 4$ and 8, the wider scope recorded for the three additional encoders. Both scopes are given because they change some of the zero-exclusion decisions. No family adjustment covers the additional encoders as a group, because DINO ViT-S/8, ConvNeXt-Tiny and Swin-Tiny were added after the DINOv2-S/14 and WideResNet50-2 results were known.

The four-condition scope uses seeds 0 and 1 with K equal to 1 and 4 for every encoder, enabling paired encoder comparisons. The wider scope uses seeds 0, 1 and 2 with all four budgets. It is reported separately because support coverage changes some interval-exclusion decisions.

Within the four-condition scope the additional encoders split on MPDD: Swin-Tiny has positive interactions in both contrasts with intervals above zero, DINO ViT-S/8 has positive points in both but its Balanced replacement interval spans zero, ConvNeXt-Tiny has none, and DINOv2-S/14 and WideResNet50-2 remain positive. On BTAD all three additional encoders have a positive I_{Equal} with an interval above zero, while for I_{Balanced} only ConvNeXt-Tiny and Swin-Tiny do. In the twelve-condition scope DINO ViT-S/8's MPDD Balanced replacement interval also clears zero, while Swin-Tiny's BTAD Balanced replacement interval no longer does. Across both scopes the only statement that survives concerns ConvNeXt-Tiny: it is the single encoder whose MPDD Equal-weight replacement interaction is negative and lies below the DINOv2-S/14 interaction with a paired interval that excludes zero, at -0.712 points in the four-condition scope and -0.733

in the twelve-condition scope. The paired differences between additional encoders and DINOv2-S/14 differences exclude zero in few cells, and which cells they are depends on the scope: in the four-condition scope ConvNeXt-Tiny on MPDD Equal-weight replacement, DINO ViT-S/8 and Swin-Tiny on BTAD Equal-weight replacement, and ConvNeXt-Tiny and Swin-Tiny on BTAD Balanced replacement; in the twelve-condition scope ConvNeXt-Tiny on MPDD in both contrasts, Swin-Tiny on MPDD Equal-weight replacement, DINO ViT-S/8 on BTAD Equal-weight replacement and ConvNeXt-Tiny on BTAD Balanced replacement. We therefore read the extra encoders as a transfer check whose ordering is scope-sensitive, not as a ranking of encoders.

Single-branch sanity is verified for all three additional encoders, each with forty-eight stored single-branch units and a passing single-branch gate. DINO ViT-S/8 spans pixel AUROC 0.916 to 0.992 with a mean of 0.967, ConvNeXt-Tiny spans 0.887 to 0.984 with a mean of 0.953, and Swin-Tiny spans 0.906 to 0.998 with a mean of 0.952. These units cover the MPDD study revision and the corrected BTAD revision, so the interaction estimates do not rest on fused-construction rows alone.

A separate extension varies the support set rather than the encoder. Eight seeds were run, of which seeds 0 to 2 keep their own query encodings and seeds 3 to 7 share a single query block, so the latter isolate the support-set contribution; the same test images are scored under every seed and only the normal references change. Table 17 summarizes the per-seed interaction, and Figure 5(b) draws the same series.

Table 17. Descriptive variation across eight support seeds.

Series	Mean point	SD across seeds	Same sign at every seed	Seeds excluding zero (95%)	Support / test uncertainty
MPDD I_{Equal}	+0.679	0.221	Yes	7 of 8	0.48
MPDD I_{Balanced}	+0.548	0.159	Yes	6 of 8	0.37
BTAD I_{Equal}	+0.125	0.136	No	3 of 8	0.68
BTAD I_{Balanced}	+0.036	0.129	No	0 of 8	0.65

AP percentage points. Seeds 0–2 retain separate query encodings; seeds 3–7 share a query block and vary the support set alone. The final column divides the between-seed standard deviation of the per-seed replicate means (paired at a common replicate index, so the test-image bootstrap cancels and only the support-set variation remains) by the median individual 95% bootstrap interval half-width across the eight seeds; it is therefore built on a different standard deviation from the SD across seeds column, which is the spread of the per-seed observed points. Values are descriptive, use canonical BTAD masks, and do not replace the corrected-geometry primary analysis. Figure 5(b) shows the same per-seed series.

The eight-seed variance evidence covers MPDD and BTAD only; no corresponding cross-seed table is available for MVTec AD, VisA or KolektorSDD2. For the recorded MPDD series, the mean points and between-seed standard deviations are +0.679 and 0.221 for I_{Equal} and +0.548 and 0.159 for I_{Balanced} ; both retain the same sign at every seed. The BTAD series have means of +0.125 and +0.036 with standard deviations of 0.136 and 0.129, respectively, and change sign. This variation describes sensitivity to the observed supports and complements the conditional image-bootstrap intervals; it is not an additional independent replication or a variance estimate for the other three datasets.

Correspondence sensitivity. Geometric alignment in this study is a deterministic canvas correspondence, not a learned or verified part-level correspondence. We therefore replaced the correspondence between the DINOv2-B/14 and the AnomalyCLIP visual rows in four ways on MPDD, using six categories, seeds 0 and 1, K equal to 1 and 4, four support conditions and the study's shared bootstrap stream: the canvas rule itself; a closed-form orthogonal Procrustes map fitted on the support rows and applied to the query and reference rows alike; a position permutation of the AnomalyCLIP visual rows, identical for queries and references; and a Sinkhorn optimal-transport correspondence whose cost matrix is the cross-branch cosine distance of the support descriptors, with its entropy regularizer fixed a priori at 0.1 times the interquartile range of that matrix. The transport rule mixes each AnomalyCLIP visual row over several canvas positions instead of

relabeling it. We also report the exact assignment obtained from the same cost matrix as the regularizer tends to zero.

Table 18. The same interaction under five correspondence conventions on MPDD.

Correspondence	I_{Equal} point	I_{Equal} 98.75%	I_{Balanced} point	I_{Balanced} 98.75%
Canvas rule (identity)	+0.767	[+0.193, +1.160]	+0.547	[+0.078, +0.962]
Procrustes (orthogonal)	+0.767	[+0.193, +1.160]	+0.547	[+0.078, +0.962]
Position permutation	+0.799	[+0.273, +1.283]	+0.813	[+0.310, +1.480]
OT, exact assignment	+0.834	[+0.207, +1.274]	+0.910	[+0.327, +1.462]
OT, soft mixing	+0.591	[+0.191, +1.077]	+0.570	[+0.043, +0.971]

AP percentage points; six MPDD categories, seeds 0 and 1, $K = 1$ and 4, four support conditions, 1000 paired replicates, the same bootstrap stream for every row. Intervals use a four-comparison adjustment within each setting across two datasets and two contrasts, not a simultaneous guarantee over all settings; the 98.75% bounds, and are percentile bounds of the image-level bootstrap series, formed inside each replicate and aggregated over categories and support conditions before the percentile is taken. The exact assignment is the limit of the transport rule as its regulariser tends to zero and uses the same cost matrix; the soft-mixing row fixes that regulariser a priori at 0.1 times the interquartile range of the support cost matrix. The canvas row reproduces the DINOv2-S/14 branch's MPDD interaction of Table 16 - its replicate mean is +0.695 with the same interval - and the point column of this table differs from that row only because it reports the condition-averaged difference rather than the replicate mean. On BTAD this harness scores through its own path, so its numbers belong to that chain and are not mixed with the WideResNet50-2, DINO ViT-S/8 or ConvNeXt-Tiny rows.

Table 19. Sensitivity to the transport regularizer on MPDD and BTAD.

Setting	MPDD I_{Equal}	MPDD I_{Balanced}	BTAD I_{Equal}	BTAD I_{Balanced}
OT, exact assignment (regulariser 0)	+0.834 [+0.207, +1.274]	+0.910 [+0.327, +1.462]	+0.220 [−0.044, +0.660]	+0.148 [−0.181, +0.699]
OT, regulariser 0.05 IQR	+0.580 [+0.169, +1.007]	+0.624 [+0.006, +1.037]	+0.162 [−0.080, +0.554]	+0.156 [−0.151, +0.653]
OT, regulariser 0.10 IQR (prescribed)	+0.591 [+0.191, +1.077]	+0.570 [+0.043, +0.971]	+0.115 [−0.109, +0.465]	+0.103 [−0.156, +0.561]
OT, regulariser 0.50 IQR	+0.725 [+0.327, +1.165]	+0.751 [+0.253, +1.151]	+0.061 [−0.124, +0.363]	+0.025 [−0.170, +0.353]

Observed condition means and 98.75% paired image-bootstrap intervals in AP percentage points. Seeds 0, 1; $K = 1, 4$; 1000 replicates; stride-eight pixels. Each setting has a separate four-comparison adjustment across two datasets and two contrasts. No adjustment covers the whole sweep. IQR is the interquartile range of support costs. BTAD uses the separate correspondence implementation.

Across the seven settings, the canvas rule, Procrustes, the permutation and four regularizer values including the exact assignment, all fourteen point estimates are positive and all fourteen MPDD 98.75% intervals exclude zero. Each setting has a four-comparison adjustment, without a simultaneous guarantee across the full sweep. Exact assignment gives the largest observed point estimates among these settings, and the only cell whose lower bound sits close to zero is the intermediate 0.05 regularizer value on I_{Balanced} , at +0.006 points.

Two properties of the transport rule belong with this table. The soft mixing also changes the multiset of the AnomalyCLIP visual descriptors, since each row becomes a weighted combination of roughly 15 to 40 canvas positions, with a median row-maximum weight of 0.044, whereas a position permutation and exact assignment preserve the descriptor multiset. An orthogonal Procrustes map changes the descriptor values but preserves within-branch distances when applied consistently to queries and references. The sweep is

nonetheless reported in full rather than summarized by its prescribed regularizer, so that a reader can see that the zero-exclusion judgement of the interaction is preserved under that constant. Table 19 gives the transport sweep for MPDD and BTAD. The separate encoder-scope comparison remains in Table 16.

The transport plans themselves carry almost no canvas-position information: the median diagonal mass is 0.000949 against a chance value of 0.000977, and the fitted Procrustes map sits at a Frobenius distance of about 39 from the identity, so neither solves a position-level correspondence. We also record the verification of the new variant: it recovers a known synthetic assignment exactly, it reproduces the stored patch scores when the plan is forced to the identity with a maximum absolute difference of $6.4\text{e-}07$, and the three earlier variants reproduce to at most $1.8\text{e-}06$ once the new code path is present. The intervals in this table are percentile bounds of the image-level bootstrap series, formed inside each replicate and aggregated over categories and conditions before the percentile is taken, which is the convention of the primary tables. On MPDD the canvas row reproduces the DINOv2-S/14 branch's interaction of Table 16 under that convention: its replicate mean is +0.695 against +0.695 and its interval $[+0.193, +1.160]$ is the same, the two agreeing to $2.4\text{e-}07$ in the bounds, so the 0.07-point gap between the point column of Table 18 and the DINOv2-S/14 row of Table 16 is the difference between the condition-averaged difference and the replicate mean rather than a difference between scoring chains. On BTAD this harness still scores through its own path, and its BTAD numbers must not be mixed with the WideResNet50-2, DINO ViT-S/8 or ConvNeXt-Tiny numbers.

On BTAD all four variants were run, together with the same sweep over four regularizer values. No interval excludes zero for either contrast: the canvas rule gives +0.001 points $[-0.183, +0.299]$ for I_{Equal} and -0.080 $[-0.270, +0.226]$ for I_{Balanced} , Procrustes gives +0.001 $[-0.183, +0.299]$ and -0.080 $[-0.270, +0.226]$, the position permutation gives +0.234 $[-0.058, +0.715]$ and +0.217 $[-0.112, +0.773]$, and the soft mixing at the prescribed regularizer gives +0.115 $[-0.109, +0.465]$ and +0.103 $[-0.156, +0.561]$. The regularizer sweep spans the same range: the exact assignment gives +0.220 $[-0.044, +0.660]$ and +0.148 $[-0.181, +0.699]$, the intermediate 0.05 value gives +0.162 $[-0.080, +0.554]$ and +0.156 $[-0.151, +0.653]$, and the widest mixing at 0.5 gives +0.061 $[-0.124, +0.363]$ and +0.025 $[-0.170, +0.353]$. All fourteen intervals span zero, so on this dataset neither replacing the correspondence nor changing the amount of smoothing in the transport rule changes the judgement, which was already "no evidence of a direction".

Geometry, shared-operation and numerical-stability checks. The corrected BTAD category-03 geometry was recomputed on a finer stride-four evaluation grid covering all three categories. The reproduction check against the study revision passes on four cells with a maximum absolute difference of $7.0\text{e-}08$, and the nested-support check records no violation in sixteen tests. The resulting bootstrap means and 98.75% intervals are -0.011 points $[-0.224, +0.251]$ for I_{Equal} and -0.085 $[-0.296, +0.177]$ for I_{Balanced} ; neither excludes zero, so the coarser stride-eight reading is unchanged at the finer grid. The eight-seed BTAD extension uses the historical canonical-mask convention, not the corrected-geometry convention used for the primary tables, and is descriptive. Figure S2 reports the single-condition shared-operation ablations and Figure S3 collects the geometry panels and per-image cases; neither supplies primary-scope validation of the shared operations.

Because the pipeline uses frozen encoders and no target-domain optimizer, a training loss or epoch-convergence curve is not defined. Figure S4 instead examines numerical stability of the stored bootstrap estimator by successively taking prefixes of each 1000-replicate array. Across the ten dataset-contrast series and the displayed prefixes of at least 500 replicates, the largest deviation from the final mean is approximately 0.00013 pixel AP and the largest relative interval-width deviation is 6.8%. At prefixes of at least 200, the latter can still reach 17.1%. These observations support the numerical stability of the reported 1000-replicate summaries on the tested prefixes; they do not establish model convergence or remove sampling uncertainty. They also are not a performance-stability comparison of all external configurations: no common metric-by-perturbation experiment was produced for that purpose, so no cross-method stability curve is reported. The plotted intervals are individual 95% intervals, while the adjusted inferential families remain those specified for the respective primary, validation and confirmation analyses.

4.2.4 Resources and Limits

The synchronized benchmark supplements the historical stage records with a common measurement procedure for all six configurations (Table 20 and Figure S5). All six were measured on the same NVIDIA GeForce RTX 3060 Laptop GPU under PyTorch 2.0.0+cu118 and CUDA 11.8. Its scope is deliberately restricted to the three MPDD bracket categories at seed 0 and K equal to 1 and 4. Each of the six category–budget units has one warm-up and three timed repeats, yielding 144 successful runs including warm-ups and no failures in the final measurement file. The 216 unique query images are visited twice across the two budgets. Reference preparation is included, so dividing the total by 432 would not yield query-only latency. The common hardware and measurement procedure do not make workloads identical: the methods retain their own input resolutions, augmentations, feature paths and reference-bank construction.

For each repeat, recorded stage times are summed over all six units; the table reports the median and observed range of the three resulting totals. Dual-encoder baseline under Joint matching and Dual-encoder baseline under Independent matching require 197.118 and 195.300 seconds, respectively, whereas the tested AnomalyDINO and PatchCore configurations range from 24.190 to 78.863 seconds. Dual-encoder baseline therefore has a substantial encoding cost in this scope; the small Joint–Independent matching timing difference does not support a general speed advantage. Its median allocated peak is 2376.8 MiB with both DINOv2-B/14 and AnomalyCLIP visual resident, compared with 111.6 MiB for AnomalyDINO and approximately 403–410 MiB for PatchCore. These measurements document the cost of the tested implementations rather than a universally optimized hardware ranking.

The PatchCore 128 configuration had an initial 30.527-second total in the first measurement pass. The reported 24.190-second value is the documented remeasurement in ``experiments/dynamic_fusion/representation_matching_interaction_20260914/_bench_speed_vram/recheck/``; it is retained because the original preprocessing excursion was rechecked on the same files and was not a protocol property. The table's three-repeat minimum–maximum is an observed range for this restricted procedure, not a full characterization of run-to-run variability.

The timer covers measured preprocessing, encoding and scoring stages, including support-bank work. It excludes model and dataset setup, metric evaluation and output-file writes. For Dual-encoder baseline and AnomalyDINO, the reported total is the sum of stage timers; for PatchCore, it is fit plus predict, with scoring represented by the residual after preprocessing and encoding. No independent complete-process wall-clock measurement or query-only latency distribution is available. Allocated-memory peaks are process-local PyTorch measurements in MiB. The archived 1 Hz device-wide nvidia-smi increases are separate observations affected by the device baseline and other activity; they are not substituted for these process peaks.

Table 20. Timed-stage sums and allocated GPU peaks on the restricted MPDD benchmark.

Configuration	Median time (s)	Time range (s)	Peak allocation (MiB)
Dual-encoder baseline / Joint matching	197.118	196.823–197.424	2376.8
Dual-encoder baseline / Independent matching	195.300	195.162–195.491	2376.8
AnomalyDINO	30.959	30.871–36.532	111.6
AnomalyDINO + rotation	78.863	78.794–79.012	111.6
PatchCore 128	24.190	24.080–24.331	403.1
PatchCore 224	36.931	36.557–37.162	409.8

Time: median and observed min–max across three repeat-level sums of six units. Memory: median of 18 timed unit peaks. Seed 0; $K = 1, 4$; three MPDD categories; one warm-up per unit. Timing boundaries and native-protocol differences are specified in Section 4.2.4 and Figure S5. Bold marks the Dual-encoder baseline configurations, not the fastest or best result.

Image-level metrics. Table 21 reports image AUROC and image AP alongside stride-eight pixel metrics for the two Dual-encoder baseline configurations on all four datasets, using the matched four conditions (seeds 0 and 1; $K = 1$ and 4). Image-level pooling and pixel ranking answer different questions, and no image-level interval is claimed.

Table 21. Image-level and pixel-level point estimates for the Dual-encoder baseline configurations under matched support conditions.

Dataset	Configuration	Image AUROC	Image AP	Pixel AUROC	Pixel AP
MPDD	Dual-encoder baseline / Joint matching	0.7809	0.7999	0.9665	0.3596
MPDD	Dual-encoder baseline / Independent matching	0.7848	0.8093	0.9696	0.3649
BTAD	Dual-encoder baseline / Joint matching	0.9324	0.9414	0.9725	0.6339
BTAD	Dual-encoder baseline / Independent matching	0.9321	0.9468	0.9736	0.6434
MVTec AD	Dual-encoder baseline / Joint matching	0.9572	0.9773	0.9663	0.5548
MVTec AD	Dual-encoder baseline / Independent matching	0.9582	0.9771	0.9677	0.5569
VisA	Dual-encoder baseline / Joint matching	0.9029	0.9049	0.9736	0.3634
VisA	Dual-encoder baseline / Independent matching	0.9041	0.9038	0.9761	0.3628

Seeds 0 and 1; $K = 1$ and 4; equal category means followed by equal condition means. Categories: MPDD 6, BTAD 3, MVTec AD 15, VisA 12. Pixel metrics use the retained output canvas at stride eight; BTAD uses corrected coordinates. These are current controlled-pipeline point estimates, not the common-original-region metrics of Tables 11 and 12. No image-level confidence interval is available. Bold identifies the Dual-encoder baseline configurations, not statistically optimal results.

Table S1 gives the restricted harmonised-geometry subset, Table S2 records each evaluated input and reference protocol, and Figure S6 shows configuration sensitivity in the external-method context. These supplement the controlled attribution study; they do not define a cross-method ranking.

5 Discussion

The practical implication is to inspect the source of a fusion gain before enlarging an encoder set. The duplicate control tests whether effective weights alone change the result. A representation replacement at fixed weights tests whether the new descriptor helps. Crossing that replacement with the matching rule tests whether reference selection changes its value. The MPDD and BTAD results show why these questions cannot be collapsed into a single three-branch-versus-two-branch comparison.

The interpretation remains limited in several ways. First, the primary interaction was formulated after earlier project exploration, and the datasets are not untouched confirmation sets. Second, the bootstrap conditions on the observed categories and fixed support manifests; its intervals do not characterize arbitrary future support choices. Third, only one prespecified heterogeneous encoder was added, with a smaller seed and budget scope than the DINOv2-S/14 study. Fourth, cross-seed variation is reported only for MPDD and BTAD, full-pixel intervals cover the DINOv2-S/14 analysis on MPDD and canonical-mask BTAD and are supplied with the reproduction materials rather than in the main tables, and the synchronized resource benchmark covers one machine and three MPDD categories rather than all datasets. Fifth, Figure S4 diagnoses stored-bootstrap numerical stability rather than training convergence or stability across all methods, and the harmonised

comparison is a 36-of-144-unit subset. Sixth, qualitative cases are selected extremes, and their Otsu contours are display choices rather than calibrated defect masks. Seventh, the operations that every construction shares, namely spatial mapping, branch normalization and scoring from concatenated descriptors, are held fixed rather than ablated at the primary scope, so this study does not quantify how much each of them contributes and does not present them as validated modules. Finally, the geometric alignment is deterministic canvas correspondence, not a learned guarantee that receptive fields describe exactly the same object part.

These qualifications leave the central finding conditional but interpretable. BTAD with DINOv2-S/14 separates positive representation value from uncertain matching dependence, while the WideResNet50-2 comparison provides an encoder-specific difference under matched conditions. The study's contribution is this controlled and qualified account. A future learned selector, foreground filter or dynamic fusion module would require its own implementation and evidence; none is implied by the present results.

Recent training-free extensions operate on axes other than the factorized comparison of representation replacement and matching rule studied here: sparse hyperedge aggregation (Hyper-FSAD) [23], iterative memory evolution (ReMem) [24] and multi-layer attention reweighting (DuoAD) [25]. They address detector design under their respective reference and inference settings, which is a separate question from whether the value of one representation replacement depends on the normal-reference matching rule. Adding them would complement the controlled attribution reported here, but their reference aggregation, memory update or layer weighting would also change the fixed operators under comparison. A controlled test of whether such mechanisms change the measured interaction therefore requires a separate design; the present study keeps those operations fixed and uses 0 target-trainable parameters.

6 Conclusion

This study separates the value of additional frozen visual representations from reweighting and normal-reference selection in few-shot industrial anomaly localization. Exact duplication and complementary fixed-weight replacements make the representation change explicit. Crossing these controls with joint and independent matching reveals whether its value depends on sharing a normal reference, beyond any absolute gain from the added encoder.

The results demonstrate why this distinction matters. DINOv2-S/14 shows positive matching interactions on MPDD, MVTec AD and VisA, while its interaction direction remains unresolved on BTAD despite positive absolute representation effects. WideResNet50-2 yields positive interactions on both primary datasets, and the separately specified KolektorSDD2 confirmation meets all four directional criteria. Encoder and support-set analyses identify substantial variation; correspondence transformations preserve positive MPDD interactions within the tested settings, while BTAD interval directions remain unresolved.

The resulting design guidance is to evaluate representation composition and reference selection together under explicit weight controls. Its scope remains conditional on the tested encoders, support sets and dataset provenance. Future work should extend prospective comparisons to additional production settings, test correspondence effects under broader prospective conditions, and evaluate calibrated thresholds and deployment costs across more datasets and operating conditions. These steps would connect the controlled attribution of fusion gains to practical inspection decisions.

7 Data and Code Availability

MPDD, BTAD, MVTec AD, VisA and KolektorSDD2 are distributed by their respective providers [1, 2, 3, 4, 36]. The local study archive retains support manifests, feature specifications, geometry revisions, prediction caches, per-condition metrics, bootstrap outputs and analysis scripts. A local reproduction package includes the analysis entry points, dependency records, support manifests and figure bindings. Raw datasets and feature caches remain separate. The reproduction entry point is ``docs/REPRODUCE_TO_TABLES.md``; checkpoint paths and hashes are recorded in ``docs/MODEL_WEIGHTS.md``.

Licensing is split three ways. The code is released under the MIT Licence (repository root ``LICENSE``, Copyright (c) 2026 LiYuening). The derived artefacts released with the package - derived tables, figures and the manuscript sources - carry the same licence as the code; any item that the root ``LICENSE`` does not

explicitly cover remains for the authors to confirm. Dataset licences are separate and are not covered by the code licence, and the datasets themselves are not redistributed here. The public repository holding the code and the reproduction materials is <https://github.com/USEU117/reference-matching-interaction-ad>; a permanent archive DOI for the complete study has not yet been established.

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References

- [1] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, 'MVTec AD - A comprehensive real-world dataset for unsupervised anomaly detection,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2019, pp. 9592-9600. <https://doi.org/10.1109/cvpr.2019.00982>
- [2] S. Jezek, M. Jonak, R. Burget, P. Dvorak, and M. Skotak, 'Deep learning-based defect detection of metal parts: Evaluating current methods in complex conditions,' in Proc. 13th Int. Congr. Ultra Modern Telecommun. Control Syst. Workshops, 2021, pp. 66-71. <https://doi.org/10.1109/icumt54235.2021.9631567>
- [3] P. Mishra, R. Verk, D. Fornasier, C. Piciarelli, and G. L. Foresti, 'VT-ADL: A vision transformer network for image anomaly detection and localization,' in Proc. IEEE 30th Int. Symp. Ind. Electron., 2021, pp. 1-6. <https://doi.org/10.1109/isie45552.2021.9576231>
- [4] Y. Zou, J. Jeong, L. Pemula, D. Zhang, and O. Dabeer, 'SPot-the-Difference self-supervised pre-training for anomaly detection and segmentation,' in Proc. Eur. Conf. Comput. Vis. (ECCV), 2022, pp. 392-408. https://doi.org/10.1007/978-3-031-20056-4_23
- [5] S. Damm, M. Laszkiewicz, J. Lederer, and A. Fischer, 'AnomalyDINO: Boosting patch-based few-shot anomaly detection with DINOv2,' in Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV), 2025, pp. 1319-1329. https://openaccess.thecvf.com/content/WACV2025/html/Damm_AnomalyDINO_Boosting_Patch-Based_Few-Shot_Anomaly_Detection_with_DINOv2_WACV_2025_paper.html
- [6] M. Oquab et al., 'DINOv2: Learning robust visual features without supervision,' Trans. Mach. Learn. Res., 2024. <https://openreview.net/forum?id=a68SUt6zFt>
- [7] X. Guo, Z. Chen, C. D. Castillo, H. Wang, and X. Liu, 'Sea-CLIP: Mining semantic-aware representations for few-shot anomaly detection with CLIP,' in Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV), 2026, pp. 3689-3699. https://openaccess.thecvf.com/content/WACV2026/html/Guo_Sea-CLIP_Mining_Semantic-Aware_Representations_for_Few-Shot_Anomaly_Detection_with_CLIP_WACV_2026_paper.html
- [8] Q. Zhou, G. Pang, Y. Tian, S. He, and J. Chen, 'AnomalyCLIP: Object-agnostic prompt learning for zero-shot anomaly detection,' in Proc. Int. Conf. Learn. Represent. (ICLR), 2024. https://proceedings.iclr.cc/paper_files/paper/2024/hash/d7b50b8ac2c781a12f26155f48310d8d-Abstract-Conference.html
- [9] K. Long, G. Xie, L. Ma, J. Liu, and Z. Lu, 'Revisiting Multimodal Fusion for 3D Anomaly Detection from an Architectural Perspective,' Proc. AAAI Conf. Artif. Intell., vol. 39, no. 12, pp. 12273-12281, 2025. <https://ojs.aaai.org/index.php/AAAI/article/view/33337>
- [10] Y. Xu, H. Zhang, Y. Ma, Y. Zhu, and K. M. Ting, 'SCoNE: Spherical Consistent Neighborhoods Ensemble for Effective and Efficient Multi-View Anomaly Detection,' Proc. AAAI Conf. Artif. Intell., vol. 40, no. 19, pp. 16083-16090, 2026. <https://ojs.aaai.org/index.php/AAAI/article/view/38643>
- [11] X.-R. Sheng, D.-C. Zhan, S. Lu, and Y. Jiang, 'Multi-View Anomaly Detection: Neighborhood in Locality Matters,' Proc. AAAI Conf. Artif. Intell., vol. 33, no. 1, pp. 4894-4901, 2019. <https://ojs.aaai.org/index.php/AAAI/article/view/4418>
- [12] L. Cheng, Y. Wang, and X. Liu, 'Neighborhood Consensus Networks for Unsupervised Multi-view Outlier Detection,' Proc. AAAI Conf. Artif. Intell., vol. 35, no. 8, pp. 7099-7106, 2021. <https://ojs.aaai.org/index.php/AAAI/article/view/16873>
- [13] M. Yang, J. Liu, Z. Yang, and Z. Wu, 'SLSG: Industrial image anomaly detection with improved feature embeddings and one-class classification,' Pattern Recognit., vol. 156, Art. no. 110862, 2024. <https://doi.org/10.1016/j.patcog.2024.110862>
- [14] X. Zhang, M. Xu, and X. Zhou, 'RealNet: A feature selection network with realistic synthetic anomaly for anomaly detection,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2024, pp. 16699-16708. <https://doi.org/10.1109/cvpr52733.2024.01580>
- [15] T. Defard, A. Setkov, A. Loesch, and R. Audigier, 'PaDiM: A patch distribution modeling framework for anomaly detection and localization,' in Pattern Recognition. ICPR International Workshops and Challenges, 2021, pp. 475-489. https://doi.org/10.1007/978-3-030-68799-1_35

- [16] K. Roth, L. Pemula, J. Zepeda, B. Schölkopf, T. Brox, and P. Gehler, 'Towards total recall in industrial anomaly detection,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2022, pp. 14318-14328. <https://doi.org/10.1109/cvpr52688.2022.01392>
- [17] Z. Hu, X. Zeng, Y. Li, Z. Yin, E. Meng, L. Zhu, and X. Kong, 'Few-shot anomaly detection with adaptive feature transformation and descriptor construction,' Chin. J. Aeronaut., vol. 38, no. 3, Art. no. 103098, 2025. <https://doi.org/10.1016/j.cja.2024.06.007>
- [18] S. Wei, X. Wei, Z. Ma, S. Dong, S. Zhang, and Y. Gong, 'Few-shot online anomaly detection and segmentation,' Knowl.-Based Syst., vol. 300, Art. no. 112168, 2024. <https://doi.org/10.1016/j.knosys.2024.112168>
- [19] J. Zhou, W. Wong, and F. Liao, 'One-shot unsupervised industrial anomaly detection: Enhanced performance under extreme data scarcity,' Pattern Recognit., vol. 173, Art. no. 112759, 2026. <https://doi.org/10.1016/j.patcog.2025.112759>
- [20] Y. Li, L. Tian, Y. Dai, W. Chen, L. Bao, and X. Liu, 'FastRef: Fast prototype refinement for few-shot industrial anomaly detection,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2026, pp. 43040-43049. https://openaccess.thecvf.com/content/CVPR2026/html/Li_FastRef_Fast_Prototype_Refinement_for_Few-shot_Industrial_Anomaly_Detection_CVPR_2026_paper.html
- [21] C. Lendering, E. Akdag, and E. Bondarev, 'SubspaceAD: Training-free few-shot anomaly detection via subspace modeling,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2026, pp. 28557-28566. https://openaccess.thecvf.com/content/CVPR2026/html/Lendering_SubspaceAD_Training-Free_Few-Shot_Anomaly_Detection_via_Subspace_Modeling_CVPR_2026_paper.html
- [22] L. Jiang, Y. Huang, Z. Xu, Y. Xu, H.-S. Wong, and S. Wu, 'Defect cue-preserved structural feature refinement for few-shot anomaly detection,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2026, pp. 35607-35616. https://openaccess.thecvf.com/content/CVPR2026/html/Jiang_Defect_Cue-Preserved_Structural_Feature_Refinement_for_Few-Shot_Anomaly_Detection_CVPR_2026_paper.html
- [23] G. Xie, X. He, D. Fan, S. Li, and Y. Liu, 'Hyper-FSAD: Training-Free and Language-Free Few-Shot Anomaly Detection via Sparse Hyper Matching,' arXiv:2605.10628v2, 2026. <https://arxiv.org/abs/2605.10628v2>
- [24] L. Yi, Z. Chen, G. Wu, J. Ding, X. Wang, and Z. Ning, 'ReMem: A Dynamic Memory Evolution Detector for Zero-Shot Anomaly Detection,' Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Findings, pp. 7697-7705, 2026. https://openaccess.thecvf.com/content/CVPR2026F/html/Yi_ReMem_A_Dynamic_Memory_Evolution_Detector_for_Zero-Shot_Anomaly_Detection_CVPRF_2026_paper.html
- [25] J.-Z. Tang, P.-H. Huang, M.-C. Chang, C.-F. Hsu, and J.-L. Li, 'DuoAD: Leveraging [CLS] Dual Characteristics for Training-Free Few-Shot Anomaly Detection,' arXiv:2607.23924v1, 2026. <https://arxiv.org/abs/2607.23924v1>
- [26] A. Radford et al., 'Learning transferable visual models from natural language supervision,' in Proc. 38th Int. Conf. Mach. Learn., vol. 139, 2021, pp. 8748-8763. <https://proceedings.mlr.press/v139/radford21a.html>
- [27] J. Jeong, Y. Zou, T. Kim, D. Zhang, A. Ravichandran, and O. Dabeer, 'WinCLIP: Zero-/few-shot anomaly classification and segmentation,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2023, pp. 19606-19616. <https://doi.org/10.1109/cvpr52729.2023.01878>
- [28] X. Li, Z. Zhang, X. Tan, C. Chen, Y. Qu, Y. Xie, and L. Ma, 'PromptAD: Learning prompts with only normal samples for few-shot anomaly detection,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2024, pp. 16838-16848. https://openaccess.thecvf.com/content/CVPR2024/html/Li_PromptAD_Learning_Prompts_with_only_Normal_Samples_for_Few-Shot_Anomaly_CVPR_2024_paper.html
- [29] J. Zhu and G. Pang, 'Toward generalist anomaly detection via in-context residual learning with few-shot sample prompts,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2024, pp. 17826-17836. <https://doi.org/10.1109/cvpr52733.2024.01688>
- [30] W. Ma et al., 'AA-CLIP: Enhancing zero-shot anomaly detection via anomaly-aware CLIP,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2025, pp. 4744-4754. <https://doi.org/10.1109/cvpr52734.2025.00447>
- [31] J. Zhu, Y.-S. Ong, C. Shen, and G. Pang, 'Fine-grained abnormality prompt learning for zero-shot anomaly detection,' in Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV), 2025, pp. 22241-22251. <https://doi.org/10.1109/iccv51701.2025.02065>
- [32] Z. Gu, B. Zhu, G. Zhu, Y. Chen, M. Tang, and J. Wang, 'UniVAD: A training-free unified model for few-shot visual anomaly detection,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2025, pp. 15194-15203. https://openaccess.thecvf.com/content/CVPR2025/html/Gu_UniVAD_A_Training-free_Unified_Model_for_Few-shot_Visual_Anomaly_Detection_CVPR_2025_paper.html
- [33] Y. Wang, J. Peng, J. Zhang, R. Yi, Y. Wang, and C. Wang, 'Multimodal Industrial Anomaly Detection via Hybrid Fusion,' in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2023, pp. 8032-8041. https://openaccess.thecvf.com/content/CVPR2023/html/Wang_Multimodal_Industrial_Anomaly_Detection_via_Hybrid_Fusion_CVPR_2023_paper.html

- [34] Y. Lin et al., 'Commonality in Few: Few-Shot Multimodal Anomaly Detection via Hypergraph-Enhanced Memory,' Proc. AAAI Conf. Artif. Intell., vol. 40, no. 9, pp. 7015–7023, 2026. <https://ojs.aaai.org/index.php/AAAI/article/view/37636>
- [35] X. Chen, X. Wang, Y. Wang, C. Han, and L. Duan, 'Learning Enhanced Representations via Contrasting for Multi-view Outlier Detection,' in Database Systems for Advanced Applications (DASFAA), 2023, pp. 110–120. https://link.springer.com/chapter/10.1007/978-3-031-30678-5_9
- [36] J. Božič, D. Tabernik, and D. Skočaj, 'Mixed supervision for surface-defect detection: from weakly to fully supervised learning,' Comput. Ind., vol. 129, Art. no. 103459, 2021. <https://doi.org/10.1016/j.compind.2021.103459>
- [37] J. Johnson, M. Douze, and H. Jegou, 'Billion-scale similarity search with GPUs,' IEEE Trans. Big Data, vol. 7, no. 3, pp. 535–547, 2021. <https://doi.org/10.1109/tbdata.2019.2921572>

Supplementary Protocol Tables

The harmonised-geometry table is a supplementary, restricted comparison: it uses a short side of 448 pixels, the frozen common valid region and the same rank-based pooled metric for 36 of the 144 category units (seed 0 and $K = 1$). It includes only configurations whose input rules can share that geometry. The square-stretch families SubspaceAD, WinCLIP+ and zero-shot AnomalyCLIP are excluded, so this subset is not a uniform comparison of every external method. Its intervals apply only to pixel AP, not pixel AUROC; they are marginal intervals rather than cross-method paired-difference tests. The table is descriptive and makes neither a ranking nor a state-of-the-art claim.

Table S1. Restricted harmonised-geometry comparison at seed 0 and $K = 1$.

Dataset	Configuration	Pixel AP	Pixel AUROC	95% AP interval
MPDD	Dual-encoder baseline / Joint matching	0.3123	0.9497	[0.2935, 0.3366]
MPDD	Dual-encoder baseline / Independent matching	0.3167	0.9527	[0.2980, 0.3409]
MPDD	AnomalyDINO	0.2846	0.9460	[0.2676, 0.3115]
MPDD	AnomalyDINO + rotation	0.2929	0.9471	[0.2763, 0.3195]
MPDD	PatchCore 448	0.2110	0.9466	[0.1978, 0.2272]
BTAD	Dual-encoder baseline / Joint matching	0.6174	0.9722	[0.5594, 0.6642]
BTAD	Dual-encoder baseline / Independent matching	0.6278	0.9738	[0.5701, 0.6735]
BTAD	AnomalyDINO	0.5576	0.9617	[0.4992, 0.6058]
BTAD	AnomalyDINO + rotation	0.5881	0.9655	[0.5354, 0.6315]
BTAD	PatchCore 448	0.3725	0.9587	[0.3315, 0.4290]
MVTec AD	Dual-encoder baseline / Joint matching	0.5582	0.9553	[0.5413, 0.5783]
MVTec AD	Dual-encoder baseline / Independent matching	0.5612	0.9570	[0.5446, 0.5817]
MVTec AD	AnomalyDINO	0.5611	0.9506	[0.5400, 0.5827]
MVTec AD	AnomalyDINO + rotation	0.5582	0.9497	[0.5394, 0.5811]
MVTec AD	PatchCore 448	0.5050	0.9228	[0.4860, 0.5246]
VisA	Dual-encoder baseline / Joint matching	0.3439	0.9544	[0.3166, 0.3671]
VisA	Dual-encoder baseline / Independent matching	0.3493	0.9579	[0.3215, 0.3721]
VisA	AnomalyDINO	0.3032	0.9530	[0.2754, 0.3278]
VisA	AnomalyDINO + rotation	0.3256	0.9549	[0.2978, 0.3502]
VisA	PatchCore 448	0.3228	0.9525	[0.2960, 0.3468]

Short side 448; frozen common valid region and the same rank-based pooling. This covers 36 of the 144 category-condition units. PatchCore 448 was recomputed; the other four configurations reuse compatible predictions. Additional crops may be required to respect the frozen intersection. SubspaceAD, WinCLIP+ and zero-shot AnomalyCLIP are excluded because their square-stretch input paths are outside this subset. Points use full-grid common-region pixels; intervals use stride-eight pixels and 1000 image-bootstrap replicates. Intervals apply only to pixel AP, are individual marginal 95% intervals, and are neither family-adjusted nor paired cross-method difference intervals. Pixel AUROC has point values only. This is not a ranking or a significance comparison. Bold identifies the Dual-encoder baseline configurations, not optimal results; Tables 11 and 12 remain unchanged.

The companion method-protocol table records each configuration's input geometry, rotation choice and reference-bank or coreset construction. In particular, SubspaceAD uses 256-pixel fp16 inputs rather than the

upstream few-shot script's 672-pixel setting: on the recorded 6 GB GPU, the latter did not finish 12 query images in at least 15 minutes and reached 5797 of 6144 MiB. This disclosed deviation is not presented as an official configuration.

Table S2. Input and reference protocols of the configurations in Tables 11 and 12.

Configuration	Input geometry	Augmentation	Reference construction / scoring
Dual-encoder baseline / Joint matching	DINOv2-B/14; short side 448; AnomalyCLIP visual: square 518, mapped to DINOv2-B/14	None	All normal support descriptors; shared neighbor across branches
Dual-encoder baseline / Independent matching	Same as Dual-encoder baseline / Joint matching	None	Same bank; independent branch neighbors
AnomalyDINO	Short side 448; aspect-preserving canvas	None	Frozen DINOv2 patch memory
AnomalyDINO + rotation	Same canvas	Support rotations	Rotated reference descriptors augment the patch memory
PatchCore 128	Resize 144; center crop 128	None	Local compact memory/coreset configuration; 256-dimensional projected features
PatchCore 224	Resize 256; center crop 224	None	Recorded upstream-style memory/coreset configuration; 1024-dimensional projected features
SubspaceAD 256	Square stretch 256; fp16	None	Normal-support subspace; documented local setting
WinCLIP+	Square stretch 240	No rotation	Frozen visual-language model, prompt ensemble and normal-reference features
AnomalyCLIP zero-shot	Square stretch 518	No rotation	Auxiliary-domain-trained prompt learner; no target support bank

These are the evaluated configurations, not interchangeable official recipes. SubspaceAD uses 256-pixel fp16 inputs without augmentation; its upstream few-shot script specifies 672 pixels and `aug_count = 30`. On the recorded 6 GB GPU, the 672-pixel attempt did not complete 12 query images in at least 15 minutes and reached 5797/6144 MiB. The local 256 setting is therefore explicitly a deviation. AnomalyCLIP uses inherited auxiliary-domain-trained prompt checkpoints, with MVTec-trained prompts for MPDD, BTAD and VisA and VisA-trained prompts for MVTec AD; target-domain evaluation is zero-shot. It has 36 units rather than the 144 support-conditioned units of the other configurations. Encoders are frozen and no target-domain gradient optimization is used. Memory, coreset or subspace construction remains preparation computation. Bold marks the Dual-encoder baseline configurations, not optimal results.

Supplementary Method Figures

(a) Primary branches and native feature widths

DINOv2-B/14	AnomalyCLIP visual	DINOv2-S/14	WideResNet50-2
ViT-B/14; 448	ViT-L/14; 518	ViT-S/14; 448	layer2 + layer3
32 × 32 native grid	37 × 37 native grid	32 × 32 native grid	aligned to base grid
768 dimensions	768 dimensions	384 dimensions	1536 dimensions

(b) Exploratory extra-slot branches

DINO ViT-S/8	ConvNeXt-Tiny	Swin-Tiny
384 dimensions	576 dimensions	576 dimensions

DINO ViT-S/8, ConvNeXt-Tiny, and Swin-Tiny are exploratory checks after DINOv2-S/14 and WideResNet50-2; all encoders remain frozen.

(c) Common alignment and distance

Every branch maps to the DINOv2-B/14 canvas grid by bilinear interpolation. align_corners = False; then unit-normalize each feature at each position. $d_b = 1 - \text{cosine}$	
BTAD-03 geometry 32 × 42 canvas; coordinate-correct AnomalyCLIP visual re-grid	Frozen inference No target-domain training, PCA, coreset, or weight search

Figure S1. Frozen encoder branches and their shared scoring path. (a) DINOv2-B/14 and DINOv2-S/14 provide the DINOv2 branches; AnomalyCLIP visual provides the visual branch; WideResNet50-2 concatenates layer2 and layer3. The 448-pixel square input and 32 by 32 grid illustrate the square primary canvas; the BTAD-03 canvas is 32 by 42, and KSDD2 uses its separately specified geometry. (b) DINO ViT-S/8, ConvNeXt-Tiny and Swin-Tiny are exploratory substitutions selected after DINOv2-S/14 and WideResNet50-2. (c) Bilinear mapping to the DINOv2-B/14 grid with align_corners = False and per-position unit normalization define the common distance path. Encoders, weights and reference identities remain frozen; no target training, coreset, PCA or weight search is introduced.

Supplementary Results Figures

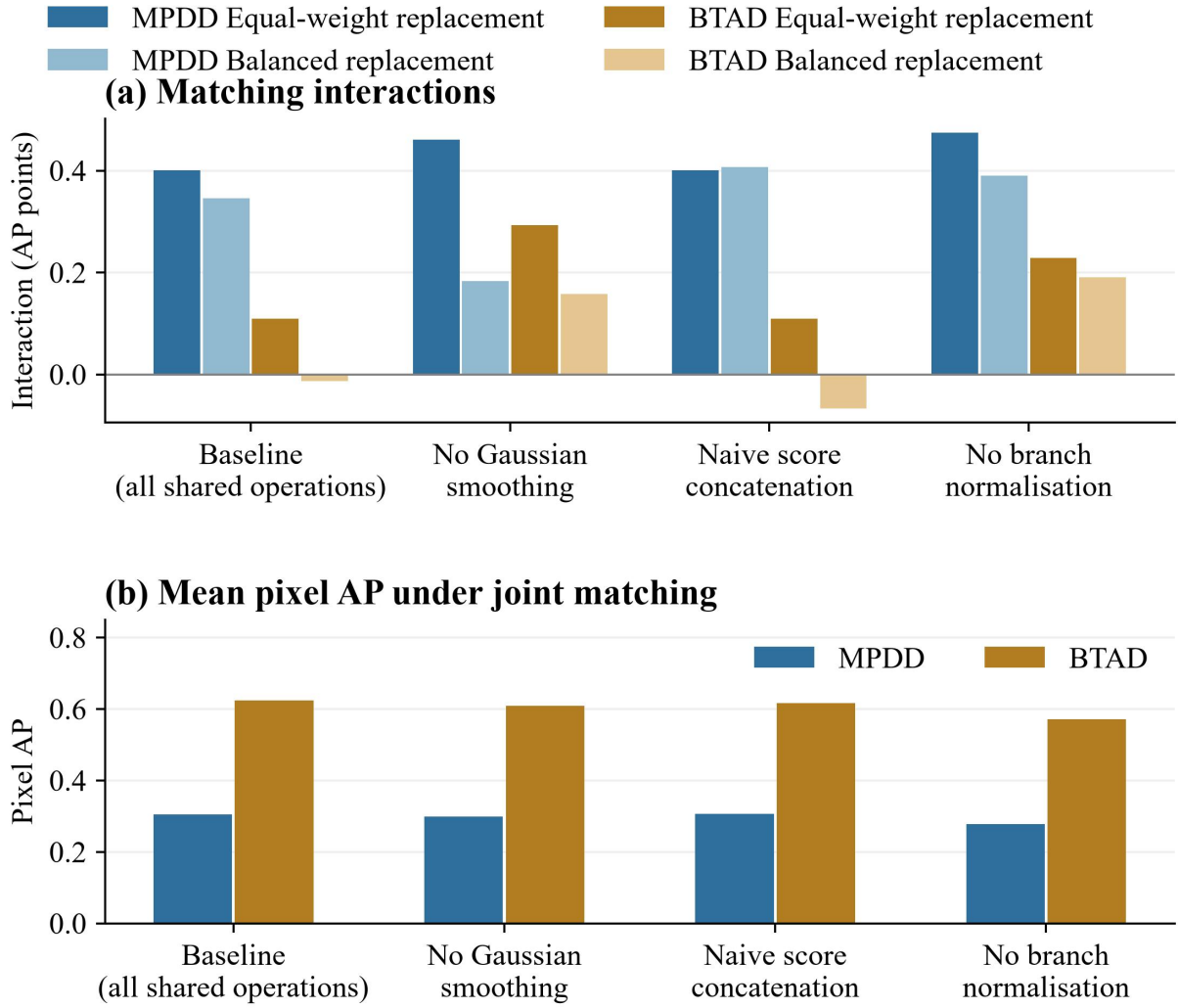


Figure S2. Exploratory shared-operation ablations. (a) The interaction under each ablation, shown as dataset means. (b) The absolute mean pixel AP that the same ablations move, over the four constructions under the joint rule. The operations every construction shares are removed one at a time: the no-smoothing ablation drops the Gaussian smoothing of the patch scores, the no-normalization ablation drops the per-branch normalization and scores with squared Euclidean distance, and the naive-concatenation ablation replaces score-level fusion with naive concatenation, which can move only the Balanced replacement contrast because the coefficients equal the weights for equal slots. The scope is a single condition, seed 0 with $K = 1$, and one run per ablation, so no interval exists and none is drawn; the panel is therefore exploratory and does not show that a shared operation has been validated.

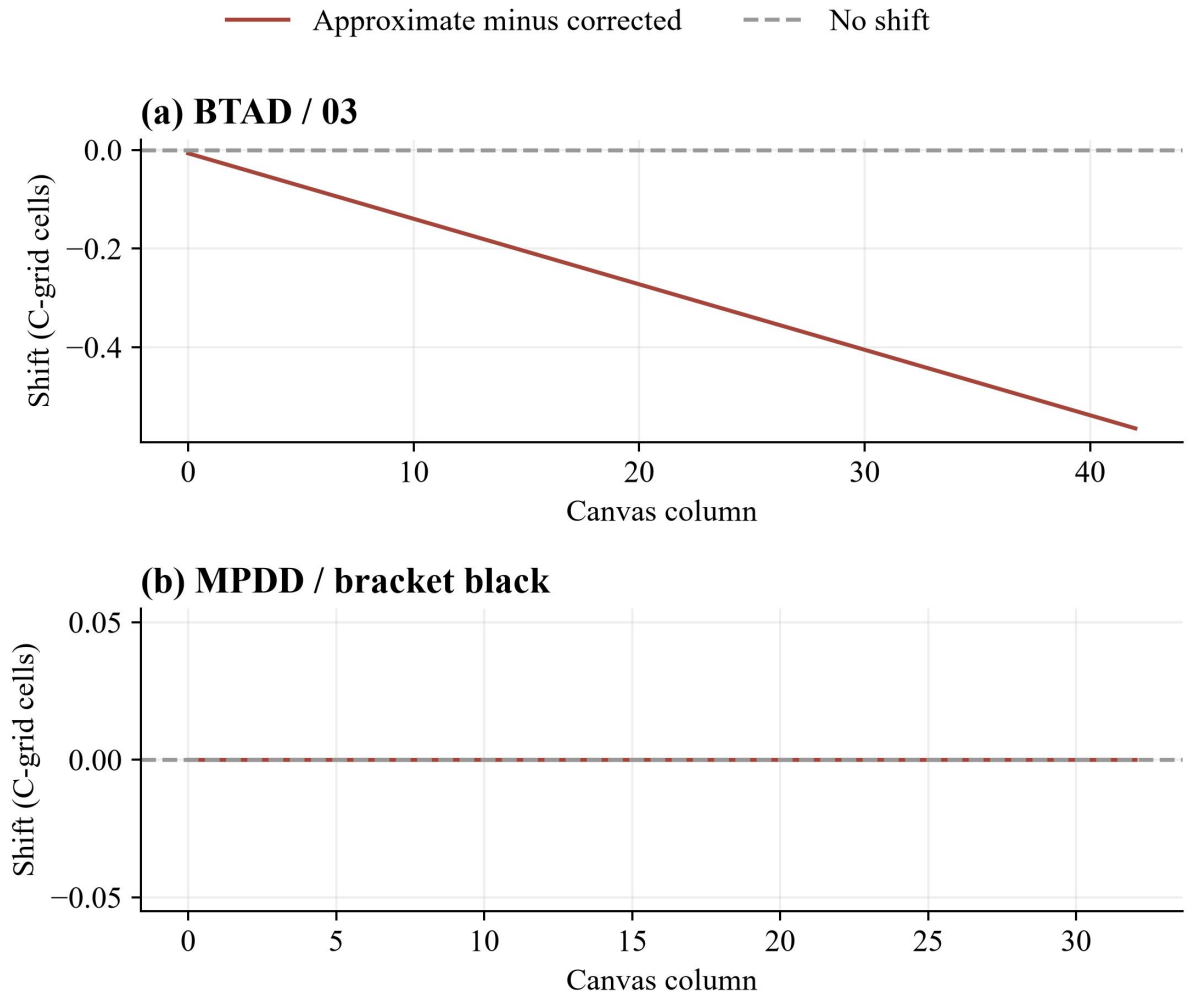


Figure S3. Supporting geometry audit. Approximate minus coordinate-corrected AnomalyCLIP visual-grid positions for BTAD category 03 and the square MPDD bracket-black example. The non-square canvas introduces a horizontal displacement; the square example has zero displacement. These are coordinate diagnostics, not additional performance measurements.

DINOv2-B/14 + DINOv2-S/14 canvas
 PatchCore 224 (centre crop)

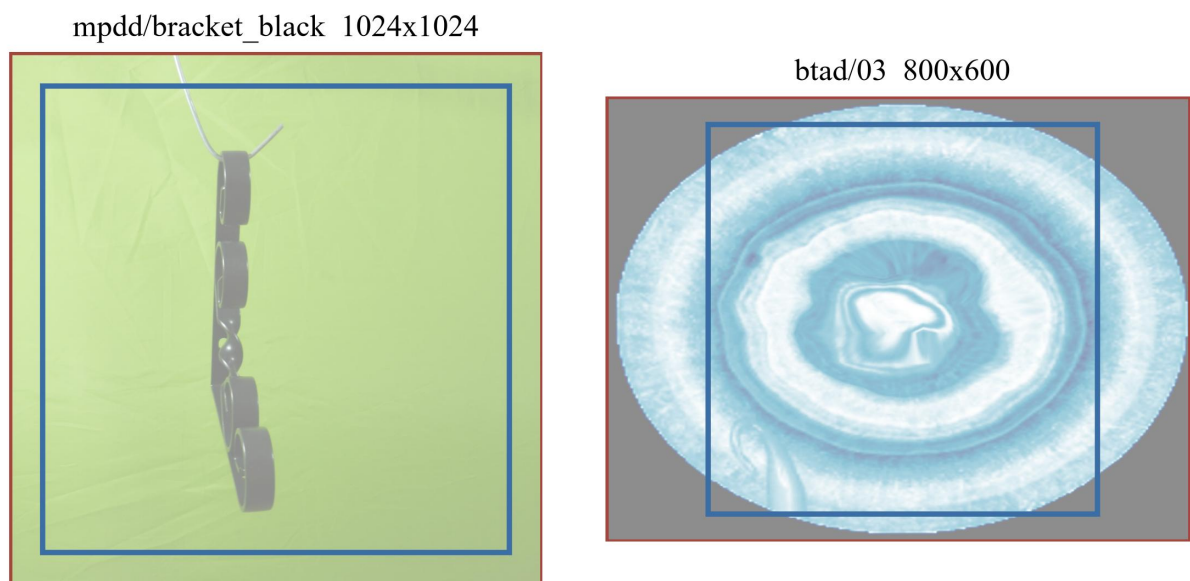
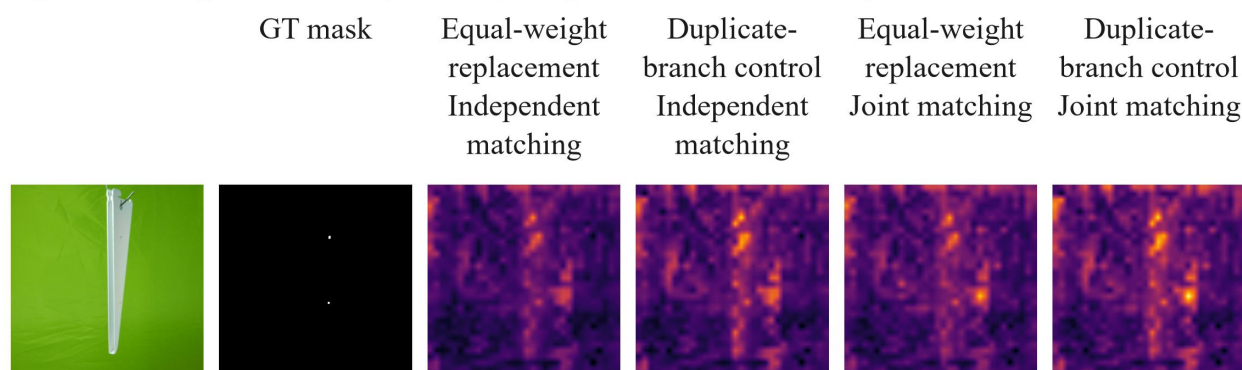


Figure S3 continued. The DINOv2-B/14 and DINOv2-S/14 canvas and PatchCore-224 center crop in original image coordinates for the same two geometry examples. Rectangles are computed from the frozen preprocessing transforms.

mpdd/bracket_white idx3 Equal-weight replacement interaction positive extreme (+0.719)



mpdd/bracket_white idx7 Equal-weight replacement interaction negative extreme (-1.000)

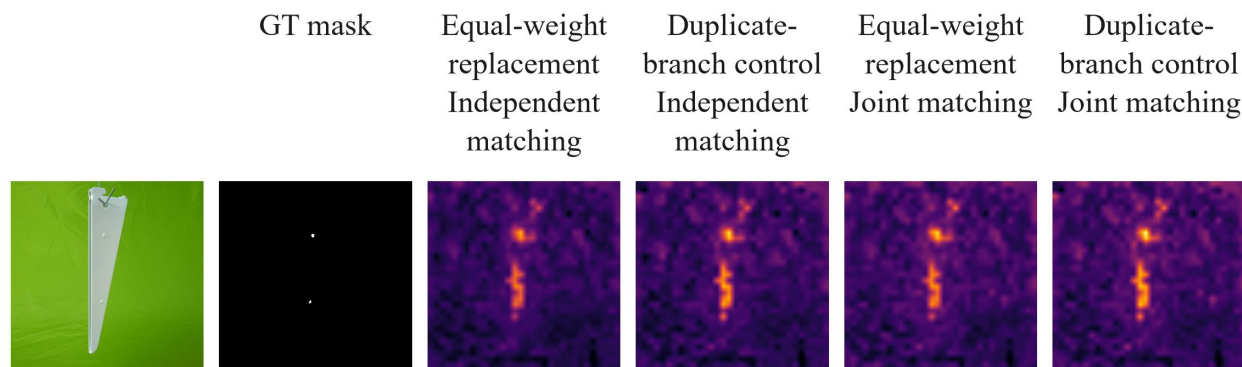
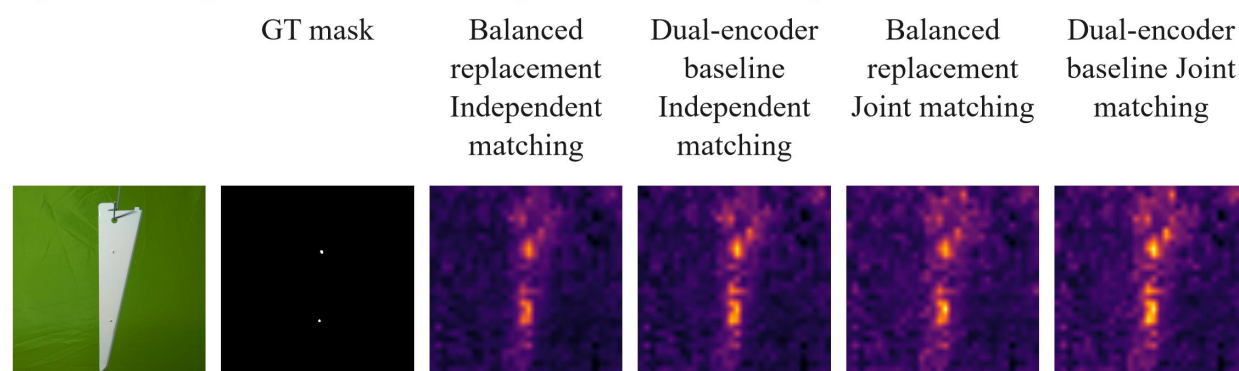


Figure S3 continued. Two of the eight frozen per-image interaction cases at seed 0 and $K = 4$. The original query, ground-truth display and four contrast maps are shown in aligned columns. The four raw patch-score maps in each row use one shared color range; they illustrate the contrast before final smoothing. The displayed interaction is in AP units, not percentage points. Cases are selected extremes and do not estimate population performance.

mpdd/bracket_white idx0 Balanced replacement interaction positive extreme (+0.800)



mpdd/bracket_white idx7 Balanced replacement interaction negative extreme (-1.000)

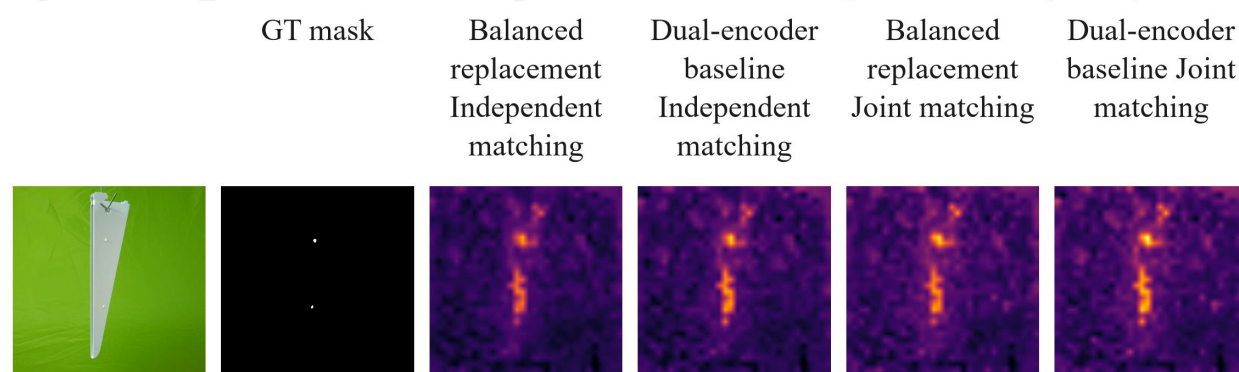
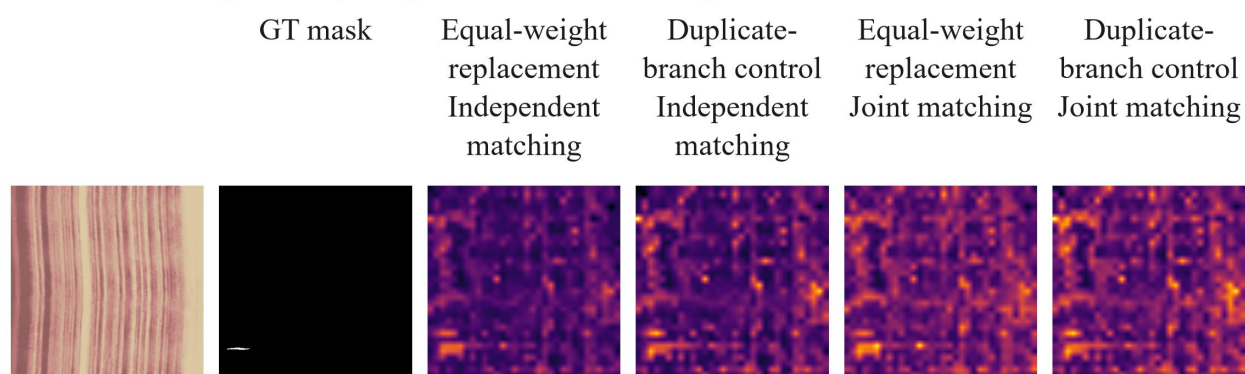


Figure S3 continued. Two of the eight frozen per-image interaction cases at seed 0 and $K = 4$. The original query, ground-truth display and four contrast maps are shown in aligned columns. The four raw patch-score maps in each row use one shared color range; they illustrate the contrast before final smoothing. The displayed interaction is in AP units, not percentage points. Cases are selected extremes and do not estimate population performance.

btad/02 idx89 Equal-weight replacement interaction positive extreme (+0.565)



btad/02 idx135 Equal-weight replacement interaction negative extreme (-0.200)

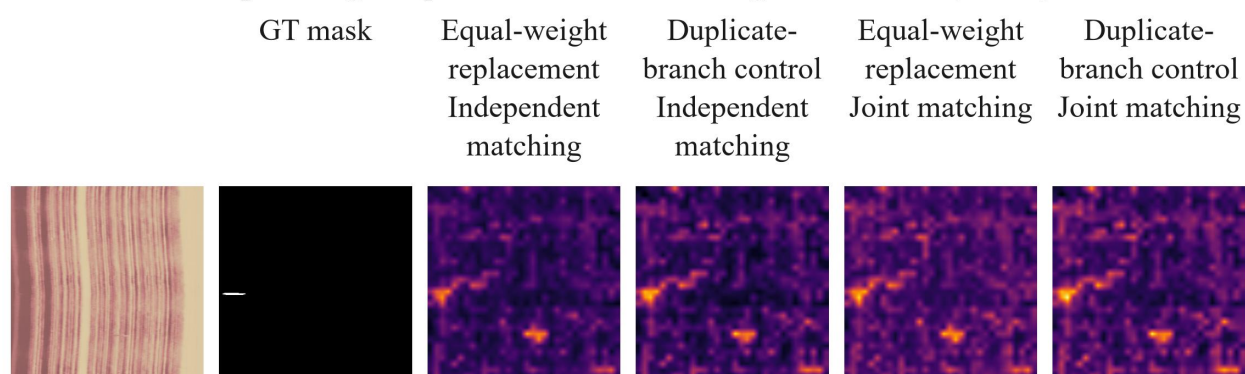
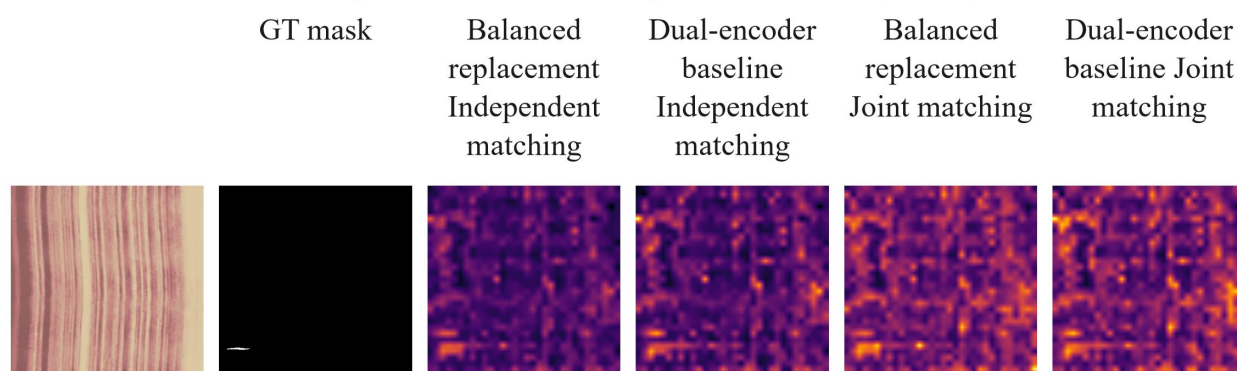


Figure S3 continued. Two of the eight frozen per-image interaction cases at seed 0 and $K = 4$. The original query, ground-truth display and four contrast maps are shown in aligned columns. The four raw patch-score maps in each row use one shared color range; they illustrate the contrast before final smoothing. The displayed interaction is in AP units, not percentage points. Cases are selected extremes and do not estimate population performance.

btad/02 idx89 Balanced replacement interaction positive extreme (+0.657)



btad/02 idx135 Balanced replacement interaction negative extreme (-0.415)

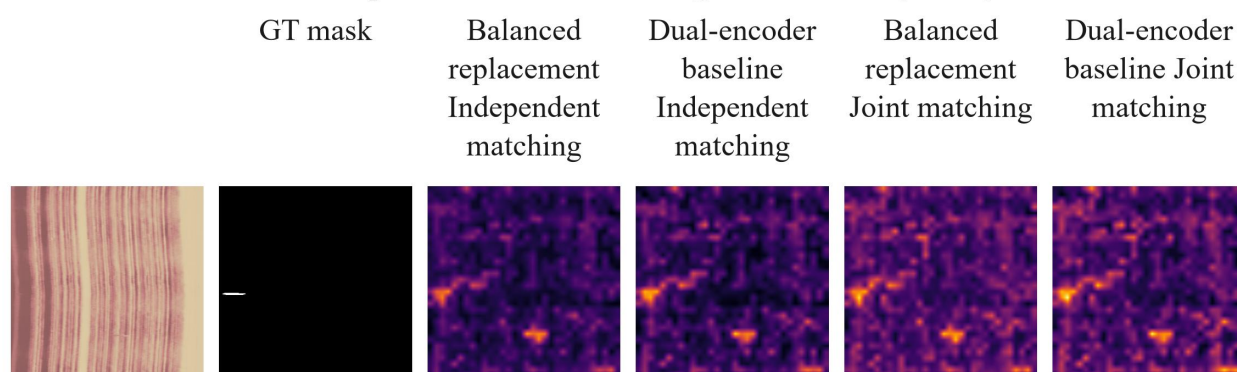


Figure S3 continued. Two of the eight frozen per-image interaction cases at seed 0 and $K = 4$. The original query, ground-truth display and four contrast maps are shown in aligned columns. The four raw patch-score maps in each row use one shared color range; they illustrate the contrast before final smoothing. The displayed interaction is in AP units, not percentage points. Cases are selected extremes and do not estimate population performance.

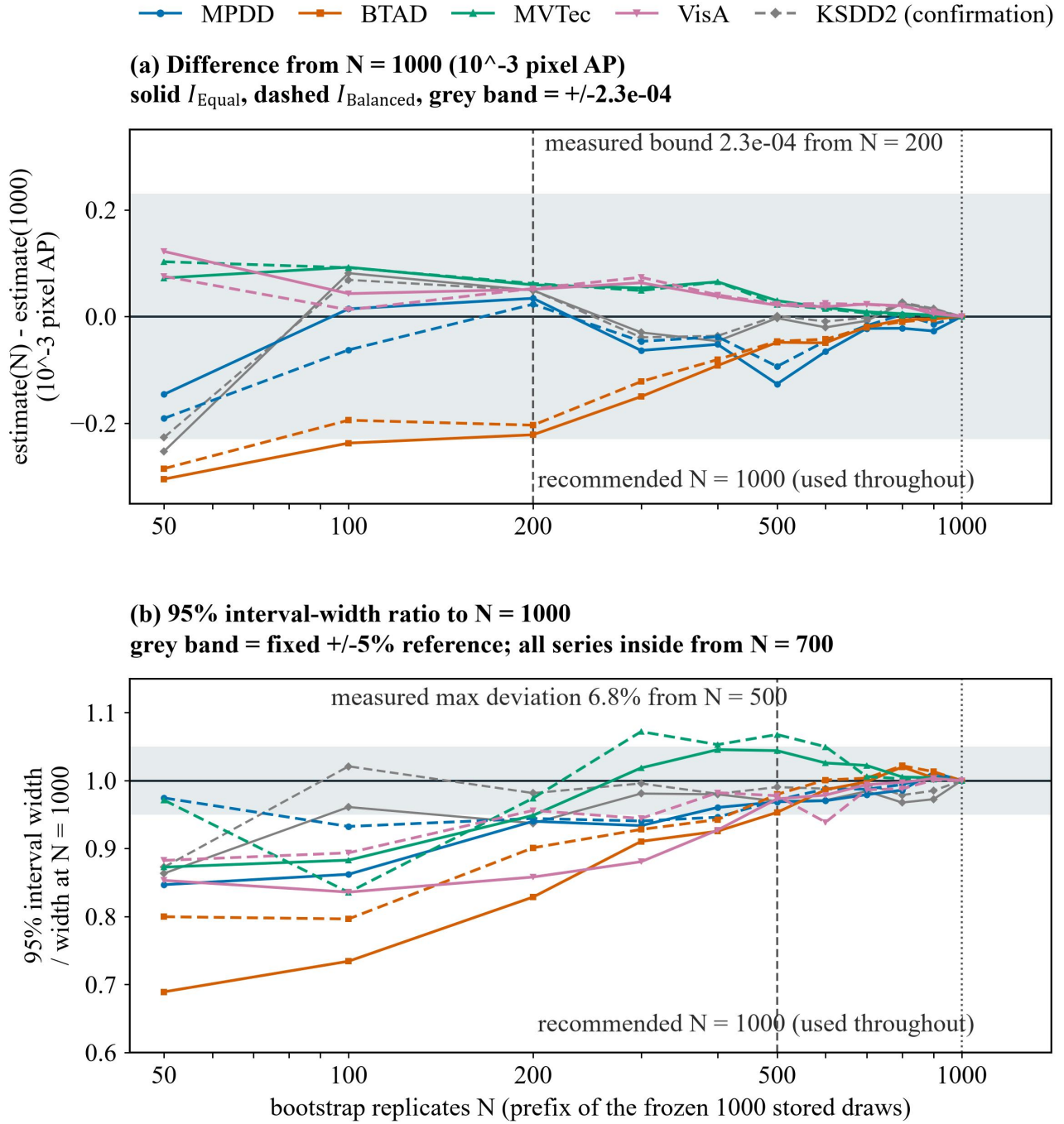


Figure S4. Numerical stability of stored bootstrap estimates, not model training or an all-method stability comparison. The same prefixes of 50–1000 stored replicates underlie both pages; the prefixes are dependent and 1000 is a reference value, not a true parameter. (a) Change from the 1000-replicate estimate (10^{-3} pixel AP); the grey band is the measured $N \geq 200$ bound. (b) Individual 95% interval width relative to its 1000-replicate width. The fixed $\pm 5\%$ reference band is not a prespecified pass criterion: the measured maximum deviation for $N \geq 500$ is 6.8%, and every series remains inside the band only from $N = 700$ on this grid. KSDD2 is separate confirmation, outside the four-dataset family; every value plotted here, its endpoints included, is the mean of the stored replicate distribution, so it is the bootstrap mean rather than the condition-averaged observed difference that the confirmation table reports as its point column. These individual intervals do not replace the adjusted primary intervals.

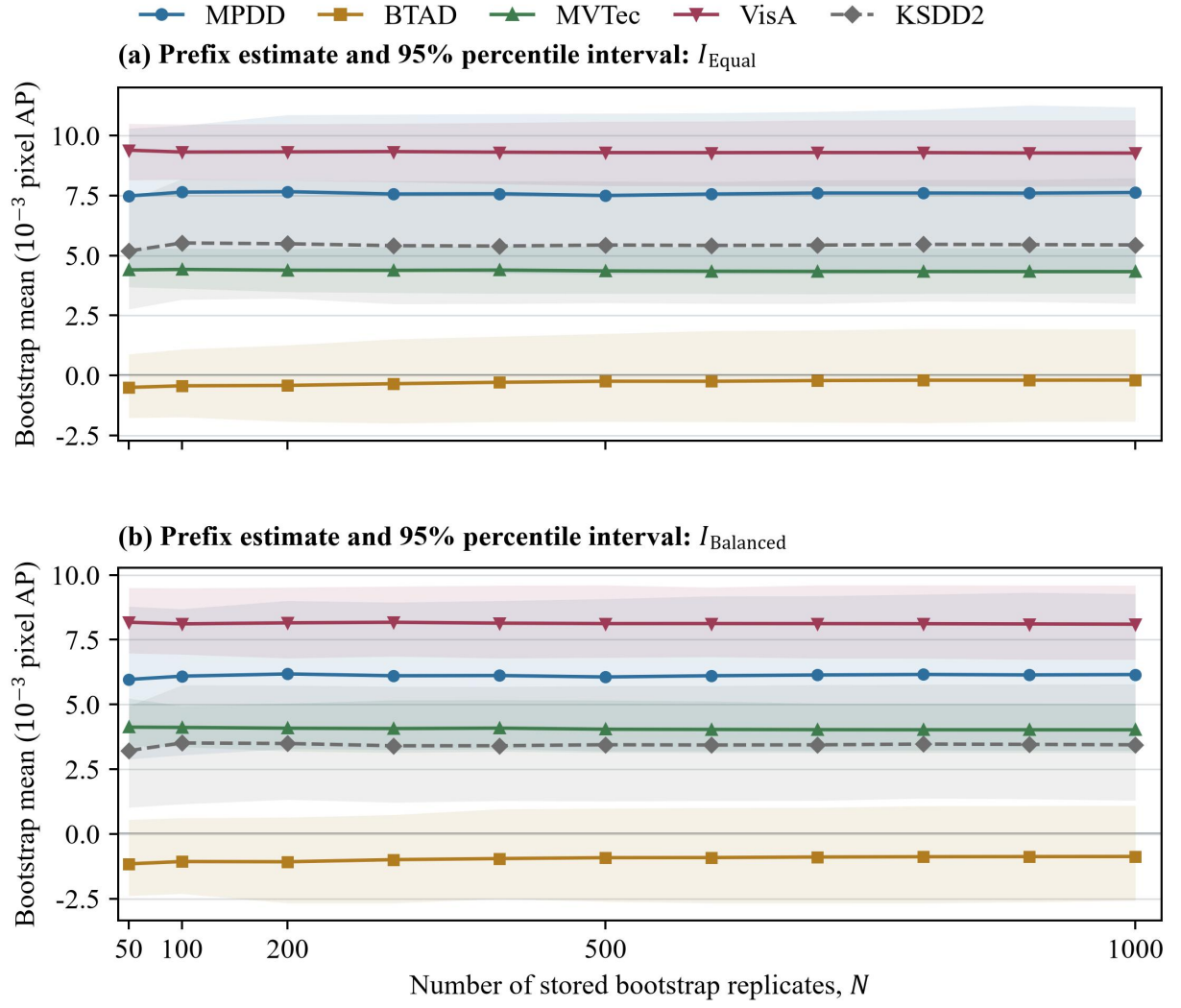


Figure S4 continued. The same stored prefixes shown at their absolute scale: prefix means and individual 95% percentile intervals of the two interaction contrasts, in 10^{-3} pixel AP. On the displayed prefixes of at least 500 replicates the largest mean deviation from the 1000-replicate value is approximately 0.00013 pixel AP and the largest relative interval-width deviation is 6.8%, while at prefixes of at least 200 the latter can still reach 17.1%. These are individual 95% intervals and do not replace the multiplicity-adjusted intervals used for the principal conclusions.

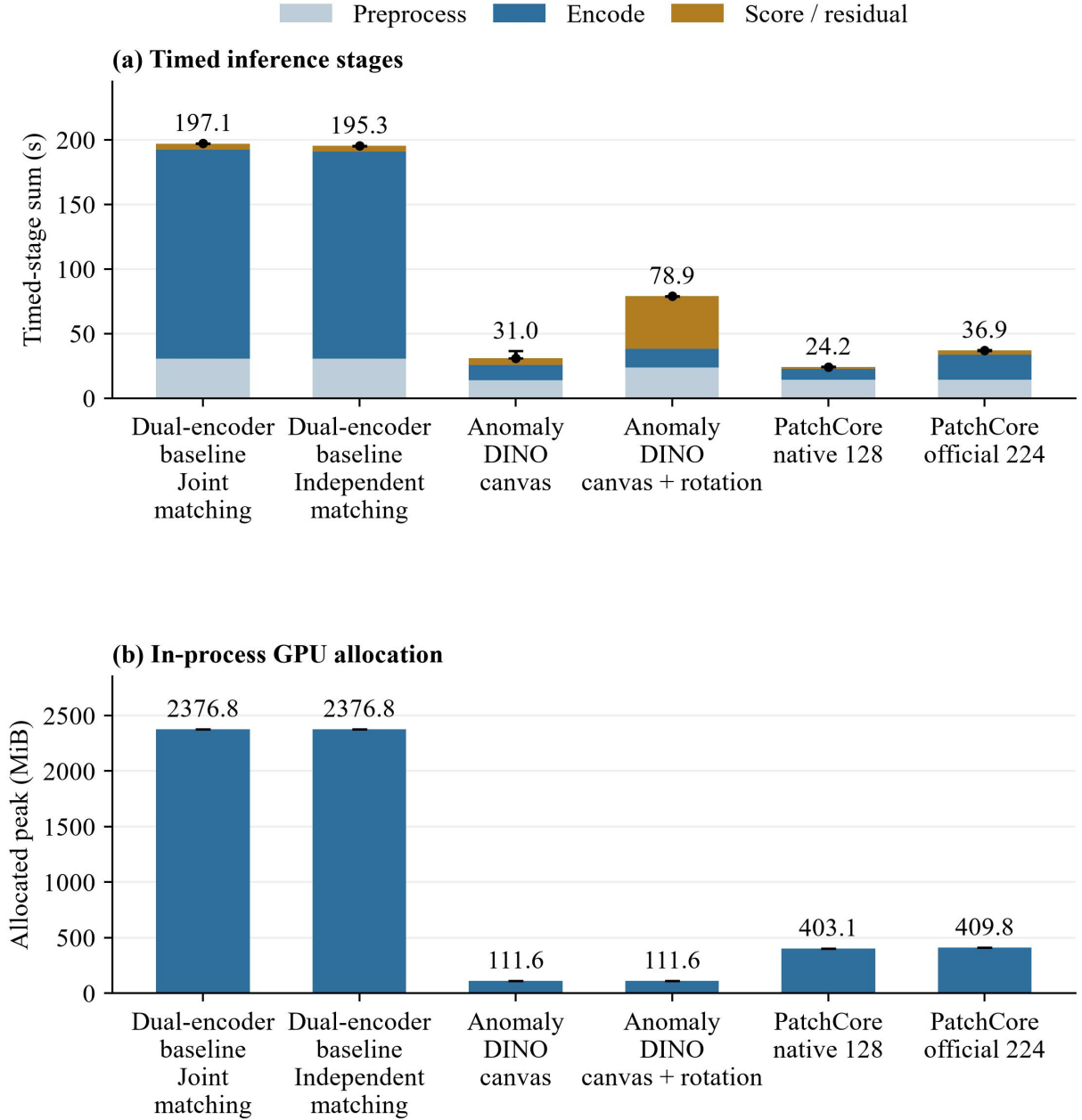


Figure S5. Timed inference stages and in-process GPU allocation on three MPDD categories (bracket_black, bracket_brown and bracket_white), seed 0 and $K = 1, 4$: six units, 216 unique queries and 432 query visits. One warm-up and three timed repeats are run per unit. (a) Bars stack the medians of repeat-level stage sums across six units; circles and whiskers show the median and observed min–max of their total across three repeats, not confidence intervals. Stage medians need not sum exactly to the median total. (b) Bars show the median PyTorch allocated-memory peak across 18 timed unit runs, with observed min–max. MiB denotes 2^{20} bytes. The Dual-encoder baseline holds DINOv2-B/14 and AnomalyCLIP visual in the same process. The AnomalyDINO entries use the method's no-rotation or reference-rotation protocol, and the PatchCore entries use the indicated native input resolution. All six configurations use the same instrumentation, but retain their native preprocessing and model protocols. Timing includes reference/query preprocessing, encoding and scoring; model loading, dataset setup, metric evaluation and result writes are excluded. PatchCore score time is the fit-plus-predict residual after measured preprocessing and encoding. These are timed-stage sums, not independently timed complete-process wall-clock or query-only latency.

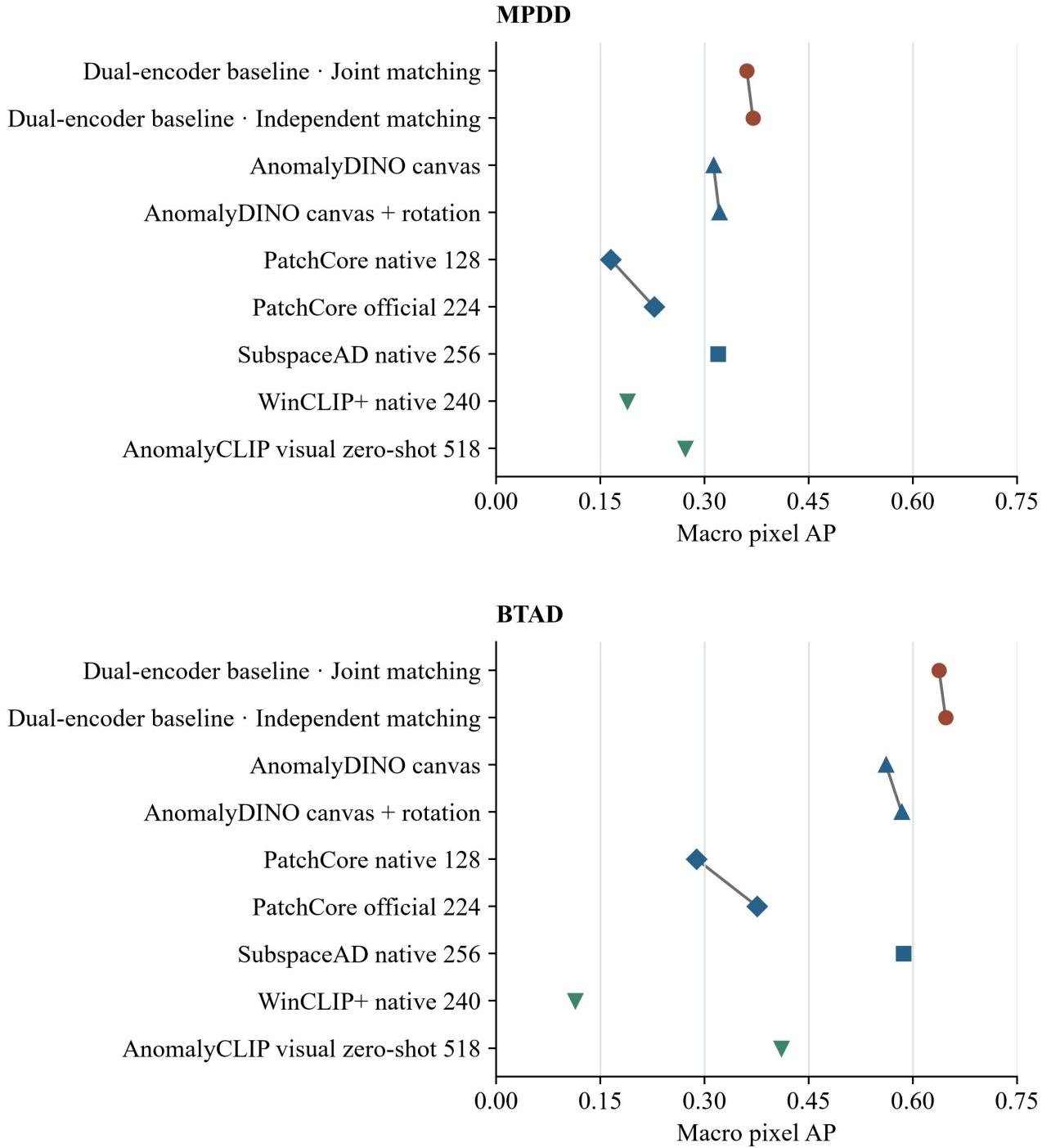


Figure S6. Sensitivity of the external-method context to the evaluated configuration. Each point is a dataset macro pixel AP from Table 12. Connectors link the Dual-encoder baseline under joint and independent matching, AnomalyDINO without/with reference rotation, and the two PatchCore configurations. Eight configurations average seeds 0 and 1 with $K = 1$ and 4; AnomalyCLIP has one support-free evaluation and is not paired with those four conditions. SubspaceAD uses the local 256-pixel fp16 setting. Fixed family grouping is not a performance ordering. These native or documented local configurations retain different input geometries and implementations. Lines describe configuration changes; no confidence interval or cross-method significance test is shown. The harmonised 448-pixel subset is reported separately in Table S1.

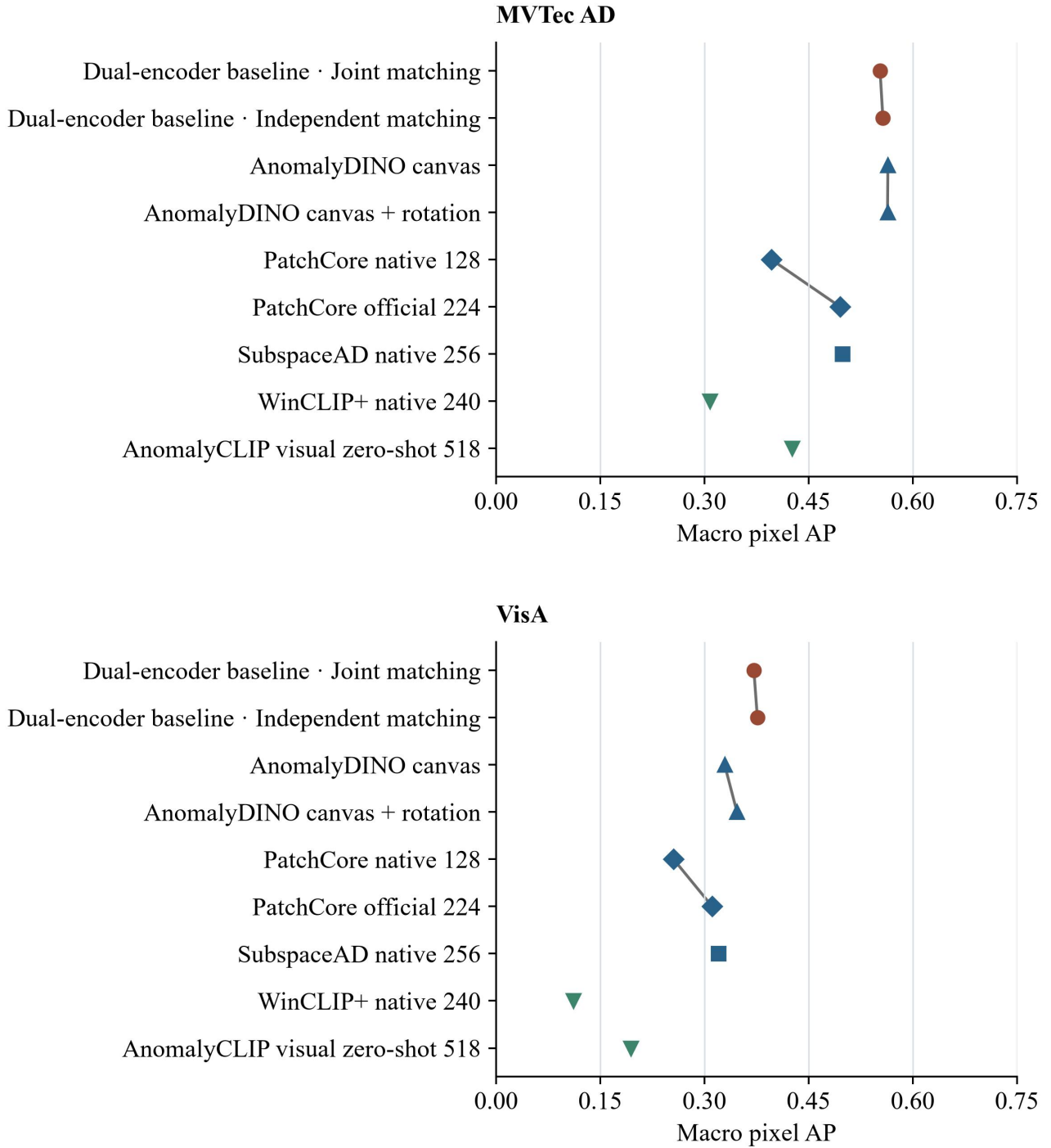


Figure S6 continued. MVTec AD and VisA under the same conventions. Values reuse the frozen external comparison summaries; no prediction or evaluation was rerun. Dataset roles remain those of the study, with VisA conservatively retained as in-domain. Table S1 covers only the five configurations compatible with its shared geometry at seed 0 and $K = 1$, not all methods plotted here.

The accompanying figure deck includes the complete set of 36 category-level comparisons with the native methods, using the same shared-region evaluation as Table 11. These selected examples show query images, ground truth and method score maps, but do not supply a predicted contour for every method. The two-output presentation of a continuous map plus a predicted contour applies only to the five main-text Dual-encoder baseline cases in Figures 6 and 7. The category-level panels supplement the aggregate results; they are not an independently sampled performance estimate. Their source records preserve the sample identities, per-image AP and common-region geometry.